5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO (5G20B)

Network Engineering Information

- Doc ID
- Version number 2.9
- Norbert Lukasiewicz, Krzysztof Wisniowski
- Approved
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Revision history and metadata

Document ID: DXXXXXXXXX
Document location:
Organization:

Version	Description of Changes	Date	Doc Owner	Doc Status	Reviewed by	Approver	Approval Date
0.1	Initial version covering Introduction and Dimensioning impact	15-08-2020	Norbert Łukasiewicz	Draft			
0.2	Update	08-10-2020	Norbert Łukasiewicz	Draft		Jacek Kotwiński	DD-MM-YYYY
0.3 – 0.9	Updates	20-10-2020	Norbert Łukasiewicz Krzysztof Wiśniowski	Draft	CFAM team PM NetEng members	Jacek Kotwiński	DD-MM-YYYY
1.0 – 1.9	Updates	20-10-2020	Norbert Łukasiewicz Krzysztof Wiśniowski	Release	CFAM team PM NetEng members	Jacek Kotwiński	DD-MM-YYYY
2.0	Update	14.12.2020	Norbert Łukasiewicz				
2.1	Update	18.12.2020	Norbert Łukasiewicz				
2.2	Update	05.02.2021	Norbert Łukasiewicz				
2.3	FGCR-11190 / FGCR-7659 added	17.02.2021	Norbert Łukasiewicz				

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Revision history and metadata

Document ID: DXXXXXXXXX
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Version	Description of Changes	Date	Doc Owner	Doc Status	Reviewed by	Approver	Approval Date
2.4	5GC001866 interworking + PM section update + minor updates	23.02.2021	Norbert Łukasiewicz				
2.5	New MoRSE simulation added based on MoRSE aligned with 5G20B product implementation	24.02.2021	Norbert Łukasiewicz				
2.6	Update to a new template	25.02.2021	Norbert Łukasiewicz				
2.7	Update of perfomrance related chapters	03.03.2021	Norbert Łukasiewicz				
2.8	Update of perfomrance related chapters	17.03.2021	Norbert Łukasiewicz				
2.9	Updates in Introduction, KPI&Counters, Technical Details, Feature Performance	20.04.2020	Norbert Łukasiewicz Krzysztof Wiśniowski				

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Feature scope matrix

List of features, subfeatures and CRs considered for this presentation

Sub-feature ID	Sub-feature name	Sub-feature description	Sub-feature dependency
5GC002200-A	Enable basic split panel 8x4 SU- MIMO	1. Add new enumeration to *dlMimoMode* in NIDD parameter, and pass it from CP to UP; All the functionality of this feature is controlled by the new enumeration. 2. Add the functionality of how to construct patternID from UE's CSI feedback for PMI beam and subpanel beam for CSI-RS. 3. Add 1 Cell specific CSI-RS, transmit via subpanel beam. 4. CP check UE capability and always configure 1 CSI-RS resource and N1=4, N2=1, extended precoding for the UE. Add one NIDD parameter to control maximum number of CSI-RS resource that can be configured to one UE. Only 1 can be configured for this subfeature. Only NIDD: *ssBurstSetPeriod* not longer than NIDD: *csiReportPeriodicity* is supported in this subfeature (note: the unit of the two parameters shall be united before comparation). 5. Consider UE capability of max 4/2 layer, as well as no UE capability available in SA mode. 6. Calculate beam weight /PatternID in CPRT and download it to OAM and then to RU, pass patternID to L2. 7. Transmit PDSCH using one/two PMI beam which is from UE's CSI feedback. 8. Add 3GPP standard 8 port precoding and port selection precoding for MU MIMO in the later subfeature; Add normalization of the precoding here too. 9. Transmit PDCCH via one x-pol PMI beam which comes from full panel. Add the logic of choosing one PMI beam from UE's CSI feedback. 10. Add updated dl power control in CP and UP. Add one separate NIDD parameter for cell specific CSI-RS power boost. 11. QCL in MAC CE for PDCCH is unchanged, still use SSB index to map TRS. Still configure PUCCH for 2 beam report. QCL for CSI-RS is not configured, i.e. the current approach is kept. 12. Only gen 4 CPRI RU, only 1 basic SISO configure in 20B. Only RU with 8 cross polar TRX columns. 13. Add PUCCH UCI retrieving for CRI/PMI/LI, only consider PUCCH format 2. 14. L1 part PDCCH precoding [1 0 1 0] and [0 1 0 1] with and L2-L1 interface update.	5GC000605 DL SU adaptive 4x4 MIMO (open loop) and its spill over features: 5GC001870, 5GC001856 (closed loop). 5GC001866 Delivery of beamset realization to gNB (OAM feature)

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Feature scope matrix

List of features, subfeatures and CRs considered for this presentation

Sub-feature ID	Sub-feature name	Sub-feature description	Sub-feature dependency
5GC002200-B	Further split panel SU MIMO functionality	1. Additional subpanel split for RUs with different number of TRX columns/rows, i.e. 8/4 cross polar TRX columns, 4/2/1 cross polar TRX rows. N1, N2 is relevant to the split method. 2. Additional subpanel beams with different direction and width. 3. CP configure the best possible CSI-RS resource number and relevant N1,N2 for the UE, based on UE capability, subpanel capability and operator configuration from NIDD parameter. Simultaneous capability is handled here. Interworking with DRX and Measurement Gap is considered. 4. Calculate beam weight for different CSI-RS resource number, different RU and different N1N2.	5GC000605 DL SU adaptive 4x4 MIMO (open loop) and its spill over features: 5GC001870, 5GC001856 (closed loop). 5GC001866 Delivery of beamset realization to gNB (OAM feature).
5GC002200-C	EIRP control with PMI beam	1. Calculate the beam response for orthogonal DFT beams. 2. Expansion of PMI beams in terms of DFT beams. 3. Calculate the beamforming gain for each PMI beam based on the beam response for orthogonal DFT beams and the correlation between PMI beam and DFT beams.	5GC001329 EIRP monitor 5GC002052 EIRP control 5GC002300 Actual EIRP control extensions. The EIRP monitor and EIRP control shall be impacted by PMI beams.
5GC002200-D	Additional optimization to beamforming functionality	1. RRC reconfigure UE to max 1 PDSCH layer, (N1,N2)=(4,1) to limit uci bit for CSI feedback as 10 bit for cell edge UE(link quality deteriorates). RRC reconfigure UE to max PDSCH layer that UE capability supported when the link quality becomes better. 2. Add separate NIDD parameter for cell CSI-RS period to improve the performance of Gap and DRX interworking as well as timeliness of CSI report.	5GC002200-A 5GC002200-B

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Feature scope matrix

List of features, subfeatures and CRs considered for this presentation

Sub-feature ID	Sub-feature name	Sub-feature description	Sub-feature dependency
5GC002200-E	Interworking with MU MIMO, FDM, different baseband block configuration	1. L2 choose Rank 2 Port Selection(PS) for PDSCH for MU paired UE if needed. 2. Consider different subcell type configuration for with/without MUMIMO. 3. Interworking with FDM is considered.	5GC000524 DL MU-MIMO for Digital Beamforming. This feature will calculate Multi users' orthogonality using beam pattern.
5GC002200-F	Previous subfeatures for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO	Previous subfeatures for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO.	5GC002200-A 5GC002200-B 5GC002200-C 5GC002200-D 5GC002200-E 5GC000534 eCPRI FR1 Beamforming.

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FGCRs

List of features, subfeatures and CRs considered for this presentation

FGCR ID	Sub-feature name	FGCR description	Link to JIRA
FGCR-11190	AEQD: Enable 2200 4x8 CSI-RS Codebook based beamforming in absence of 1866 Delivery of beam set realizations	FR1 TDD 3rd Gen CPRI RU AEQD does not support 5GC001866 Delivery of beam set realization to gNB feature in 5G20B schedule. Support was moved out from 5G20B earlier with FGCR-7659 as RF DU schedule was not fitting to 5G20B. 5GC001866 is mandatory precondition for 5GC002200 Support 4 x 8 port CSI-RS codebook for SU-MIMO. So before this FGCR AEQD cannot support 5GC002200. Vodafone team escalated, as for them 5GC002200 with AEQD is mandatory feature in 5G20B. Due to that, workaround solution was invented to support 5GC002200 with AEQD with modified 5GC001866 functionality. This CR introduces the WA solution in 5G20B.	Link
FGCR-7659	5G20B Operability descopes (Move content from 5G20B to 5G21A/5G21B)	**5GC001866** can't be supported for all requested RU types in 5G20B. With this CR support of following RU is postponed to 5G21A: Varinat | Chip | Program | Type | Ext antennas | cmWave / mmWave | Protocol | Sales Item Name | RF HW family group | 1866 Sub-feature mapping AEQN | Beamer FPGA | AEQN | 32T32R | NA | cmWave | CPRI | AEQN AirScale MAA 32T32R 96AE n78 80W (CPRI) | cMAA_Gen4.0_32T32R | 1866-A AEHA | ** | 5G Gen3 Sub6 GHz | 64T64R | NA | cmWave | CPRI | AEHA AirScale MAA 64T64R 128AE n41 200W | 5Ggen3_LowF_64T64R | 1866-E AEQD* | ** | 5G Gen3 Sub6 GHz | 64T64R | NA | cmWave | CPRI | AEQD AirScale MAA 64T64R 128AE n78 200W | 5Ggen3_LowF_64T64R | 1866-E AZQG | ** | Pulsar | 8T8R | 8 | cmWave | CPRI | AZQG AirScale RRH 8T8R n78 320W (CPRI) | cMAA_Gen4.0_8T8R | 1866-J AZQH | ** | Pulsar | 8T8R | 8 | cmWave | CPRI | AZQH AirScale RRH 8T8R n78 320W (CPRI) | cMAA_Gen4.0_8T8R | 1866-J AZQL | ** | Pulsar | 8T8R | 8 | cmWave | CPRI | AZQL AirScale RRH 8T8R n78 320W (CPRI) | cMAA_Gen4.0_8T8R | 1866-J note that for some units, only CPRI mode is postponed while eCPRI is kept. * - for AEQD see FCGR-11190 above	Link

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5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO



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Introduction



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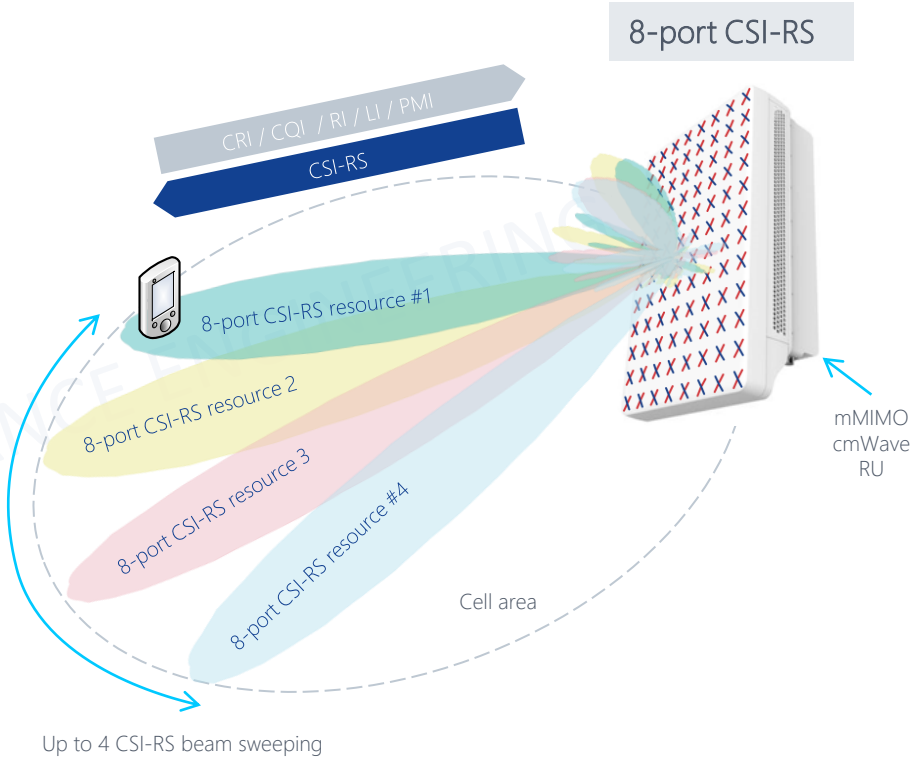
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Introduction

5GC002200 in nutshell

The feature introduces the following functionalities:

- **Cell specific CSI-RS acquisition** with **8 ports**.
- Type I Single Panel **Codebook based beamforming.**
- **Closed loop SU-MIMO**, and UE report **cri-RI-LI-PMI-CQI** based on selected best beam.
- Support **CSI-RS beam sweepping** and **up to 4** beamformed **CSI-RS resources**.
- Feature is applicable for **64TRX, 32TRX, 8TRX TDD FR1** digital beamforming capable radios (**CPRI/eCPRI**), supported by **5GC001866** feature
- Interworking with **beam vertical tilt**, **MU-MIMO**, **EIRP control** and **DRX features**.



CQI (Channel Qulity Information)
PMI (Precoding Matrix Indicator)
CRI (CSI-RS Resource Indicator)
SSBRI (SS/PBCH Resource Block Indicator)
LI (Layer Indicator)
RI (Rank Indicator) an/or L1-RSRP

Cell specific CSI-RS helps to reduce signalling overhead in loaded cells

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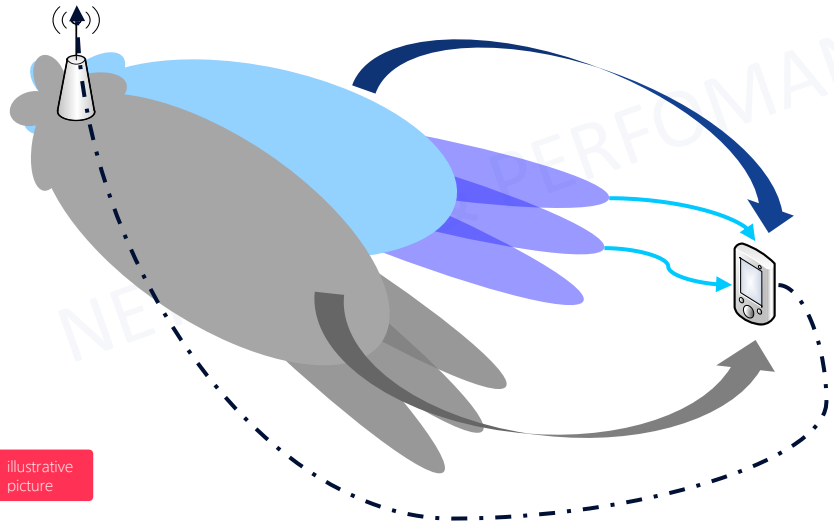
Before 5GC002200

Introduction

5GC000533 Digital Beamforming for CPRI RUs

In 5GC000533 Digital Beamforming for CPRI Based RUs, beam management is based on:

- transmission of **SSB beams** and **CSI-RS for beam refinement**
- measurement and **reporting (resource ID and RSRP) of best beams** by UE



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measure

- SBRI-RSRP (SSB beams)
- CRI-RSRP (refined beams)

report

- 2 best SBRI-RSRP or
- 2 best CRI-RSRP

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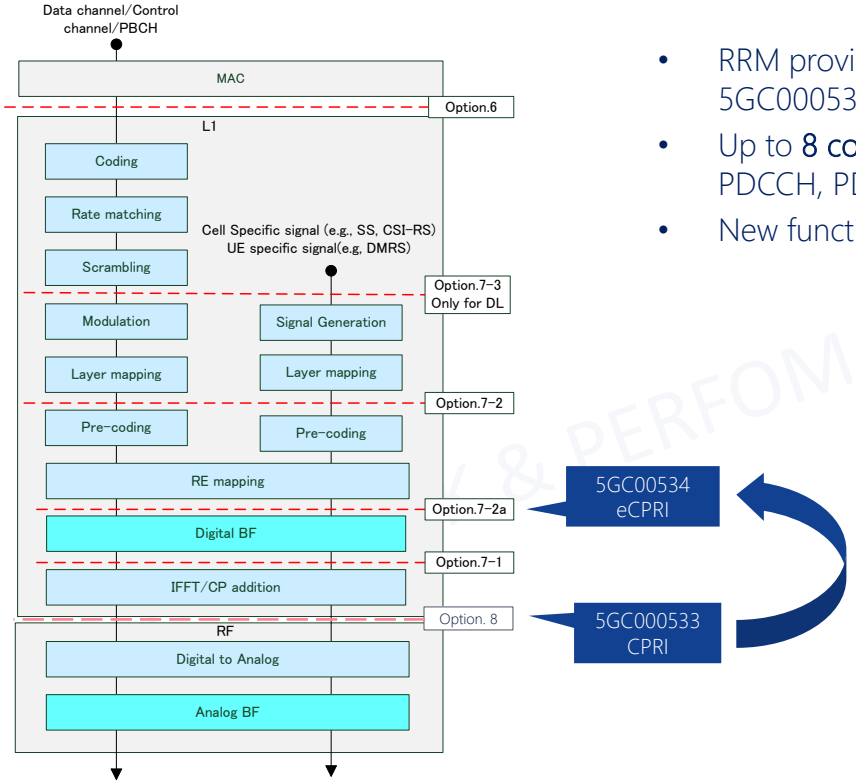
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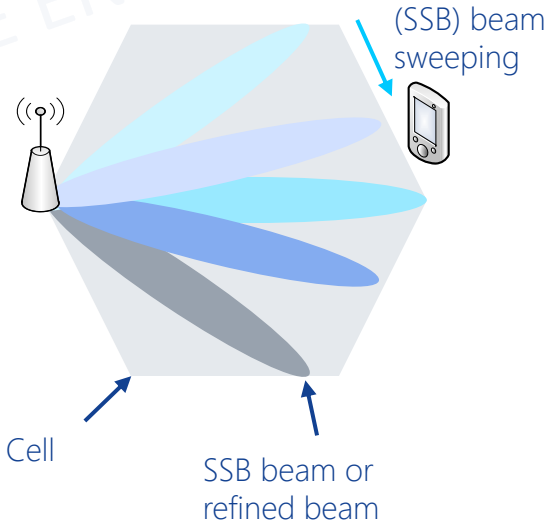
Before 5GC002200

Technical Details

5GC000534 Digital Beamforming for eCPRI based RUs



- RRM provides **support for eCPRI connected RUs** by adapting the feature 5GC000533 to the new interface.
- Up to **8 coarse (SSB) beams**, **up to 32 refined beams** on sub6GHz for PDCCH, PDSCH, PUCCH, PUSCH.
- New function **split for L1** is introduced.



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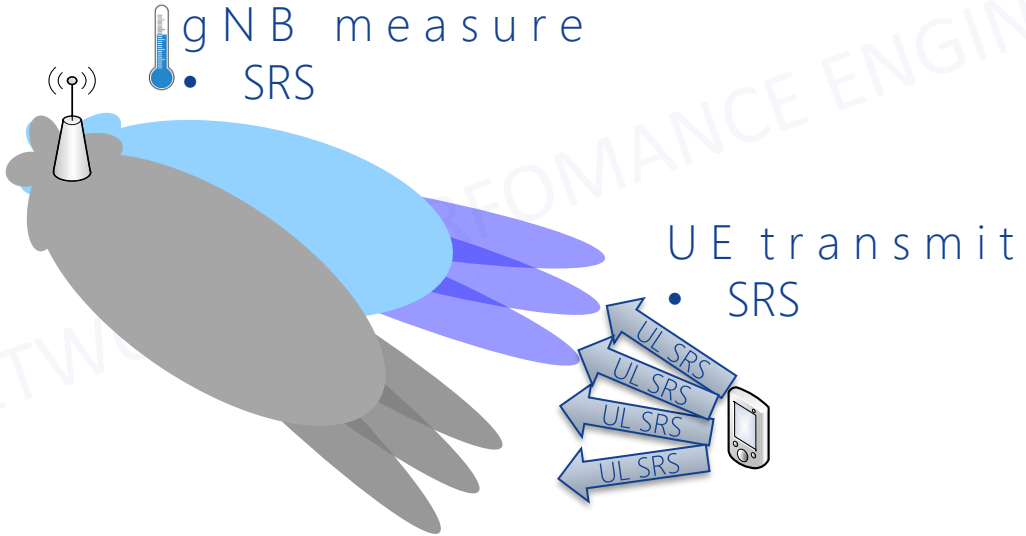
Before 5GC002200

Introduction

5GC001258 Sounding based beamforming for GoB sub 6GHz for CPRI

In 5GC001258 Sounding based beamforming for GoB sub 6GHz for CPRI, beam management is based on:

- SRS evaluation on the refined beams



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Feature not available from 5G20B onwards

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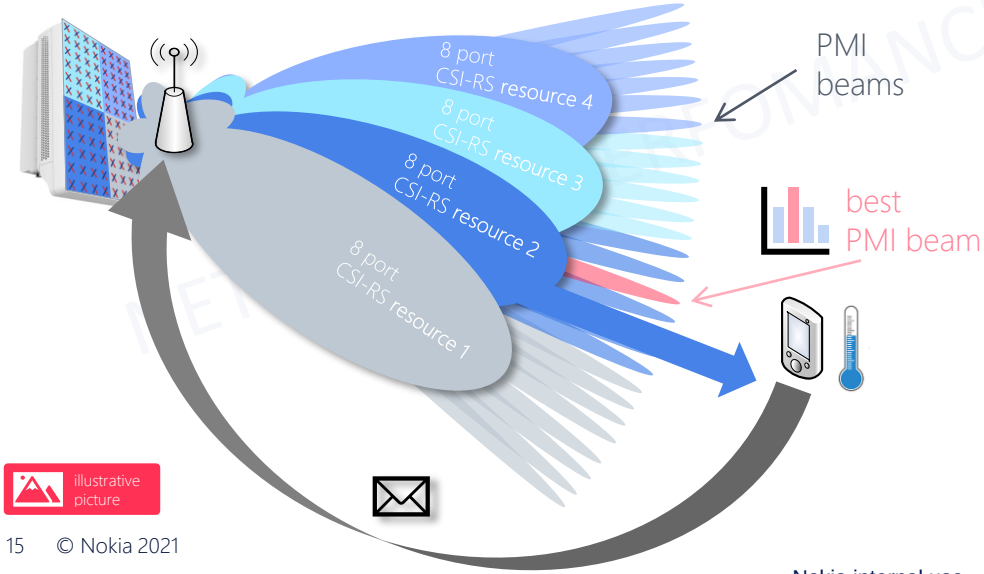
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After 5GC002200

Introduction

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO (1/2)

- Support **up to 4 beamformed CSI-RS resources**, where each CSI-RS **resource conveys 8 ports**.
- Firstly a **UE determines the best CSI-RS resource from plain power measurements** and hence chooses one of the 4 beams, e.g. it chooses beam with CSI-RS resource #2.
- This is a very robust choice since the CSI-RS are beamformed **and decision is only on power** without the need for error prone sophisticated math.
- Secondly a **UE choses a PMI beam using the 8-port codebook** e.g. in this case the red pencil beam within the resource 2 beam.
  The **UE reports CRI/CQI/RI/LI/PMI based on the selected best CSI-RS beam** using an 8-port codebook.
- **PBCH/PSS/SSS are transmitted on legacy SSB beams**, feature is DL specific.



UE measure
- CSI-RS (CSI-RS beams)

UE report
- CRI/RI/PMI

gNB select
- best PMI beam (up 2) for PDSCH

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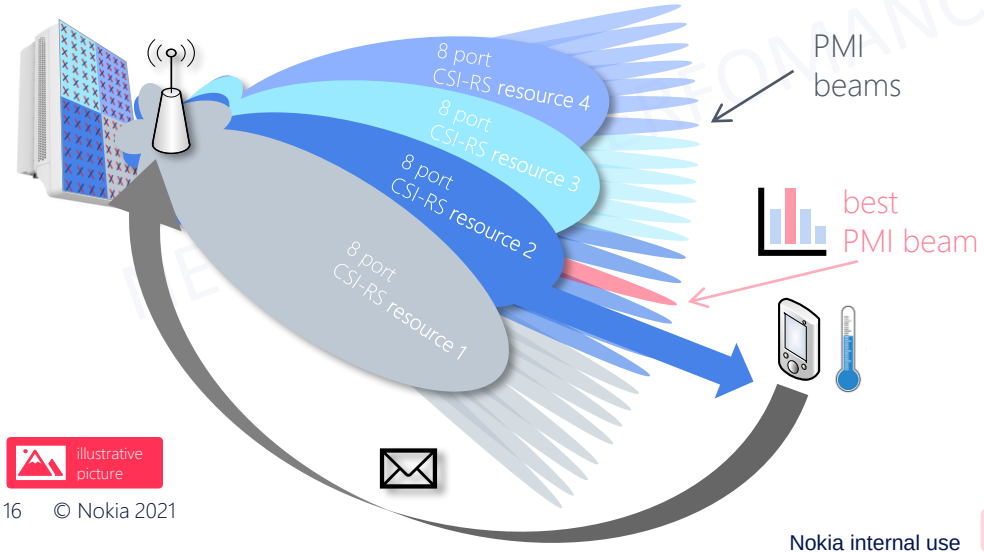
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After 5GC002200

Introduction

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO (2/2)

- 5GC002200 supports DL closed loop MIMO based maximum four 8-ports cell specific CSI-RS for channel acquisition.
- Before this feature, the baseline is DL closed loop MIMO based One 4-ports UE specific CSI-RS for channel acquisition.
- This feature only changes transmissions for DL PDCCH/PDSCH, with no touch for UL channels and beams for SSB.
- 5GC002200 changes beams for PDCCH/PDSCH because it adopts one mechanism that precoding matrices based on PMI info from CSI feedback is part of the beam weight vector.
- This feature does not change beams for SSB, common channels like SIBs, paging.
- This feature is independent with SSB/refined beam report, which can still be configured to UE (DL channels for UE can fall back to refined beams in case of e.g. missing CSI-RS report) and the beam index from UE's beam report can still be adopt for UL channels or DL common channels and etc.



illustrative picture

UE measure
- CSI-RS (CSI-RS beams)

UE report
- CRI/RI/PMI

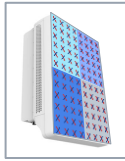
gNB select
- best PMI beam (up 2) for PDSCH

- interworking with refined beams in scope of pending PR571608

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Introduction

Available beams before & after introduction of 5GC002200

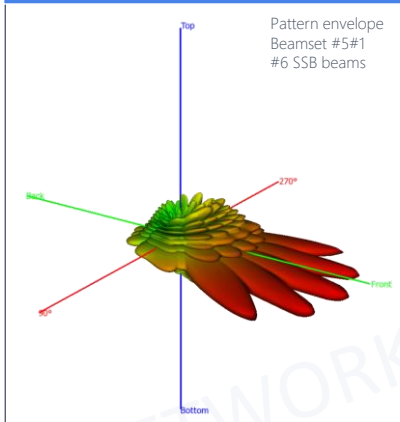
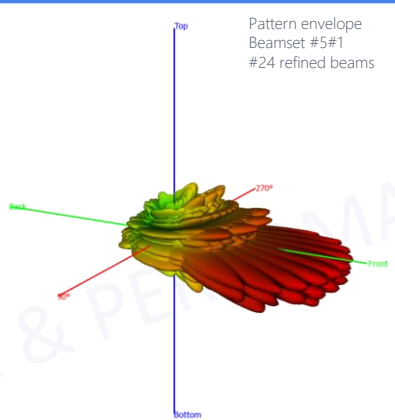
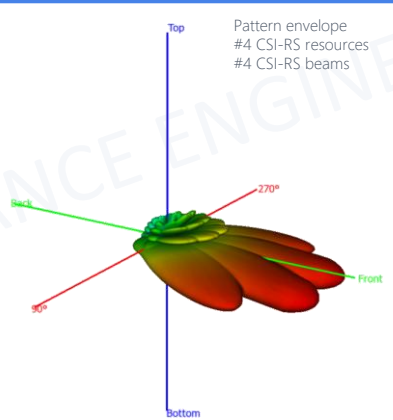
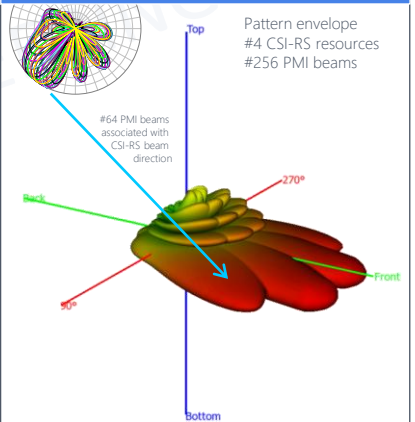


Example beams:
- Radio Unit AEHC 64TRX
- Beamset #5#1
- Panel split (2,2)
- #4 CSI-RS resources
- Beam patterns envelope

illustrative picture

After introduction of 5GC002200

Before introduction 5GC002200

SSB/Coarse Beams	Refined Beams	CSI-RS Beams	PMI Beams
			
• SSB beams are mandatory and remain after 5GC002200 is activated • SSB beams sending Synchronization/Broadcast/PDSCH • Up to 8 beams (beamset specific)	• Refined beams are optional (see restrictions in Known Issues subchapter). • DL channels for UE can fall back to refined beams in case of e.g. missing CSI-RS report. • Up to 32 beams (beamset specific).	• CSI-RS beams sending Reference Signals • Up to 4 beams (RU/UE capability /maxCsirsNumber specific)	• PMI beams sending PDSCH • Up to 256 beams (RU/UE capability /maxCsirsNumber specific)

UE interworking issues

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Introduction
Before & after

Before

- 2x2/4x2/4x4 MIMO
- UE specific CSI-RS
- One 2 or 4 antenna port CSI-RS resource is used for periodic channel state measurements.
- UE is configured with unique CSI-RS resource, where both resource location and resource sending periodicity / slot offset are dedicated to it.

After

- 8x4 MIMO
- Cell specific CSI-RS
- CSI-RS resource(s) used for periodic channel state measurements are pre-allocated at cell level and shared by all the UEs served by this cell.
- Improved 4x4 MIMO rank 3/4 throughput.
- Improved downlink average/cell edge throughput.
- Better spectral efficiency.
- Reduced CSI-RS for CSI Overhead for large amount of RRC connected UEs.
- Robust choice of user beams by 2 stage approach.

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Technical Details

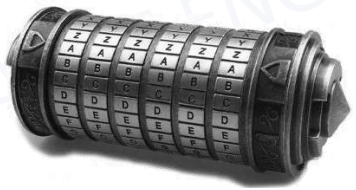


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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Rel.15
Codebook



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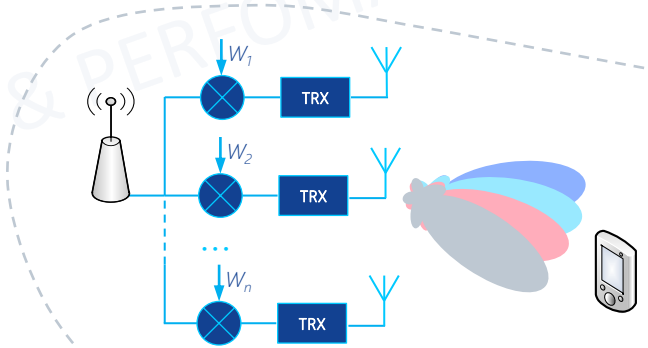
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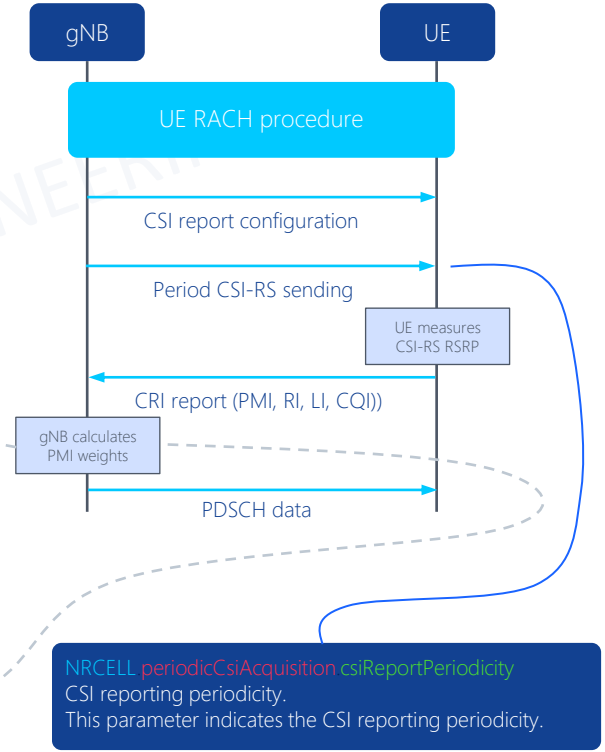
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Technical Details
Weighting principles

- In downlink beamforming **gNodeB can weight transmitted signal** to generate directional narrow beam towards UE.
- The transmitted signal is weighted using different weights w (w_1, …, w_n) on different antenna ports to change the **amplitude and phase** of the signal.
- After the signal energy radiated by the antennas is combined and beam is generated **towards the UE**.
- The more the antennas, the narrower the beam and the **more flexible** the beam direction.





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Technical Details

Rel.15 Codebook (1/3)

- The Rel15. 'CLASS A' codebook adopts the dual-stage codebook structure, what means that each precoding matrix W can be decomposed into two matrices: W1 and W2 so that:

$$W = W_1 \cdot W_2$$

- Where:
  - W_1 is considered to be related to long term properties of multipath channel. It is also considered to be wideband. Due to that W_1 is considered to be reported less frequently
  - W_2 is considered to be related to short term properties of multipath channel. It is also considered to be sub-band (narrowband). Due to that W_2 is considered to be reported more frequently

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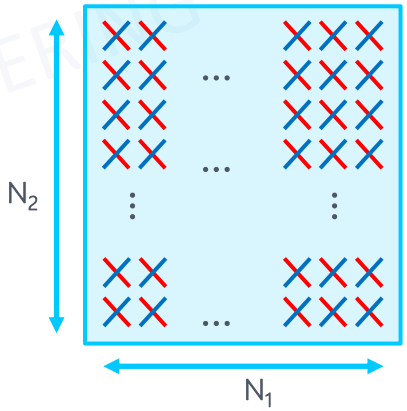
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Technical Details

Rel.15 Codebook (2/3)

Rel.15 'CLASS A' codebook is configurable with following parameters:

- N_1 - antenna port number per polarization in horizontal direction
- N_2 - antenna port number per polarization in vertical direction
- O_1 – oversampling factor in horizontal direction
- O_2 - oversampling factor in vertical direction
- Codebook Config - defines the construction of W_1, which applies to different antenna deployments (valid only when $N_2 > 1$)



- in 3GPP 38.214 number of CSI-RS ports P_{CSI-RS} is calculated as $2N_1N_2$
- both O_1 and O_2 are not strictly defined in 3GPP

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Technical Details
Codebook (3/3)

illustrative picture

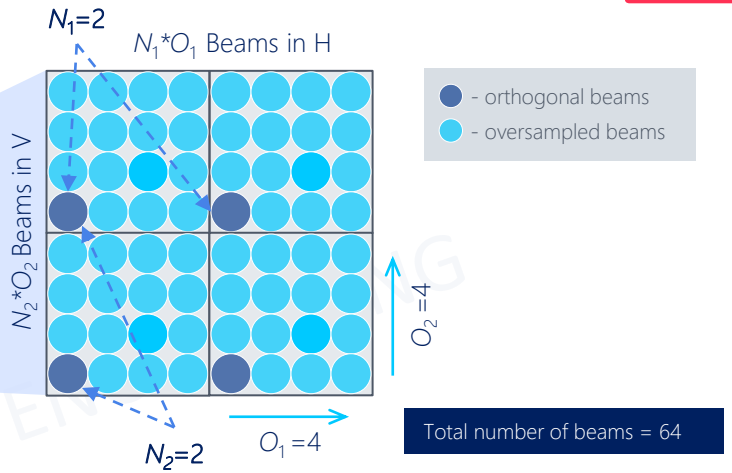
3GPP: 38.214 Type 1 Single Panel Codebook:
Supported configurations of (N_1, N_2) and (O_1, O_2)

Number of CSI-RS antenna ports, P_{CSI-RS}	(N_1, N_2)	(O_1, O_2)
4	(2,1)	(4,1)
8	(2,2)	(4,4)
	(4,1)	(4,1)
12	(3,2)	(4,4)
	(6,1)	(4,1)
16	(4,2)	(4,4)
	(8,1)	(4,1)
24	(4,3)	(4,4)
	(6,2)	(4,4)
	(12,1)	(4,1)
32	(4,4)	(4,4)
	(8,2)	(4,4)
	(16,1)	(4,1)

Configurations supported by 5GC002200

N_1O_1 - total number of beams in horizontal domain

N_2O_2 - total number of beams in vertical domain



Example for 1 CSI-RS resource

- Codebook is constructed in such a manner that some beams (the basic set) are strictly mutually orthogonal
- Total number of orthogonal beams is $N_1 \cdot N_2$
- In order to increase number of beams, phase shifting technique also called oversampling technique is used
- Thus oversampling factors O_1 and O_2 says what is the granularity and the total number of beams

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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

DL precoding and beamforming architecture



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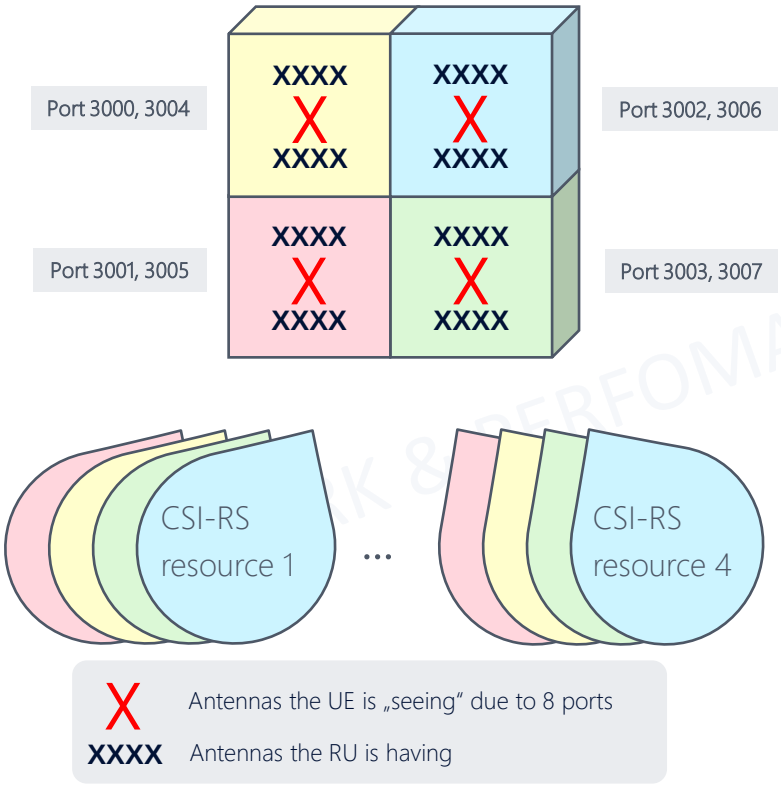
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Technical Details

Basic principles of the 2_2 panel split



- Each CSI-RS port is transmitted over 8 TX i.e. 2 rows and 4 columns using beamforming over the columns and rows.
- This delivers decent CSI-RS power due to coherently summing up the power of the antennas.

- PMI within CRI report determines:
  - PDSCH transmission via one or two PMI beams:
    - 1 layer – one PMI beam
    - 2 layers – one or two PMI beams
    - 3 or 4 layers – two PMI beams
  - Single beam transmission for PDCCH.

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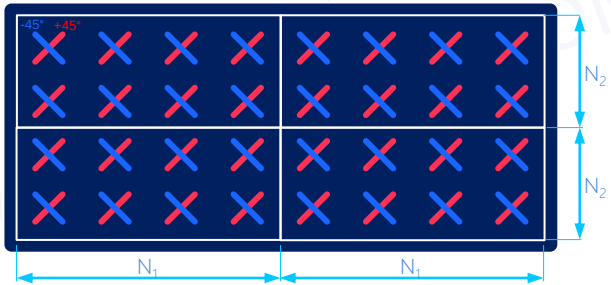
illustrative picture

Technical Details

Panel split – supported configurations

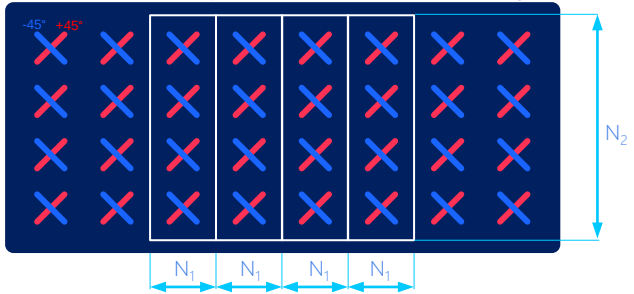
- The precoding matrix **based on 8 ports CSI feedback is 8xR**, where R is the number of layers and is determined by Rank Indicator.
- The **8 data streams** after precoding will be transmited **via 4 sub-panels and 2 polarizations** of Radio Unit.
- The precoding matrix is constructed based on N_1 and N_2, which is configured by high layer parameter n_1-n_2 respectively.
- As defined in *3GPP 38.214* 8 port CSI feedback supports $(N_1,N_2)=(2,2)$ or $(N_1,N_2)=(4,1)$ and the number of CSI-RS ports: $P_{CSI\text{-}RS}=2*N_1*N_2=8$.
- This feature links (N_1, N_2) to sub-panel relevant position, i.e. N_1 indicates the **number of sub-panels in one row**, N_2 indicates the **number of sub-panels in one column**.

2_2 Cross Split



The feature supports cross split of one RU. if RU has more than 1 TRX rows (noted as 2_2 split). 2_2 split introduces 2 sub-panels per row and 2 rows in total $(N_1,N_2)=(2,2)$.

4_1 Horizontal Split with Extended Precoding



For RUs with more than 4 TRX columns, feature supports horizontal extended precoding, where CSI-RS is transmited via 4 of the TRX columns and PDSCH is extended to more than 4 TRXs.

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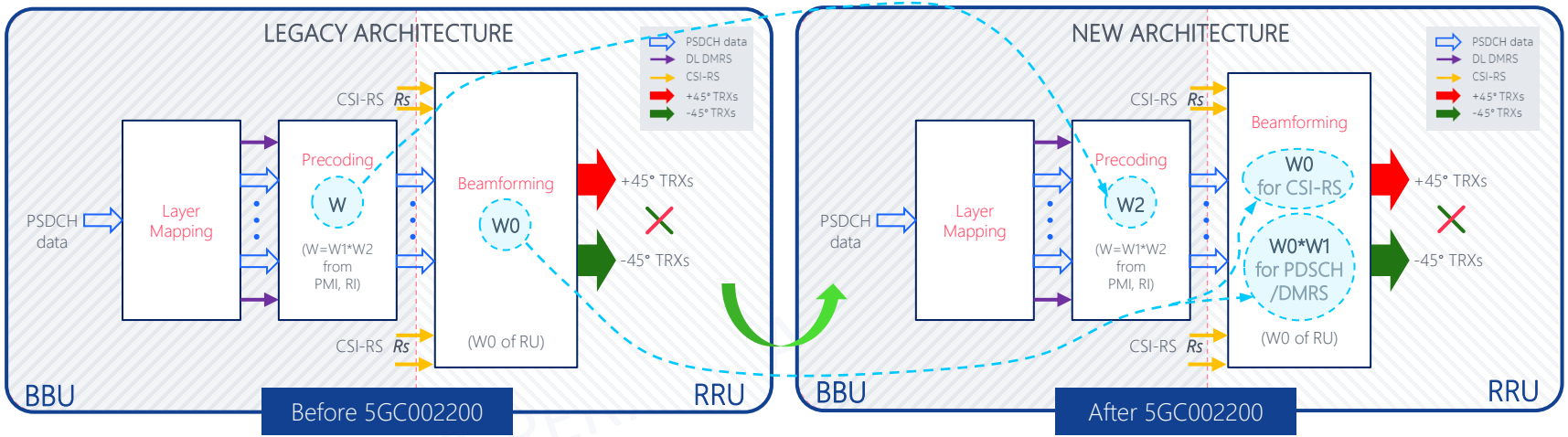
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illustrative picture

Technical Details

DL precoding and beamforming architecture change (1/2)



- Whole **precoding is done at BBU** side and **beamforming is done at RRU side**.
- Overall **8 fronthaul streams are needed** for 5GC002200 feature.
- The **feature splits 8xR precoding matrix W to two parts: 8x4 (W1) and 4xR (W2)**, to keep the maximum fronthaul streams at 4 per UE (2 of the fronthaul streams are empty for rank 1). After port selection of rank 2 for MU-MIMO, 2 of the fronthaul streams are also empty for rank 2.
- W2 will be handled as precoding at L1, **W1 will be combined to beamforming weight W0 at RRU side**.
- **New beamforming weight and precoding matrix are introduced by this feature**, and here define **new beamforming weight vector combining W1 and W0 as PMI beam**.
- The **dimension of PMI beam weight vector is exactly the same as legacy full panel beamforming weight vector** and **is relevant to RU's TRX number for one polarization**.

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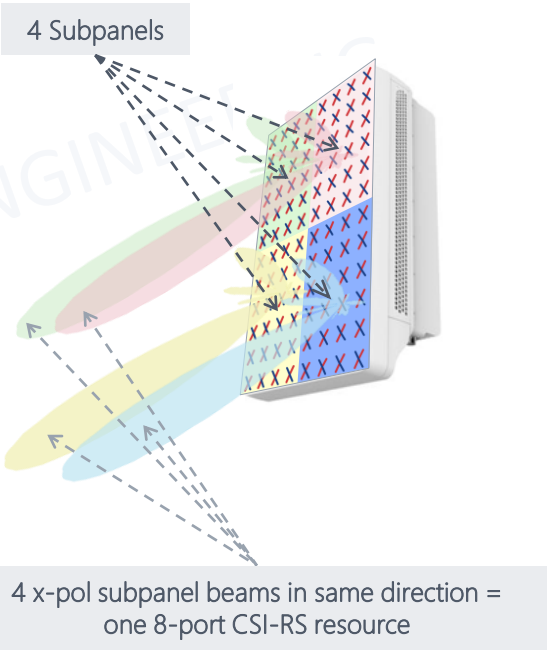
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illustrative picture

Technical Details

DL precoding and beamforming architecture change (2/2)

- The data streams after 8 port precoding matrix is transmitted **via 4 subpanels and 2 polarizations of one RU** - this point is not changed with the new precoding and beamforming architecture, only the new architecture achieves this in two steps.
- Each subpanel has **more than 1 TRX** which allows adding subpanel beamforming effect.
- This feature supports **maximum 4 subpanel beams (4 beam directions)** based on UE capability, RU capability and operator configuration.
- Each 8-port CSI-RS resource is transmitted **via 1 subpanel beam (1 beam direction)**.
- All the 4 cross polarized subpanels have **the same beam direction at the same time for one UE** and the beam direction for one UE are decided via its CRI field in CSI feedback.
- Feature supports **4/2/1 directions** to cover the whole cell.



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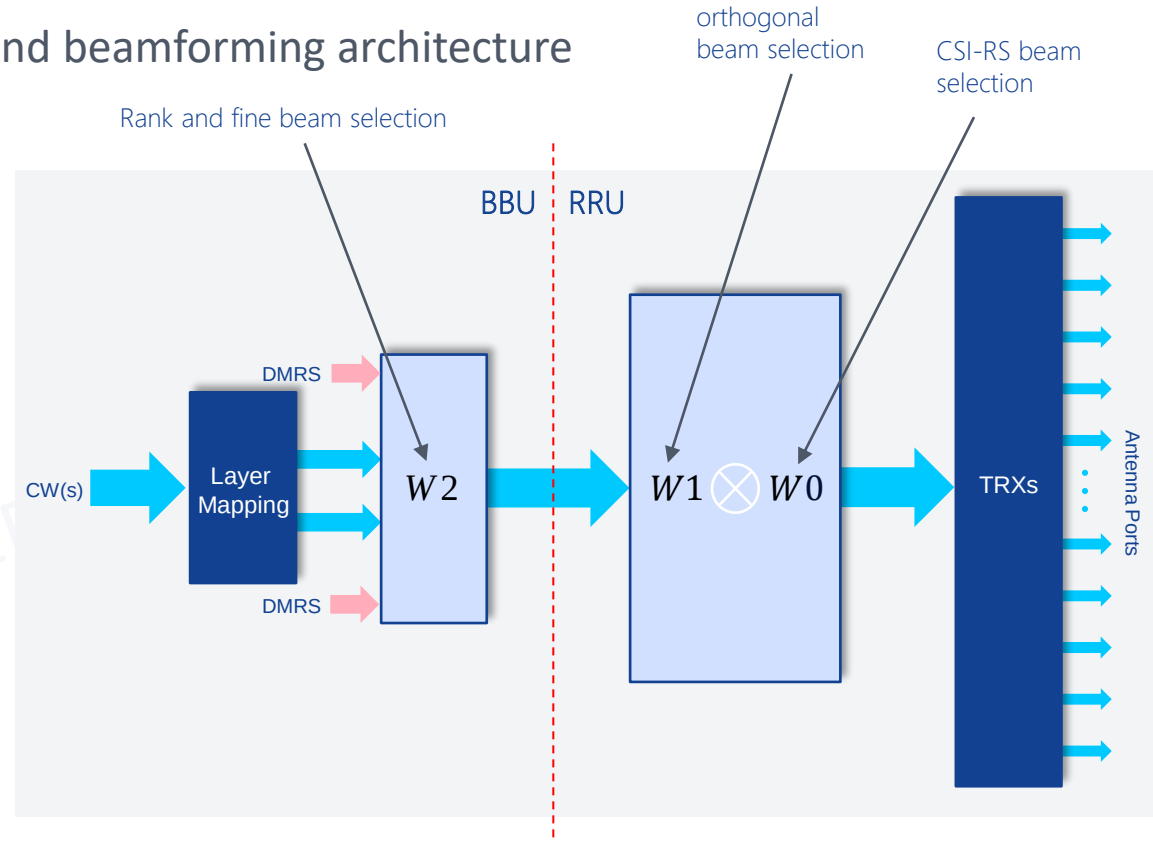
Summary of new DL precoding and beamforming architecture

- The **PMI** reported from UE **can be split** as:

  $W=W1*W2$

  where:

  - *W1* - is the 2D DFT beam proposed by UE based in N1*N2 configured.
  - *W2* - is the beam selection and co-phasig for 2 polarizations.
- For split panel solution, **each CSI-RS port** is mapped to **one sub-panel** with its **own beam**, normally same beam (*W0*) for all ports.
- The ***W1* reported from UE** can be **combined with the beam *W0*** used for split panel
  - The streams required to send to Radio Unit by BB over fronthaul will then reduce to number of layers to max 4 only.



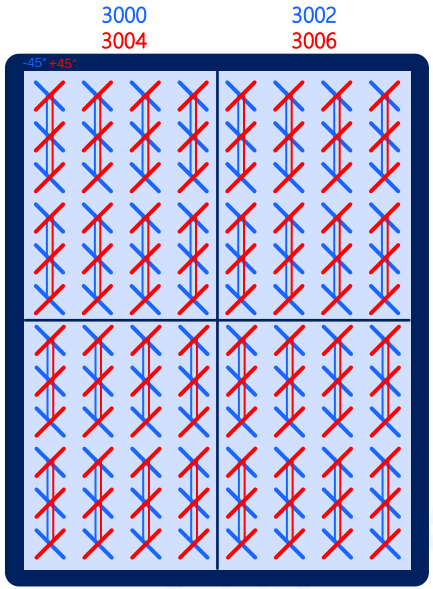
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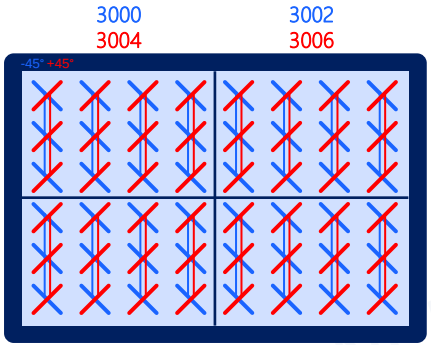
illustrative picture

Technical Specification

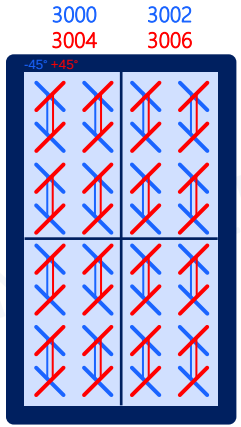
Examples 8 CSI-RS port configurations for selected 64TRX/32TRX/8TRX Radio Units



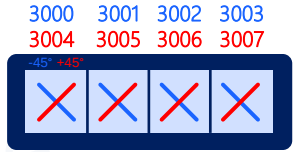
64TRX Radio Units e.g. AEHC



32TRX Radio Units e.g. AEQN



32TRX Radio Units e.g. AAHF



8TRX Radio Units e.g. AZQG

Up to 4 CSI-RS resources to enable 4 beam sweeping of each split panel

Up to 4 CSI-RS resources

Up to 2 CSI-RS resources

1 CSI-RS resource with sector beam

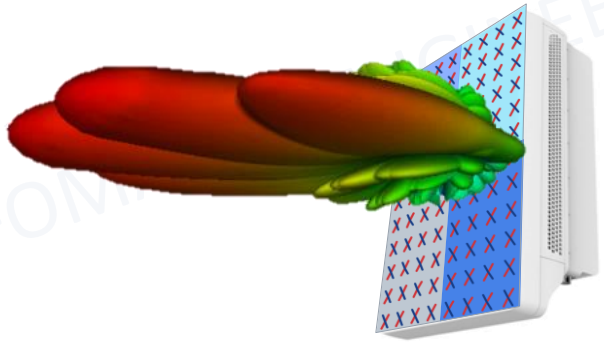
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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

CSI-RS aspects



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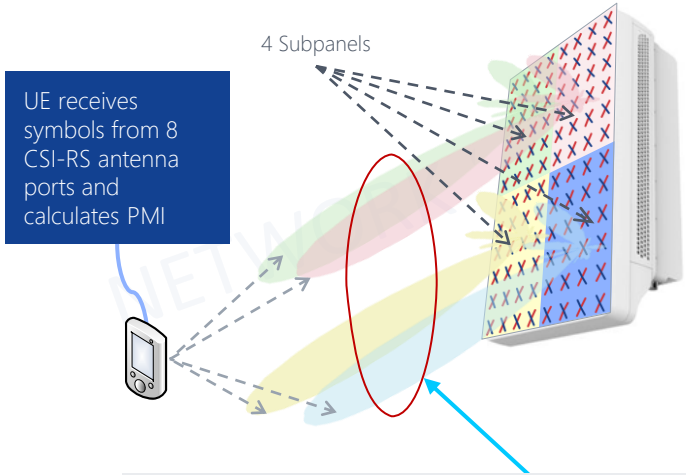
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Technical Details

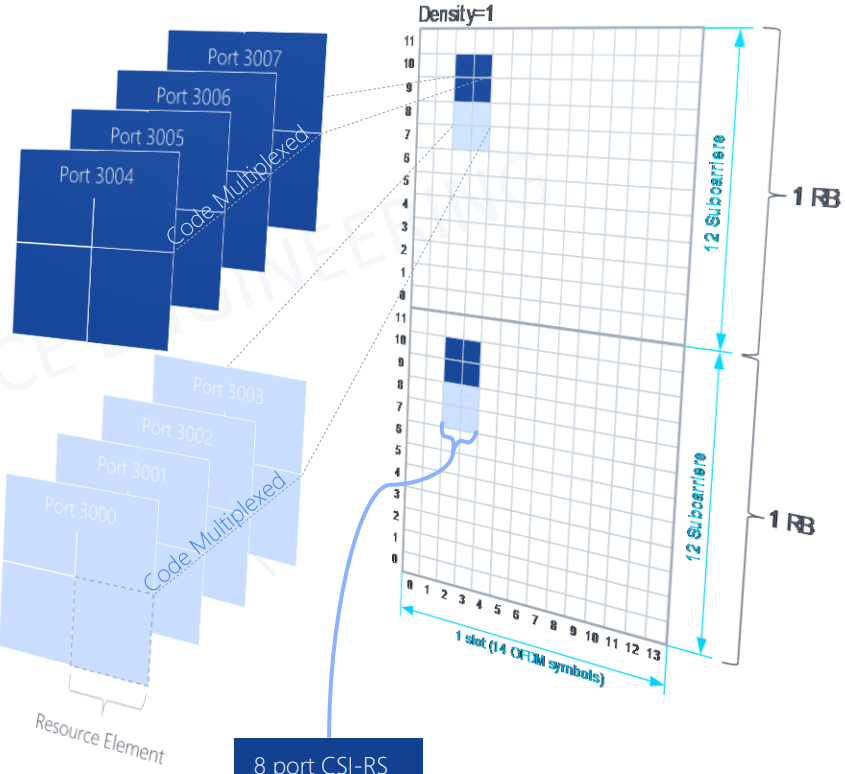
CSI-RS resource explained (1/3)

- In a given slot **1 CSI-RS resource** is transmitted.
- For the CSI-RS port transmission RU is split into **4 subpanels**.
- Each subpanel transmits **1 cross-polarized beam**.
- Each beam carries **2 CSI RS ports** (1 per polarization).
- Subpanel beams of the same CSI-RS resource are transmitting in the **same direction**.



4 x-pol subpanel beams in same direction = 1 CSI-RS resource = 1 CSI-RS beam

CSI-RS Location within a Slot
(row8: CDM4-FD2-TD2)



8 port CSI-RS resource occupies 2 symbols

illustrative picture

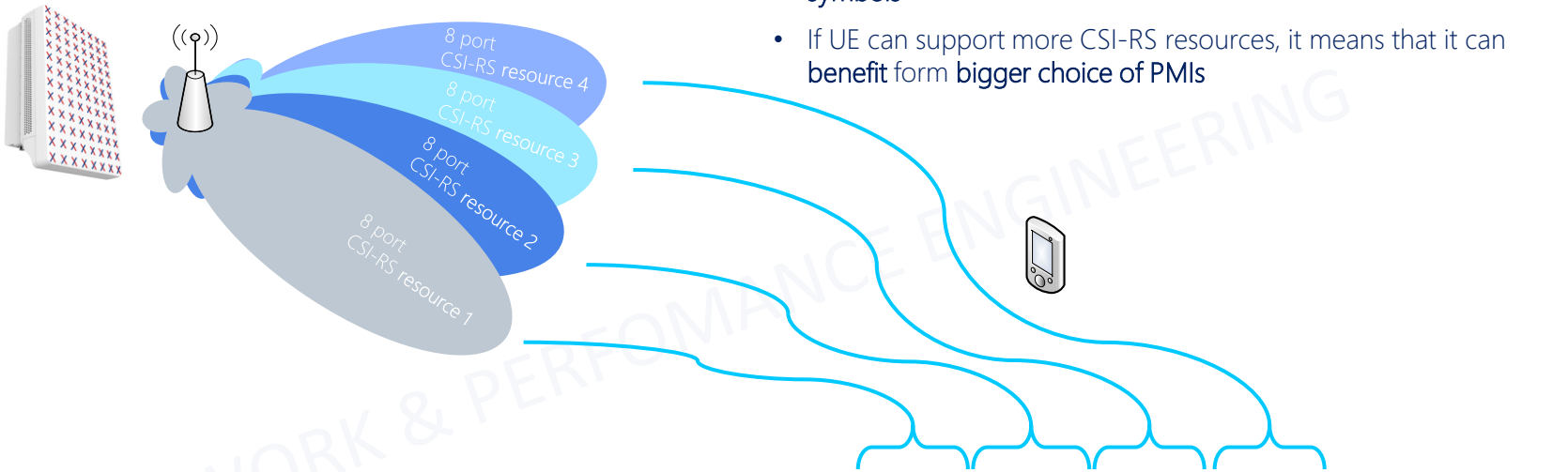
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Technical Details

CSI-RS resource explained (2/3)

illustrative picture

- More CSI-RS resources can be configured on other CSI-RS symbols
- If UE can support more CSI-RS resources, it means that it can benefit form bigger choice of PMIs



example special slot with CSI-RS symbols

- 5GC002200 reuses the same CSI-RS resource pool as UE specific CSI-RS.
- The pool consists two candidate list:
  - Primary list: the DL symbols that are currently not used for PDSCH;
  - Secondary list: all DL symbols not in primary list.
- 5GC002200 will find #4,#2,#1 from the above lists, primary first and try to add slot offset based on cell id mod 6 to avoid inter cell interference.

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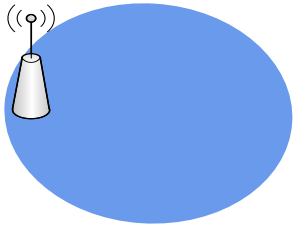
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Technical Details

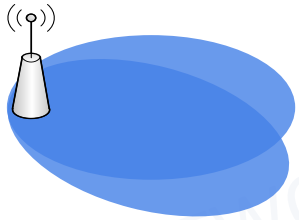
CSI-RS resource explained (3/3)

CSI-RS **resource set 1**



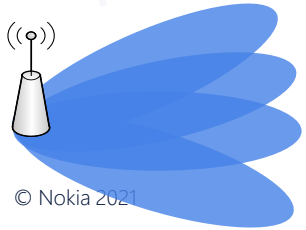
Resource set 1:
- 1CSI-RS resource
- Panel split 4,1
- 16 PMI beams

CSI-RS **resource set 2**



Resource set 2:
- 2 CSI-RS resources
- Panel split 2,2
- 128 PMI beams

CSI-RS **resource set 3**



Resource set 3:
- 4 CSI-RS resources
- Panel split 2,2
- 256 PMI beams

- CSI-RS resources are grouped in the **CSI-RS resource sets**, and configured to the UE
- Each resource set can have different number of **CSI-RS resources**
- Also, UEs may support different number of CSI-RS resources
- Therefore 5GC002200 defines **3 CSI-RS resource sets**

Maximum number of the CSI-RS resources is configurable with **NRCELL.periodicCsiAcquisition.maxCsirsNumber** parameter.

Please note that **multiple** resource sets **can be transmitted**.

UE will be configured with the **resource set** appropriate to its **reported capabilities** of the maximum numer of supported CSI-RS resources.

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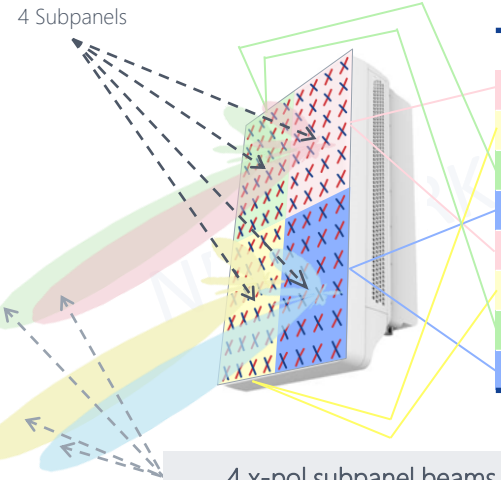
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illustrative picture

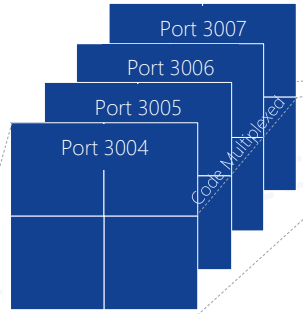
Technical Details

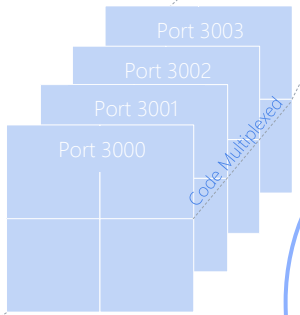
Example of one CSI-RS resource mapping to antenna port

- If **dlMimoMode is set to 8x4**, periodic CSI-RS resource for channel state feedback is sent on **dedicated subpanel beams**.
- To distinguish the totally **8 antenna ports**:
  - 4 X-polarization subpanel beams are needed.
  - **Each X-polarization beam serves 2 antenna ports** - one antenna port per polarization.

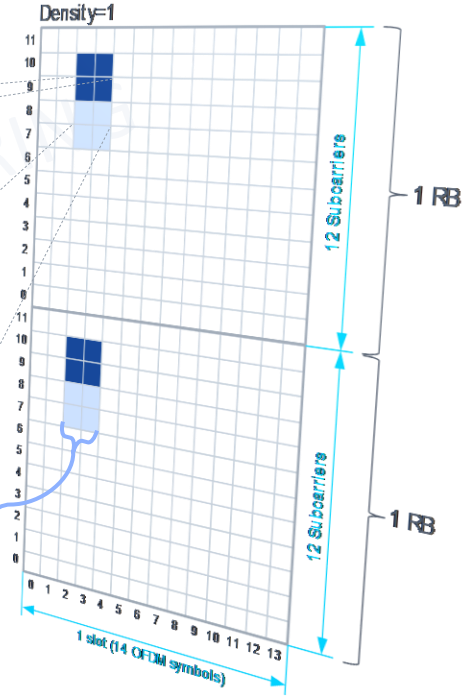


	antenna port #	beam #	polarization
	3000	1	+45°
	3001	2	+45°
	3002	3	+45°
	3003	4	+45°
	3004	1	-45°
	3005	2	-45°
	3006	3	-45°
	3007	4	-45°





CSI-RS Location within a Slot



8 port CSI-RS resource occupies 2 symbols

4 x-pol subpanel beams in same direction = 1 CSI-RS resource = 1 CSI-RS beam

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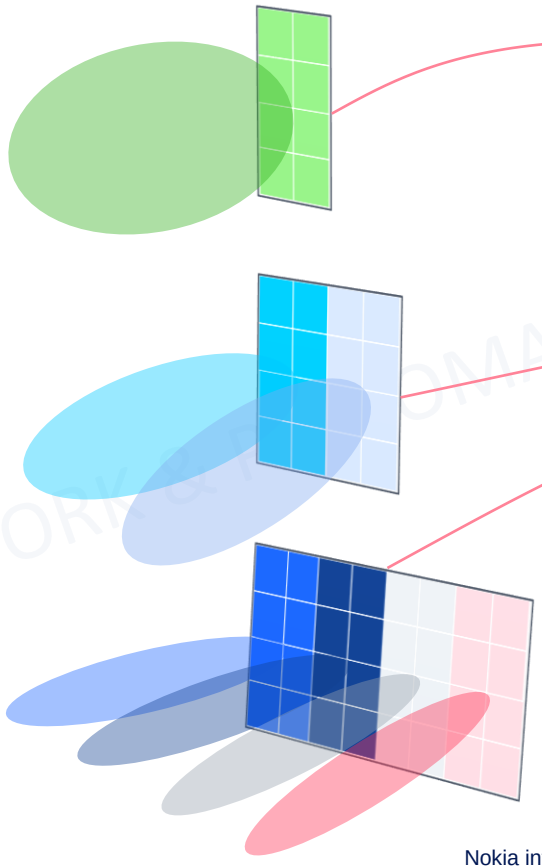
Technical Details

Example of multiple CSI-RS resource configuration

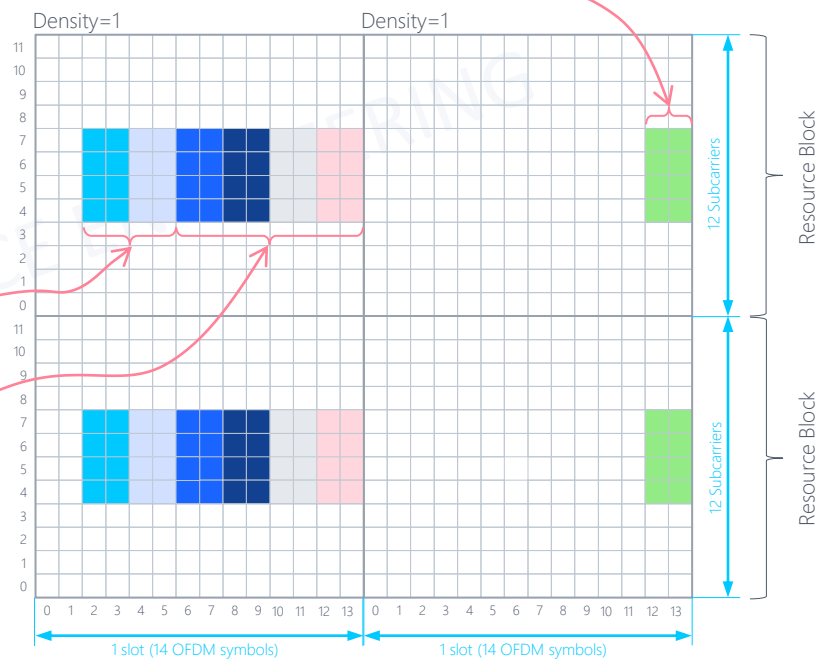
Resource set 1:
- 1CSI-RS resource
- 1 CSI-RS beam
- Panel split 4,1

Resource set 2:
- 2 CSI-RS resources
- 2 CSI-RS beams
- Panel split 2,2

Resource set 3:
- 4 CSI-RS resources
- 4 CSI-RS beams
- Panel split 2,2



CSI-RS location within a slot



7 CSI-RS resources configured in cell if *maxCsirsNumber* = 4

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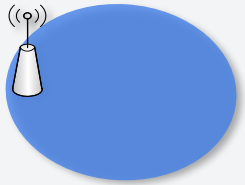
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illustrative picture

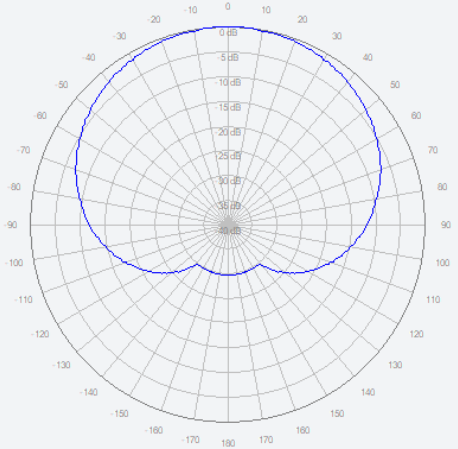
Technical Details

Example of CSI-RS resource beam horizontal patterns (RU AEQA)

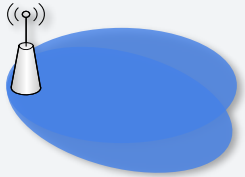
CSI-RS resource set 1



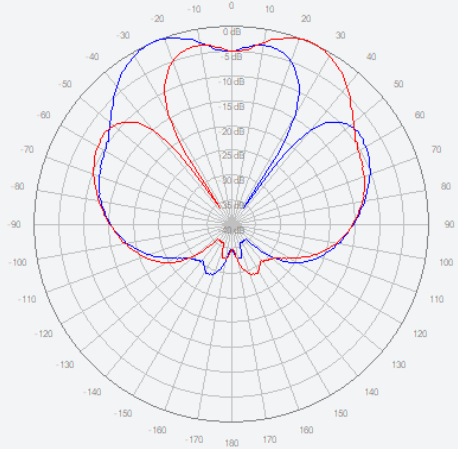
Resource set 1:
- 1CSI-RS resource
- Panel split 4,1



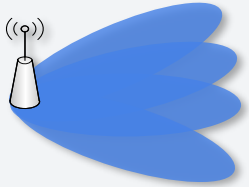
CSI-RS resource set 2



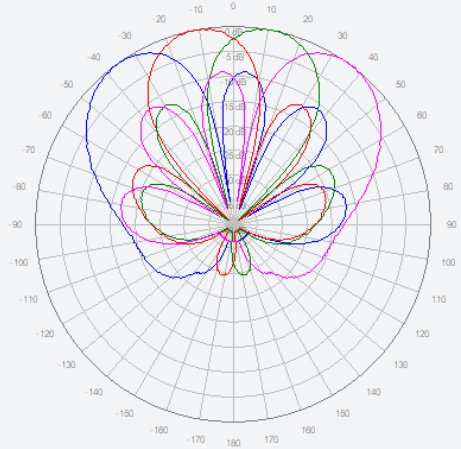
Resource set 2:
- 2 CSI-RS resources
- Panel split 2,2



CSI-RS resource set 3



Resource set 3:
- 4 CSI-RS resources
- Panel split 2,2



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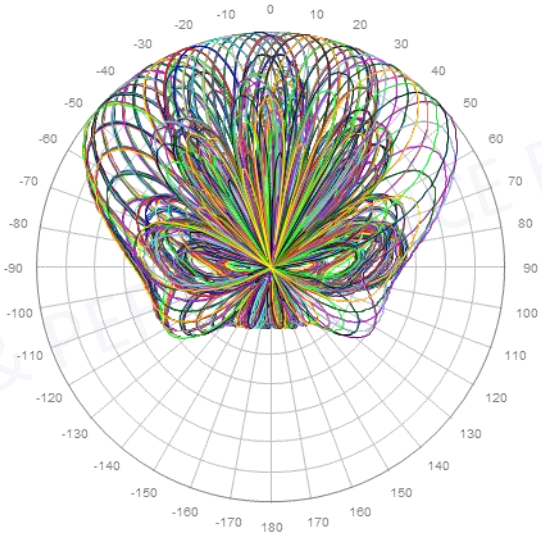
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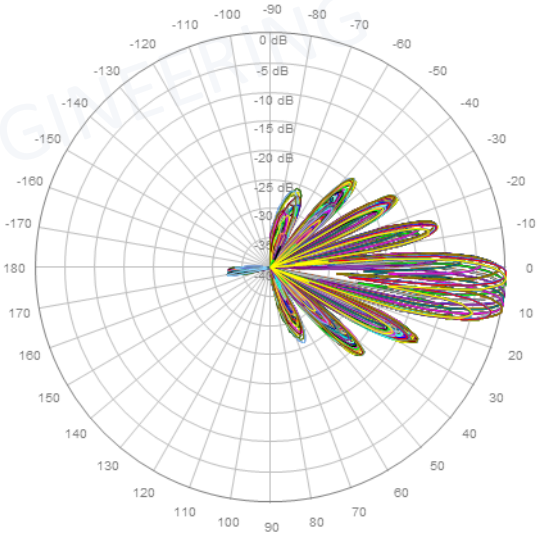
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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

PMI beams





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Technical Details
PMI beams (1/2)

NRCELL.periodicCsiAcquisition.**maxCsirsNumber** is used to set the number of sub-panel beams.
If **maxCsirsNumber** value is:
1: N_{Bi} = 1 is supported
2: N_{Bi} = 1,2 is supported
4: N_{Bi} = 1,2,4 is supported

Beam Set of PMI beams is described by (N_{Bi}, N_1, N_2), where

- N_{Bi} is number of **subpanel beams**
- N_1 is number of antenna ports per polarization in horizontal direction
- N_2 is number of antenna ports per polarization in vertical direction

Each direction of subpanel beams leads to **64 PMI beams** for (N_1, N_2) =(2,2) and to **16 PMI beams** for (N_1, N_2) =(4,1).

For 64TRX/32TRX RUs:

t	CSI-RS beam configuration	# of PMI beams
0	1 sector beam config, (N_1, N_2) =(4,1)	16
1	2 sector beam config, (N_1, N_2) =(2,2)	2x64=128
2	4 sector beam config, (N_1, N_2) =(2,2)	4x64=256

For 8TRX RUs:

t	CSI-RS beam configuration	# of PMI beams
0	1 sector beam config, (N_1, N_2) =(4,1)	16

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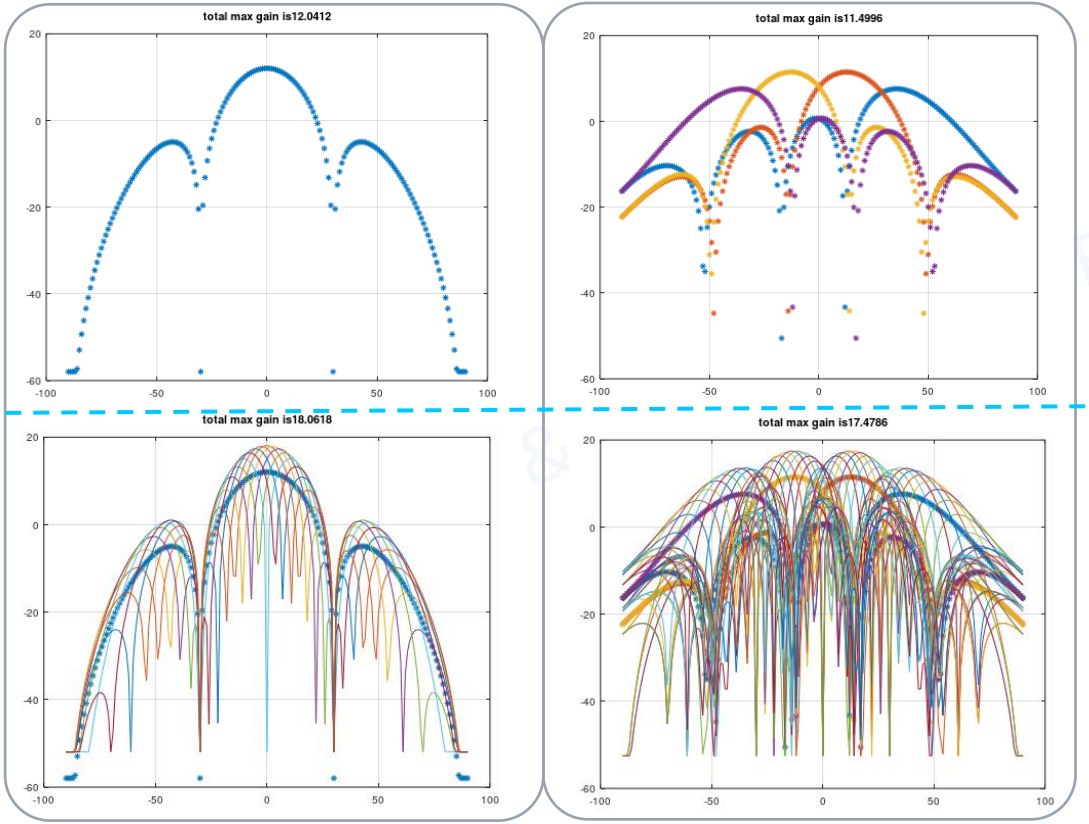
illustrative picture

Technical Details

PMI beams (2/2) - example

One CSI-RS beam case:

4 CSI-RS beam case:



CSI-RS beams

PMI (PDSCH) beams

- The PMI beam is used **for PDSCH** transmission.
- The PMI beam gain outline is related with sector beam.
- The CSI-RS beam and PMI beam can be **prestored in RFU** same as other beams.
- **If one CSI-RS** beam (resource) is used, the **total PMI beam** number is **16** - in horizontal plane.
- **If 4 CSI-RS beams** (resources) are used (4 beams are in horizontal), there are total **256 PMI beams** - 32 beams in horizontal and 8 beams in vertical.

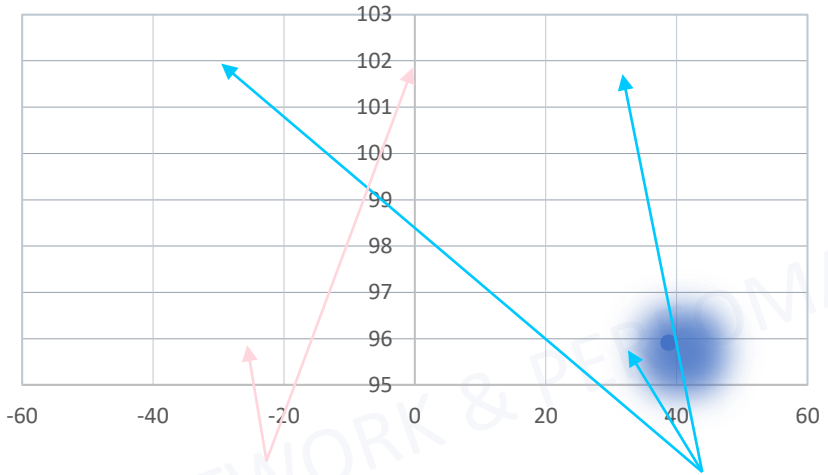
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Technical Details

Illustration of PMI beams vs. SSB/Refined beams design

SSB beamset #3#1 with refined beams

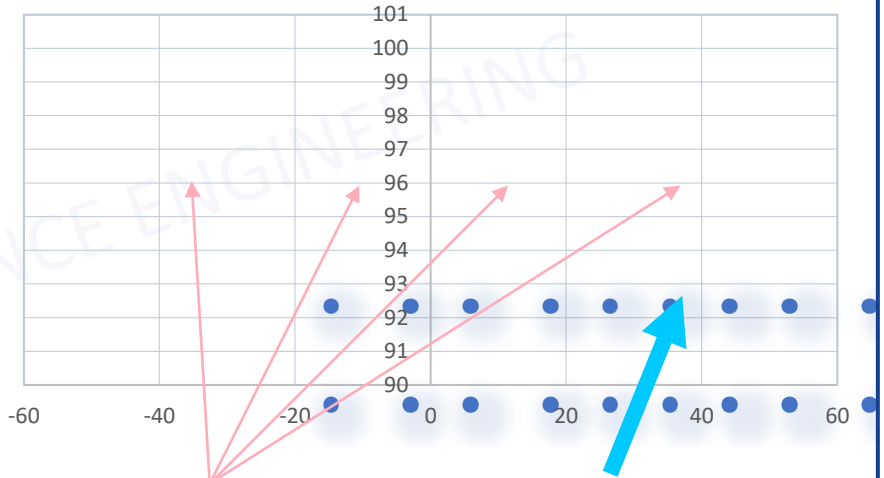


SSB beams
sending Synchronization/Broadcast/ PDSCH

Note: SSB beams are mandatory and **remain** after 5GC002200 is activated

Refined beams
sending PDSCH in legacy 5GC000533/534/1258 features
Optionally present, may be used together with 5GC002200

4 CSI-RS beams with 2_2 panel split



CSI-RS beams
sending Reference Signals

PMI beams
sending PDSCH

Before 5GC002200

After 5GC002200

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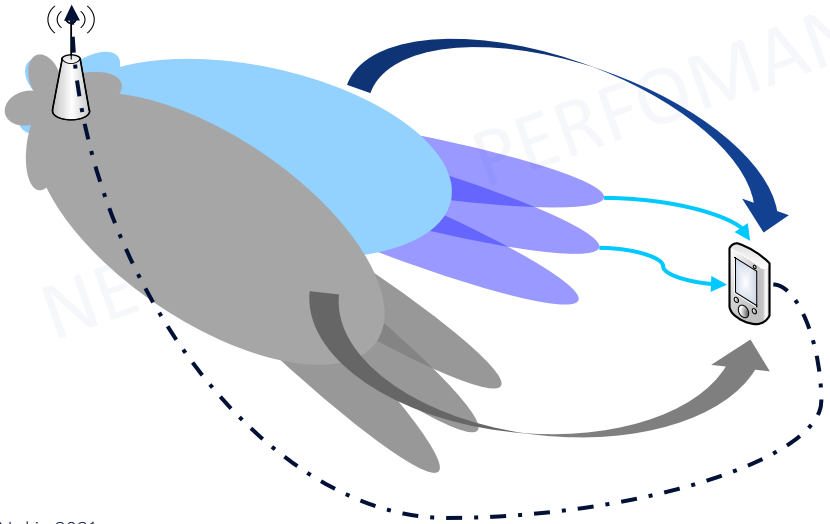
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Before 5GC002200

Technical Details

How UE selects the PMI beam (1/2)

- Before 5GC002200 in Digital Beamforming SSB/refined beam selection is done based on the measurements of the received power of the reference symbols of the beams,which are eventually used for data transmission.
- This means that each beam used for PDSCH needs to carry reference symbols (SSB or CSI-RS for beam management). This limits the numer of the available beams.
- Power measurement of the reference symbol ignores the knowledge about channel coefficients – there's much more to be learned from reference symbols.



measure
- SBRI-RSRP (SSB beams)
- CRI-RSRP (refined beams)

report
- 2 best SBRI-RSRP
- 2 best CRI-RSRP

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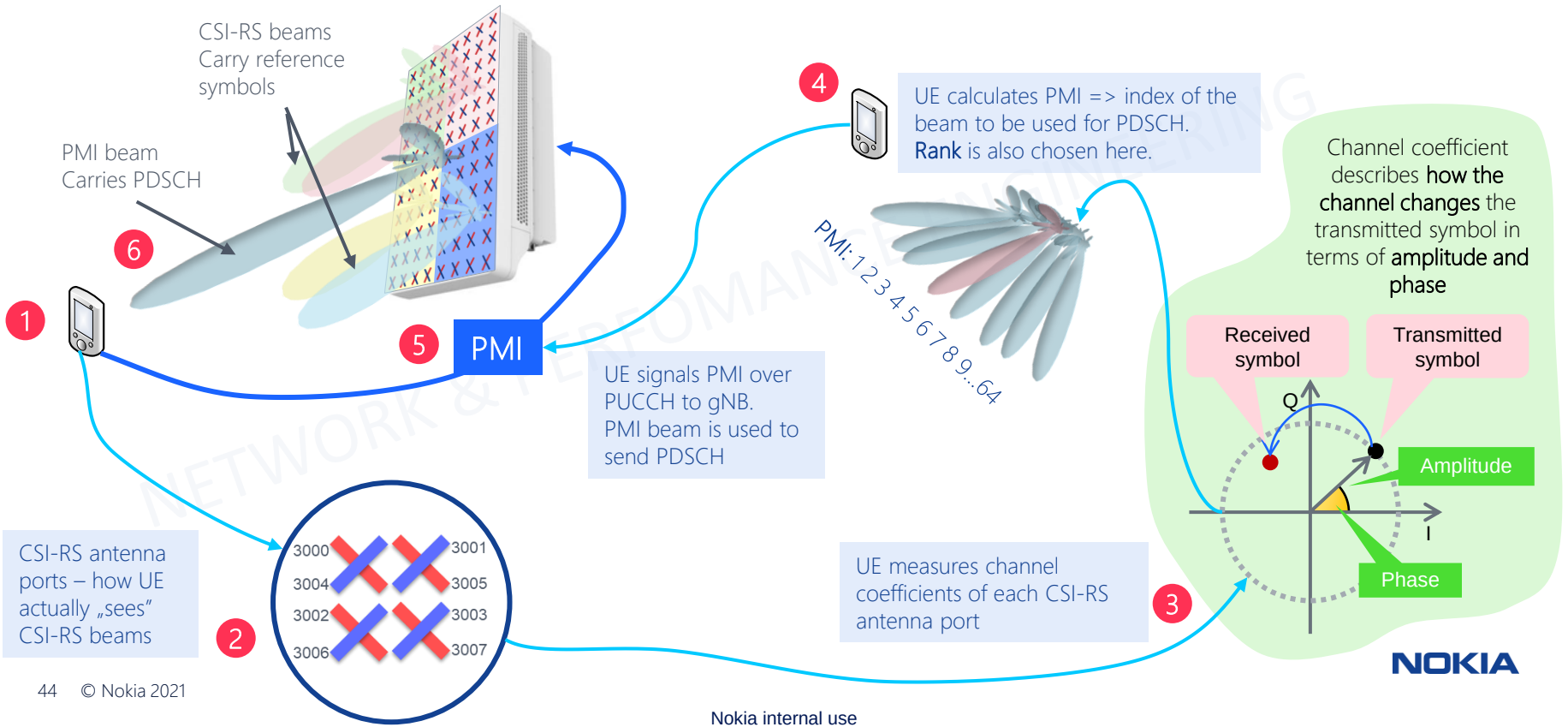
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Technical Details

How UE selects the PMI beam (2/2)

After 5GC002200

In 5GC002200 PMI beam selection is done based on the **channel coefficient** measurements.
This measurement allows the UE **to calculate the Precoding Matrix Indicator (PMI)** of the optimal beam(s).



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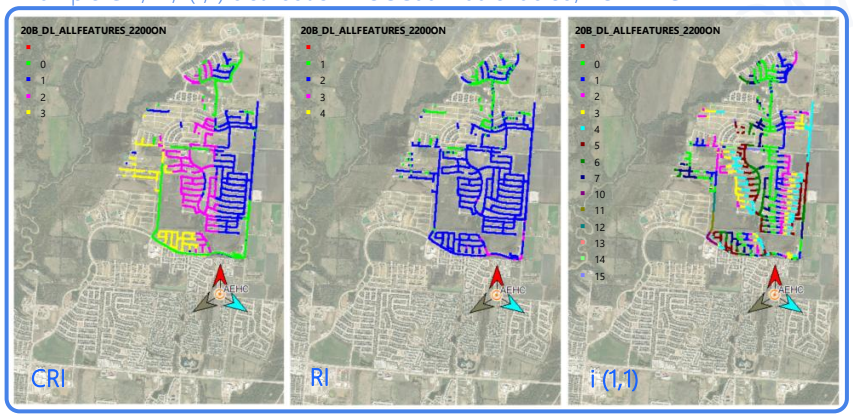
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Technical Details

UE report and best beam selection

- UE sends CSI-RS report to gNB, that determines beamforming weights from the Codebook to form PMI beams that convey PDSCH data.
- The following UE report parameters impacting weights selection :
  - **CRI:** CSI-RS Resource Indicator
  - **i(1,1), i(1,2), i(1,3), i(2)** - PMI indices
  - **RI:** Rank Indicator
  - **LI:** Layer Indicator

Example CRI, RI, i(1,1) distribution – 5GC002200 enabled, RU AEHC



UE reported parameters that determine the best PMI beam

If					Then		
RI value	i11 value	i12 value	i13 value	LI value	Beam #1 reported?	Beam #2 reported?	Best beam?
0	Any	**Any** (2x2) or **N/A** (4x1)	N/A	N/A	Yes	No	Beam #1
1	Any	**Any** (2x2) or **N/A** (4x1)	Any	0	Yes	Yes	Beam #1
1	Any	**Any** (2x2) or **N/A** (4x1)	Any	1	Yes	Yes	Beam #2
2	Any	**Any** (2x2) or **N/A** (4x1)	[Before PR562904] **>0** (always) [PR562904] **Any** [End PR562904]	0,2	Yes	Yes	Beam #1
2	Any	**Any** (2x2) or **N/A** (4x1)	[Before PR562904] **>0** (always) [PR562904] **Any** [End PR562904]	1	Yes	Yes	Beam #2
3	Any	**Any** (2x2) or **N/A** (4x1)	[Before PR562904] **>0** (always) [PR562904] **Any** [End PR562904]	0,2	Yes	Yes	Beam #1
3	Any	**Any** (2x2) or **N/A** (4x1)	[Before PR562904] **>0** (always) [PR562904] **Any** [End PR562904]	1,3	Yes	Yes	Beam #2

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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

UE capability aspects



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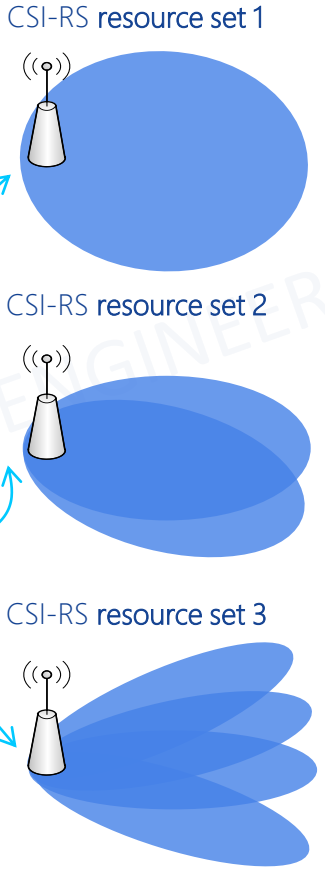
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Technical Datails

UE capabilities aspects (1/2)

- Combined reporting of sub-panel beam (CRI) with legacy CSI (CQI, RI and PMI) is **subject of UE capability**.
- UE supports either **1, 2 or 4 CSI-RS resources** and respective CRI reporting.
- Different resources are defined for the different sets of UEs:
  - Single cell wide sub-panel beam for UEs supporting only a single resource.
  - 2 resources, each covering half of the cell for UEs, supporting 2 resources.
  - 4 resources with quarter-sector beams for UEs supporting 4 resources.
- If all variants are supported in a cell, **7 CSI-RS resources** are provided by the cell.



Resource set 1:
- 1CSI-RS resource
- Panel split 4,1
- 16 PMI beams

Resource set 2:
- 2 CSI-RS resources
- Panel split 2,2
- 128 PMI beams

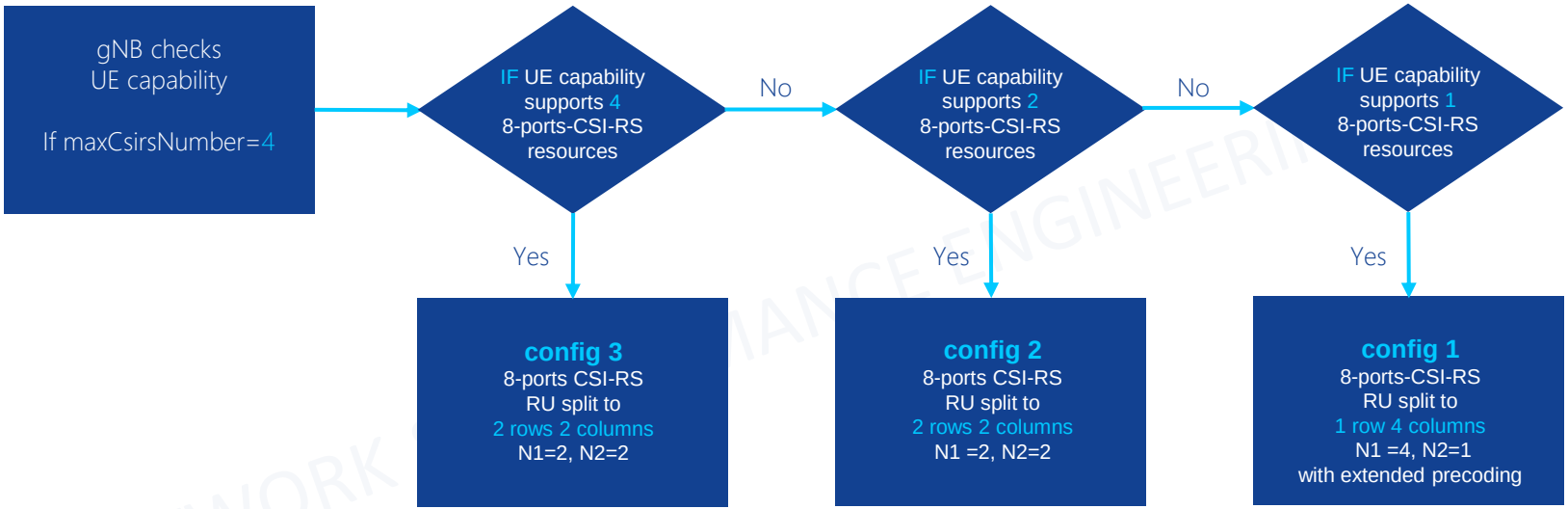
Resource set 3:
- 4 CSI-RS resources
- Panel split 2,2
- 256 PMI beams

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Technical Details

UE capabilities aspects (2/2) – UE configuration



codebookParameters
Indicates the codebooks and the corresponding parameters supported by the UE.

Parameters for type I single panel codebook (type1 singlePanel) supported by the UE, which is mandatory to report:

- *supportedCSI-RS-ResourceList*;
- *modes* indicates supported codebook modes (mode 1, both mode 1 and mode 2);
- *maxNumberCSI-RS-PerResourceSet* indicates the maximum number of CSI-RS resource in a resource set.

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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Interworking with other features



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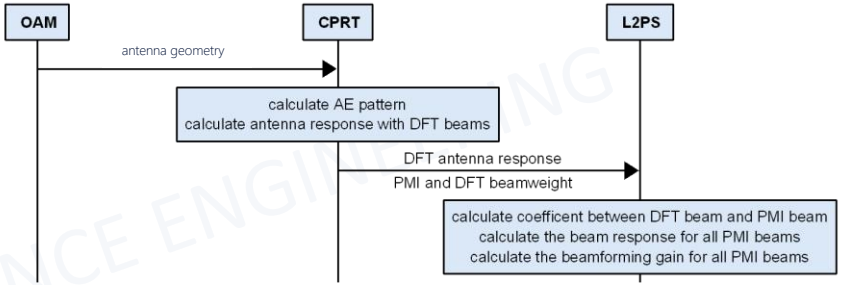
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Technical Details

Overview of interaction with EIRP control features

- For EIRP control, the beamforming **gain for all used beams must be known by L2-PS**.
- Before introduction of 5GC002200, the beamforming weights are generated offline. Accordingly, the **beamforming gain for beams are also calculated offline**.
- Both beamforming weights and beamforming gain **are stored as file** and then CPRT reads these weights and gains during cell setup.
- In 5GC2200, the **beamforming weights are generated in gNB during cell setup**.
- Accordingly the **beamforming gain for these weights will be calculated in gNB** too instead of being generated offline.
- After beamforming gain calculation for each PMI beam, **other procedure is same** as the functionalities defined in EIRP control features (5GC 001392/002052/002300).
- For 8TRX RRHs with external antenna, the antenna geometry can not be available, the beamforming gain for new introduced PMI beam can not be calculated, so the **antenna max gain from CM is used directly**.



1. OAM indicates antenna geometry to CPRT.
2. CPRT calculates the AE pattern according to 3GPP 38.901 with some adaptation.
3. CPRT calculates the antenna response for each DFT beams, for each segment.
4. CPRT sends DFT beams response for each segment and PMI projection coefficient to L2-PS.
5. L2-PS calculates the projection coefficient: between PMI beams and DFT beams.
6. L2-PS calculates the beamforming gain for each beam in each segment based on the DFT beams response and PMI beam projection coefficient.

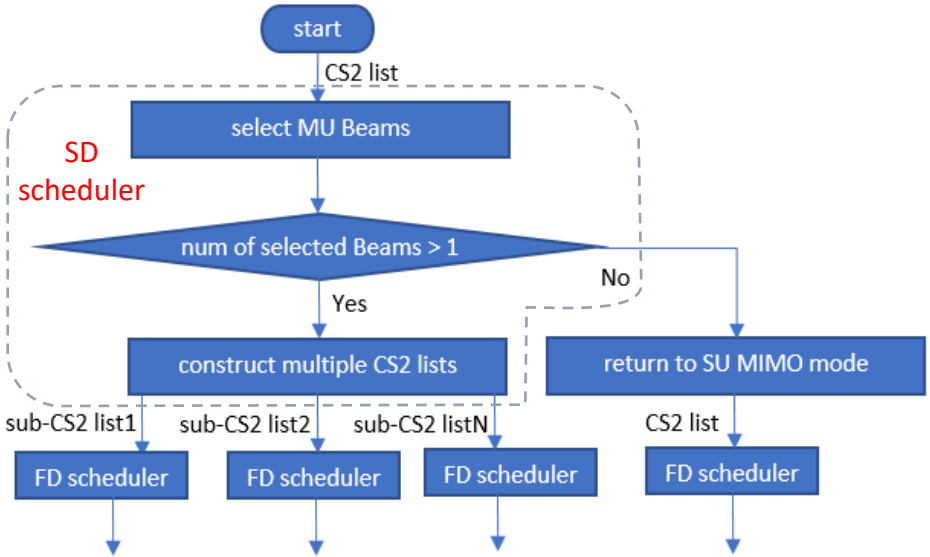
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Technical Details

Overview of interaction with 5GC000524 DL MU-MIMO for Digital Beamforming

- PMI beams are used for MU-MIMO.
- Correlations between PMI beams are calculated and stored during initialization.
- Calculate PMI beam indices for UEs based on their PMI feedback.
- Estimate UE's correlations based on stored correlations between their PMI beams.



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Technical Details

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Sub-feature split summary



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Technical Details

5GC002200 sub-feature split

5GC002200 feature consists of the following sub-features:

- 5GC002200-A: Enable basic split panel 8x4 SU- MIMO
- 5GC002200-B: Further split panel SU MIMO functionality
- 5GC002200-C: EIRP control with PMI beam
- 5GC002200-D: Additional optimization to beamforming functionality
- 5GC002200-E: Interworking with MU MIMO, FDM, different baseband block configuration
- 5GC002200-F: Previous sub-features for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO

5GC002200-A: Enable basic split panel 8x4 SU- MIMO

1. Add new enumeration to *dlMimoMode* in NIDD parameter, and pass it from CP to UP; All the functionality of this feature is controlled by the new enumeration.
2. Add the functionality of how to construct patternID from UE's CSI feedback for PMI beam and subpanel beam for CSI-RS.
3. Add 1 Cell specific CSI-RS, transmit via subpanel beam.
4. CP checks UE capability and always configure 1 CSI-RS resource and N1=4, N2=1, extended precoding for the UE. Add one NIDD parameter to control maximum number of CSI-RS resource that can be configured to one UE. Only 1 can be configured for this subfeature. Only NIDD: *ssBurstSetPeriod* not longer than NIDD: *csiReportPeriodicity* is supported in this subfeature (note: the unit of the two parameters shall be united before comparation).
5. Consider UE capability of max 4/2 layer, as well as no UE capability available in SA mode.
6. Calculate beam weight /PatternID in CPRT and download it to OAM and then to RU, pass patternID to L2.
7. Transmit PDSCH using one/two PMI beam which is from UE's CSI feedback.
8. Add 3GPP standard 8 port precoding and port selection precoding for MU MIMO in the later subfeature; Add normalization of the precoding here too.
9. Transmit PDCCH via one x-pol PMI beam which comes from full panel. Add the logic of choosing one PMI beam from UE's CSI feedback.
10. Add updated dl power control in CP and UP. Add one separate NIDD parameter for cell specific CSI-RS power boost.
11. QCL in MAC CE for PDCCH is unchanged, still use SSB index to map TRS. Still configure PUCCH for 2 beam report. QCL for CSI-RS is not configured, i.e. the current approach is kept.
12. Only gen 4 CPRI RU, only 1 basic SISO configure in 20B. Only RU with 8 cross polar TRX columns.
13. Add PUCCH UCI retrieving for CRI/PMI/LI, only consider PUCCH format 2.
14. L1 part PDCCH precoding [1 0 1 0] and [0 1 0 1] with and L2-L1 interface update.

* - informative: spillover part of subfeature 5GC002200-D content to 5G21A (CB007157)

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Technical Details

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- 5GC002200-F: Previous sub-features for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO

5GC002200-B: Further split panel SU MIMO functionality

1. Additional subpanel split for RUs with different number of TRX columns/rows, i.e. 8/4 cross polar TRX columns, 4/2/1 cross polar TRX rows. N1, N2 is relevant to the split method.
2. Additional subpanel beams with different direction and width.
3. CP configure the best possible CSI-RS resource number and relevant N1,N2 for the UE, based on UE capability, subpanel capability and operator configuration from NIDD parameter. Simultaneous capability is handled here. Interworking with DRX and Measurement Gap is considered.
4. Calculate beam weight for different CSI-RS resource number, different RU and different N1N2.

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Technical Details

5GC002200 sub-feature split

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- 5GC002200-B: Further split panel SU MIMO functionality
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- 5GC002200-E: Interworking with MU MIMO, FDM, different baseband block configuration
- 5GC002200-F: Previous sub-features for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO

5GC002200-C: EIRP control with PMI beam

1. Calculate the beam response for orthogonal DFT beams.
2. Expansion of PMI beams in terms of DFT beams.
3. Calculate the beamforming gain for each PMI beam based on the beam response for orthogonal DFT beams and the correlation between PMI beam and DFT beams.

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5GC002200 sub-feature split

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- 5GC002200-B: Further split panel SU MIMO functionality
- 5GC002200-C: EIRP control with PMI beam
- 5GC002200-D: Additional optimization to beamforming functionality
- 5GC002200-E: Interworking with MU MIMO, FDM, different baseband block configuration
- 5GC002200-F: Previous sub-features for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO

5GC002200-E: Interworking with MU-MIMO, FDM, different baseband block configuration

1. L2 choose Rank 2 Port Selection (PS) for PDSCH for MU paired UE if needed.
2. Consider different subcell type configuration for with/without MU-MIMO.
3. Interworking with FDM is considered.

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5GC002200 sub-feature split

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- 5GC002200-B: Further split panel SU MIMO functionality
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- 5GC002200-F: Previous sub-features for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO

5GC002200-F: Previous sub-features for eCPRI interworking with enhanced FD beamforming enhancement of common channels for 4x4 MIMO

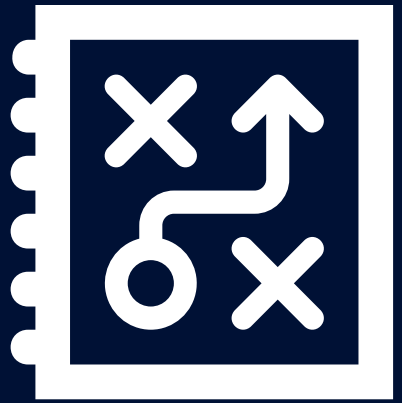
1. Previous subfeatures for eCPRI interworking with enhanced FD beamforming
2. Enhancement of common channels for 4x4 MIMO.

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Interdependencies

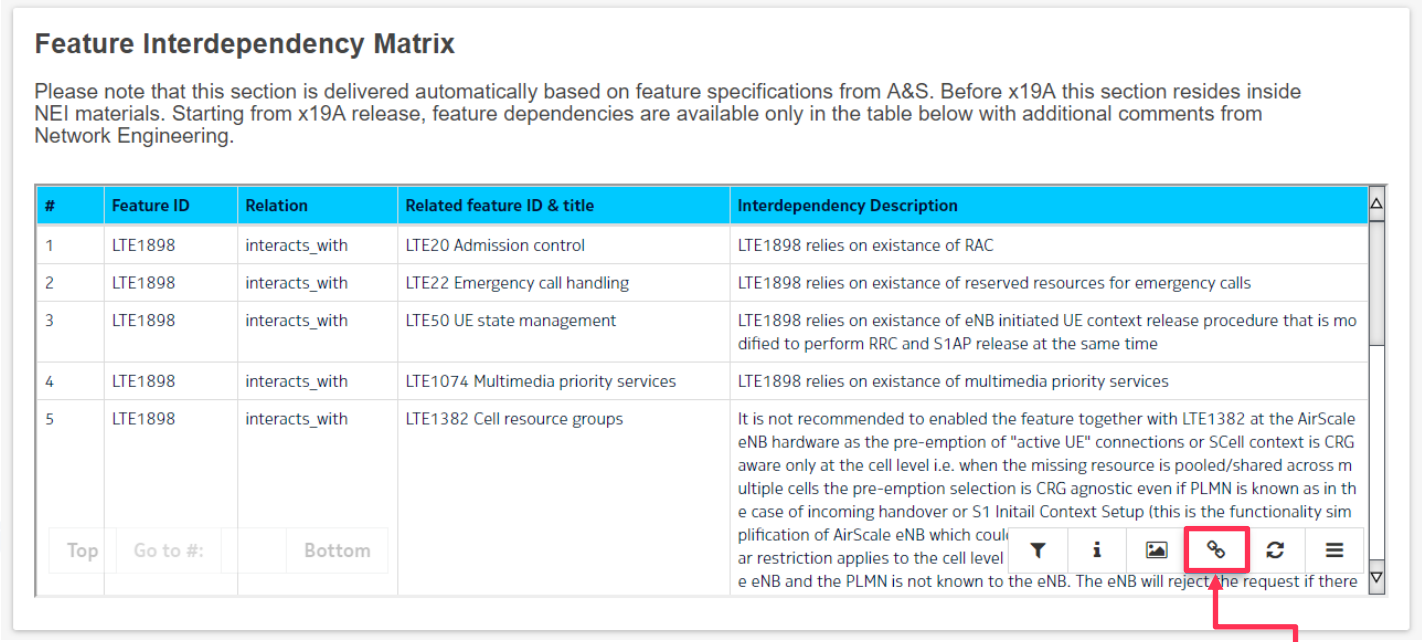


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How to find feature interdependencies in WebNEI?

- Under Network Engineering Information materials for certain feature there is dedicated section called Feature Interdependency Matrix, which describes feature relations based on R&D input with additional NetEng add-ons:



Feature Interdependency Matrix

Please note that this section is delivered automatically based on feature specifications from A&S. Before x19A this section resides inside NEI materials. Starting from x19A release, feature dependencies are available only in the table below with additional comments from Network Engineering.

#	Feature ID	Relation	Related feature ID & title	Interdependency Description
1	LTE1898	interacts_with	LTE20 Admission control	LTE1898 relies on existance of RAC
2	LTE1898	interacts_with	LTE22 Emergency call handling	LTE1898 relies on existance of reserved resources for emergency calls
3	LTE1898	interacts_with	LTE50 UE state management	LTE1898 relies on existence of eNB initiated UE context release procedure that is mo dified to perform RRC and S1AP release at the same time
4	LTE1898	interacts_with	LTE1074 Multimedia priority services	LTE1898 relies on existance of multimedia priority services
5	LTE1898	interacts_with	LTE1382 Cell resource groups	It is not recommended to enabled the feature together with LTE1382 at the AirScale eNB hardware as the pre-emption of "active UE" connections or SCell context is CRG aware only at the cell level i.e. when the missing resource is pooled/shared across m ultiple cells the pre-emption selection is CRG agnostic even if PLMN is known as in th e case of incoming handover or S1 Initail Context Setup (this is the functionality sim plification of AirScale eNB which coul ar restriction applies to the cell level e eNB and the PLMN is not known to the eNB. The eNB will reject the request if there

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Configuration Management

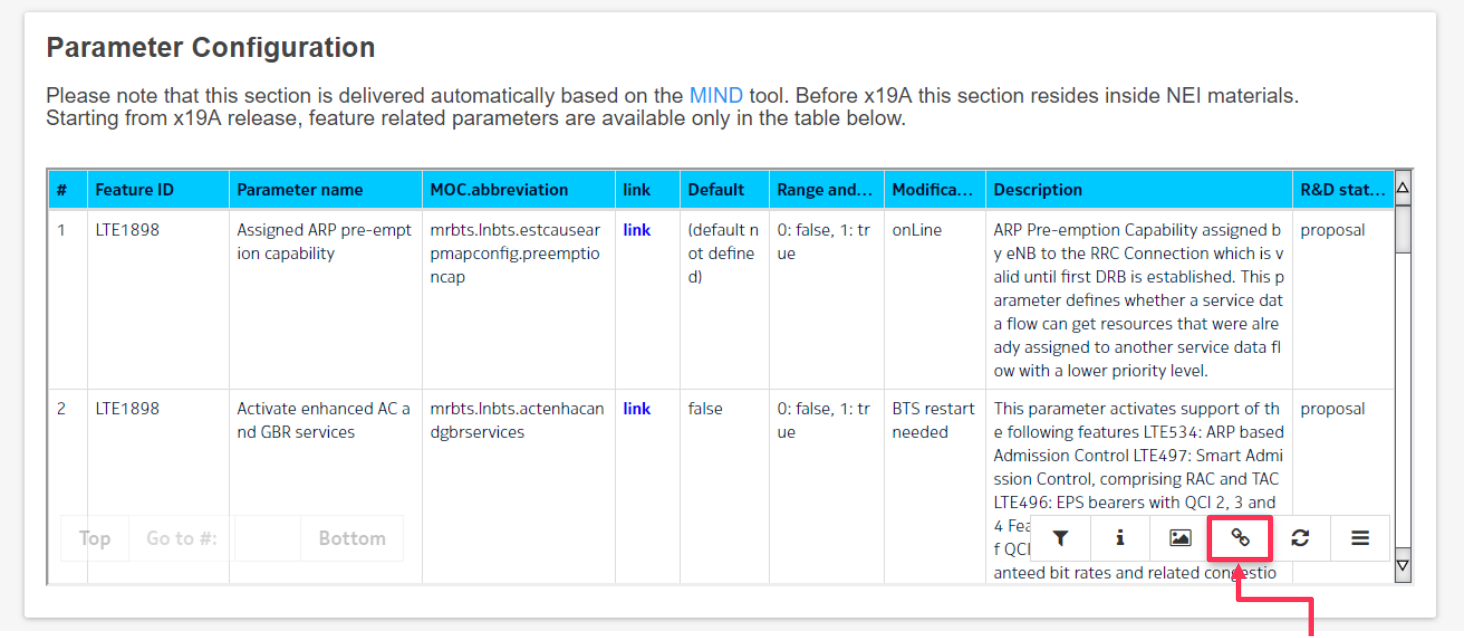


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How to find feature parameters in WebNEI?

- Under feature materials in WebNEI platform there is dedicated section called Parameter Configuration, which lists all parameters related to given functionality



Parameter Configuration

Please note that this section is delivered automatically based on the MIND tool. Before x19A this section resides inside NEI materials. Starting from x19A release, feature related parameters are available only in the table below.

#	Feature ID	Parameter name	MOC.abbreviation	link	Default	Range and...	Modifica...	Description	R&D stat...
1	LTE1898	Assigned ARP pre-emption capability	mrbts.lnbts.estcausearpmapconfig.preemptioncap	link	(default not defined)	0: false, 1: true	onLine	ARP Pre-emption Capability assigned by eNB to the RRC Connection which is valid until first DRB is established. This parameter defines whether a service data flow can get resources that were already assigned to another service data flow with a lower priority level.	proposal
2	LTE1898	Activate enhanced AC and GBR services	mrbts.lnbts.actenhacandgbrservices	link	false	0: false, 1: true	BTS restart needed	This parameter activates support of the following features LTE534: ARP based Admission Control LTE497: Smart Admission Control, comprising RAC and TAC LTE496: EPS bearers with QCI 2, 3 and 4 Fea f QCI anteed bit rates and related congestio	proposal

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Configuration Management <5GC002200>

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.dlMimoMode	Downlink MIMO Mode This parameter selects the DL MIMO mode for the PDSCH. NxM MIMO Notation: N = number of CSI-RS antenna ports used for PMI per UE; M = max number of PDSCH layers (max Rank) per UE. Configuring this parameter during cell addition might provoke gNB reset - gNB takes decision of reset necessity. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	Value	Name	[-]	30
		30	2x2 Closed Loop Spatial Multiplexing		
		40	4x4 or 4x2 Closed Loop Spatial Multiplexing		
		50	2x2 Open Loop Spatial Multiplexing		
		60	4x4 or 4x2 Open Loop Spatial Multiplexing		
		70	8x4 Closed Loop Spatial Multiplexing		

New entry introduced by 5GC002200 to enable the feature

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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.periodicCsiAcquisition.csirsPower	CSI-RS power This parameter indicates the power of the cell-specific CS-RS for CSI acquisition in terms of EPRE. Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	-60...50 [decimal]	[dBm]	-

If NRCELL.dlMimoMode is equal to 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be equal to powerControlOffsetSS+(NRCELL.ssPbchBlockPower+3dB, with powerControlOffsetSS has values [-3,0,3,6]dB, else this value is invalid. powerControlOffsetSS=0 is recommended.

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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.periodicCsiAcquisition.csirsNumberWithinReportPeriod_Set1	CSI-RS number for CSI-RS resource set1 within one report period This parameter defines the number of CSI-RS for CSI-RS resource set1 (1 CSI-RS resource in this set that is configured to UE) within one report period. Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	1...5 [decimal]	[-]	-

If NRCELL.dlMimoMode is 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be configured and shall be equal to recommended value 5.

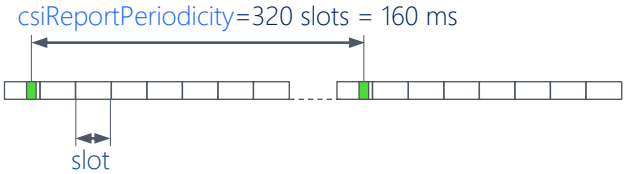
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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.periodicCsiAcquisition.csiReportPeriodicity	CSI reporting periodicity This parameter indicates the CSI reporting periodicity	Value	Name	[-]	320
		20	slots20		
		40	slots40		
		80	slots80		
		160	slots160		
		320	slots320		



For each NRCELLGRP and each NRCELL instances belonging to that NRCELLGRP instance (known via NRCELLGRP.nrCellList),
- All these NRCELL instances shall have the same value for NRCELL.periodicCsiAcquisition.csiReportPeriodicity
- If NRCELLGRP.scs = 30 (corresponding to 30 KHz) and NRCELL.prachConfigurationIndex = 90, 91, 94 or 98, this parameter shall be greater or equal to Max (NRCELLGRP.ssBurstSetPeriod * 2, 20)
- If NRCELLGRP.scs = 120 (corresponding to 120 KHz) and NRCELL.prachConfigurationIndex = 38, this parameter shall be greater or equal to Max (NRCELLGRP.ssBurstSetPeriod * 8, 160)

This parameter shall be a multiple of NRCELL.drxProfile1.drxLongCycle (NRCELL.periodicCsiAcquisition.csiReportPeriodicity(ms) = N* NRCELL.drxProfile1drxLongCycle with N=1, 2,...16) if NRCELL.actCDrx is equal to TRUE and
- If NRCELL.nrarfcn shall be<= 800000 (FR1) and NRCELL.drxProfile1.drxLongCycle is smaller or equal to 7 (corresponding to 80ms)
- If NRCELL.nrarfcn>= 2016667 (FR2) and NRCELL.drxProfile1.drxLongCycle is smaller or equal to 3 (corresponding to 40ms)

Note: (ms) above denote the corresponding parameter values converted from slot to ms as follows:
- If NRCELLGRP.scs is equal to 15 (corresponding to 15 KHz), 1slot is equal to 1ms
- If NRCELLGRP.scs is equal to 30 (corresponding to 30 KHz), 1slot is equal to 0.5ms;
- If NRCELLGRP.scs is equal to 120 (corresponding to 120 KHz), 1slot is equal to 0.125ms

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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.periodicCsiAcquisition.maxCsirsNumber	Maximum CSI-RS number This parameter defines the maximum number of cell-specific CSI-RS for channel acquisition per UE. Note that, for each UE, the number of CSI-RS actually configured is also limited by the capability of the UE. Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	1...4 [decimal]	[-]	-

If NRCELL.dlMimoMode is 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be configured and shall be equal to 1,2 or 4 according to RU/UE capability.

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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.dlQam256PowerBackoffSub6	Power backoff in DL bands below 6GHz This parameter defines the power backoff applied for UEs supporting the 256QAM scheme in DL bands of below 6GHz	0...10 [decimal]	[-]	1.5

If NRCELL.dlMimoMode is 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be configured and shall be equal to recommended value 3.

Nokia internal use

Configuration Management <5GC002200>

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.nrCellType	NR Cell type This parameter allows the operator to specify the number of fronthaul streams of a New Radio Cell. This parameter is also used for identification of Supplemental downlink cell (FR2). Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	Value	Name	[-]	-
		20	2DL0UL		
		21	2DL1UL		
		22	2DL2UL		
		24	2DL4UL		
		42	4DL2UL		
		44	4DL4UL		
		82	8DL2UL		
		84	8DL4UL		
		88	8DL8UL		
		168	16DL8UL		
		1616	16DL16UL		

If NRCELL.dlMimoMode is equal to 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), then nrCellType=4DL2UL or nrCellType=4DL4UL or nrCellType=8DL4UL is allowed.

Nokia internal use

Configuration Management <5GC002200>

Related parameters <Interworking with DRX feature>

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.{drxProfile1}	DRX Profile 1 This parameter defines the common connected DRX profile for UEs without voice DRB	-	-	[-]	-
NRCELL.drxProfile1.drxInactivityTimer	DRX Inactivity Timer This parameter defines the duration after the PDCCH occasion in which a PDCCH indicates an initial UL or DL user data transmission for the MAC entity.	Value 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 16 17 18 19 20 21 22	Name 0ms 1ms 2ms 3ms 4ms 5ms 6ms 8ms 10ms 20ms 30ms 40ms 50ms 60ms 80ms 80ms 100ms 200ms 300ms 500ms 750ms 1280ms 1920ms 2560ms	[-]	4

Nokia internal use

Configuration Management <5GC002200>

Related parameters <Interworking with DRX feature>

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.drxProfile1.drxRetransTimerDl	DRX Retransmission Timer for Downlink Maximum duration until a DL retransmission is received; per DL HARQ process. This parameter defines the DRX active time for the UE to receive potential DL retransmission data.	Value	Name	[-]	24
		0	0slot		
		16	16slot		
		24	24slot		
		33	33slot		
		40	40slot		
		64	64slot		
		80	80slot		
		96	96slot		
		112	112slot		
		128	128slot		
		160	160slot		
		320	320slot		

Nokia internal use

Configuration Management <5GC002200>

Related parameters <Interworking with DRX feature>

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.drxProfile1.drxRetransTimerUl	DRX Retransmission Timer for Uplink Maximum duration until a grant for UL retransmission is received; per UL HARQ process. This parameter defines the DRX active time for the UE to receive potential UL retransmission data.	Value	Name	[-]	24
		0	0slot		
		16	16slot		
		24	24slot		
		33	33slot		
		40	40slot		
		64	64slot		
		80	80slot		
		96	96slot		
		112	112slot		
		128	128slot		
		160	160slot		
		320	320slot		

Nokia internal use

Configuration Management <5GC002200>

Related parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}. Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.ssPbchBlockPower	SS PBCH Block Power This parameter defines the SS and PBCH powers in terms of EPRE, where - For beamforming type of radio (NRCELL.actBeamforming equal to TRUE), NRCELL.ssPbchBlockPower is defined as SS/PBCH block power per RE for 1 polarization, with number of TRX(s) per polarization. - For non-beamforming type of radio (NRCELL.actBeamforming equal to FALSE), NRCELL.ssPbchBlockPower is defined as SS/PBCH block power per RE for 1 MIMO stream/layer, with 1TRX per MIMO stream/layer. Its value shall be configured while taking into account the constraints between this parameter and the maximum power configured through NRCELL.pMax, where - In general, NRCELL.ssPbchBlockPower<= NRCELL.pMax + 10log10(Num. of TRXs per polarization) – 10log10(Num. of REs), where Num. of REs = max(ssbPbchBlock_REs, CSI-RS_REs) and CSI-RS_REs includes CSI-RS for tracking, CSI-RS for beam refinement, and CSI-RS for measurements in SSB slot. - If NRCELL.dlMimoMode is 70, (corresponding to 8x4 Closed Loop Spatial Multiplexing), cell- specific CSI-RS for measurements shall take power adjustment into consideration, so CSI-RS_REs for measurements = (actual CSI-RS RE number)*x, with x=10^(powerControlOffsetSS/10) and has values [0.5,1,2,4]. For cell-specific CSI-RS measurement with is extended precoding, x = 4*10^(powerControlOffsetSS/10) and has values [2,4,8,16]. powerControlOffsetSS=NRCELL.csirsPower - (NRCELL.ssPbchBlockPower+3dB). Only SSB transmit via both polarizations is supported here. - If CSI-RS signals for refined beams are transmitted in the same slot as SS PBCH, then NRCELL.ssPbchBlockPower<= NRCELL.pMax + 10log10(Num. of TRXs per polarization) – 10log10(Num. of REs), where Num. of REs = max(ssbPbchBlock_REs, CSI-RS_REs) and CSI-RS_REs includes CSI-RS for tracking, CSI-RS for beam refinement+240, and CSI-RS for measurements in SSB slot. These constraints depends on the RU configuration with examples as shown in below: 1- TDD CPRI cmWave RUs (e.g. AEQA): NRCELL.ssPbchBlockPower ≤ NRCELL.pMax + 10*log10(number of TRX per polarization) – 10*log10(20*12) (e.g. for AEQA: number of TRX per polarization = 32 and it is assumed that Num. of REs = ssbPbchBlock_REs with 240REs=20 PRB*12 REs/RB) 2- TDD CPRI mmWave RUs (e.g. AEUA): NRCELL.ssPbchBlockPower ≤ NRCELL.pMax – 10*log10(20*12) (e.g. for AEUA: number of TRX per polarization = 1 and it is assumed that Num. of REs = ssbPbchBlock_REs with 240REs=20 PRB*12 REs/RB) 3- FDD CPRI below 3GHz (cmWave) RUs (e.g. AHLOA): NRCELL.ssPbchBlockPower ≤ NRCELL.pMax – 10*log10(20*12) (it is assumed that Num. of REs = ssbPbchBlock_REs with 240REs=20 PRB*12 REs/RB). The actual range of values supported by NRCELL.ssPbchBlockPower also depends on the dBFS range supported by CPRI/eCPRI RUs. The range of NRCELL.ssPbchBlockPower supported is: For CPRI: The range is [max(-60, -66.3+NRCELL.pMax+10*log10(number of TRX per polarization)), min(50, NRCELL.pMax+10*log10(number of TRX per polarization))] For eCPRI: The range is [max(-60, -90.3+NRCELL.pMax+10*log10(number of TRX per polarization)), min(50, NRCELL.pMax+10*log10(number of TRX per polarization))] In order to avoid Alarm-4090, the lowest boundary value of NRCELL.ssPbchBlockPower should be set based on - NRCELL.pMax - the OAM configured InputPowerLowThreshold (defined in the BFP) and - an offset Delta value in dB with NRCELL.ssPbchBlockPower>= NRCELL.pMax + InputPowerLowThreshold – Delta(dB). The default value for Delta(dB) is 3 (dB) and the default value for InputPowerLowThreshold is -30 (dB). Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	-60...50 [decimal]	[dBm]	-

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Performance Executive Summary



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Performance Executive Summary

Please note that gains coming from simulations might not reflect final product performance

Feature monitoring

New counters	no
Related legacy counters	yes*
KPIs	yes
Activation Monitoring via CM	no

Capacity and Performance impact related aspects

	DL	UL
Cell Coverage	yes	no
Cell Throughput	yes	no
User Peak Rates	no**	no
RRC Connected Users per cell	yes	
End-to-End Delay	no	no
Power Consumption	no	no
... (e.g., # cells per BTS, # voice users per slot)	no	

Architectural enablers

Interfaces	no
Transport	no
Cloud	no

Benefits and gains

- Status: 23.02.2021 based on simulation results
- Improved 4x4 MIMO rank 3/4 reporting utilisation comparing to 5GC000533.
- Improved downlink average cell and cell edge throughput.
- Better spectral efficiency.
- Reduced CSI-RS for CSI Overhead for large amount of RRC connected UEs.
- Robust choice of user beams by 2 stage approach.

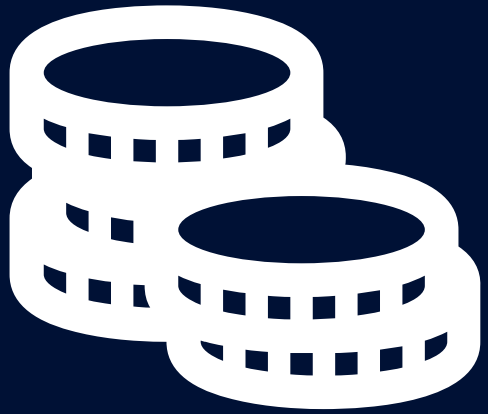
* - 5GC002200 related counters to be provided with CB007157 (01.04.21: stopped)
** - impact on User Peak Rates under investigation

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Expected Benefits & Performance Impact



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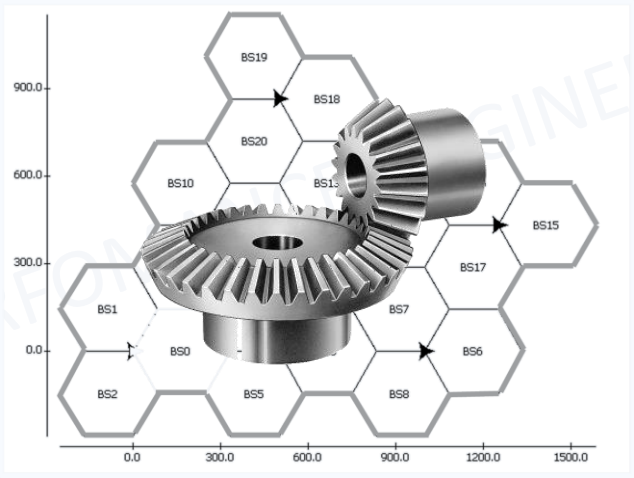
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Expected Benefits & Performance Impact

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

System Level Simulations



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Disclaimer

- The results of simulations shown in this presentation are examples only. They demonstrate trends (not absolute values) expected after feature activation. The presented simulations should be analyzed with respect to the assumptions taken. They may differ from results achievable in real networks.

- Simulation results cannot be used as commitments towards customers. If they shall be used as source for customer communication, it is mandatory to align the contents with Product Management and local Sales teams first.

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Simulation Assumptions

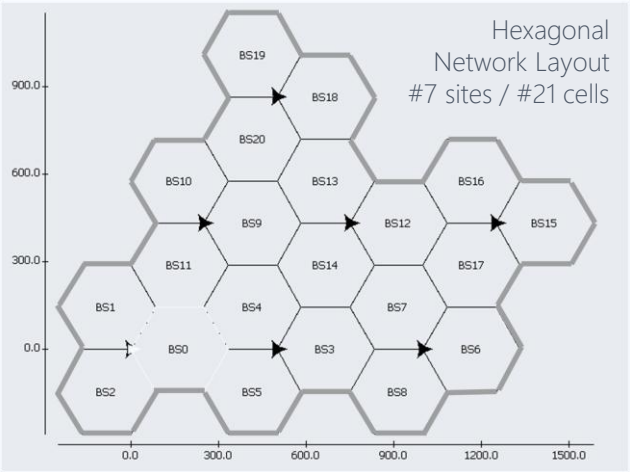
Simulation Assumptions		#21 cells / #10UE per cell
deployment scenarios	Tx antenna array	64TRx / 32TRX
	Carrier frequency and bandwidth	100MHz @ 3.5GHz, single carrier only
	Site/Cell configuration	ISD500m, BS height = 25m and mechanical downtilt 6 degree for UMA/UMA-B ISD200m, BS height = 10m and mechanical downtilt 6 degree for UMI/UMI-B
	Traffic model	full buffer
	Channel model	38.901
gNB configuration	Antenna model	AEQB: 192AE->4x8x2@ 64T64R, output 200W AEQN: 96AE->2x8x2@ 32T32R, output 80W
	TDD subframe configuration	Subframe Type defined in 5GC001208 with 2GP and 6SSB per 40ms for #6SSB; CQI report period = 40ms
	BLER target	0.1
	SSB#	6(beam set#6), product aligned, and SSB beam with 6 degree pre-downtilt AEQB-V3-Optimized-120degAzOp_Coarse(/refined)BeamSetTRX#6; AEQN-V3-Optimized-120degAzOp_Coarse(/refined)BeamSetTRX#6;
	8 port 2200 1x4x2_extH	CSI-RS 8 ports: based on sector beam on(from less sub-panels in H domain) x-pol sub-panels ; 1 Res CSI-RS precoding = extended (to 16 port) based on 8 port 3GPP codebook; beamforming = sector beam on sub-panel
	8 port 2200 2x2x2 4Res	CSI-RS 8 ports: based on 4 sector beams(from 4 sub-panels) plus x-pol; 4 Res CSI-RS precoding = 3GPP codebook; beamforming = sub-panel sector beam
	4 port 1856	CSI-RS 4 ports: based on UE selected two best beams, i.e, coarse beam (i.e. SSB) or refine beam from full panel close loop pre-coding with 3GPP codebook; port-selection up to layer 4
	Downlink MU MIMO	disabled
	DL scheduler	FDM (2537 FD-beamforming enabled): max 8 scheduled user per TTI
	Link Adaptation	5G LA CFAM aligned, up to 256QAM
UE configuration	Downlink UE receiver	4Rx IRC receiver
	UE movement	10 active users per cell random movement@3km/h

The simulation is to verify the performance in multi-cell multi-UE scenarios for 5GC002200, **based on latest MoRSE software**, which is aligned with current product implementation (5G20B)

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Simulation Scenarios

Simulation Case	Description
1856+533 (without refinement)	5GC001856 close loop with 6 SSB beams ← Baseline
1856+533 (with refinement)	5GC001856 close loop with 24 Refine beams
8port_1*4*2_ExtH	8 port CSI-RS, 64 TRX 4*8*2 or 32 TRX 2*8*2 mapping to 1*4*2 ports, PowerControlOffset 0dB or 6dB (i.e. PO0, PO6), 1 CSI-RS resources
8port_2*2*2_4Res	8 port CSI-RS, 64 TRX 4*8*2 or 32 TRX 2*8*2 mapping to 2*2*2 ports, PowerControlOffset 0dB or 6dB (i.e. PO0, PO6), 4 CSI-RS resources

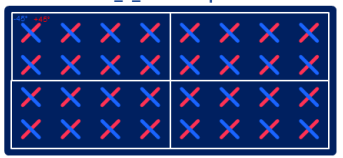


Simulation scenarios

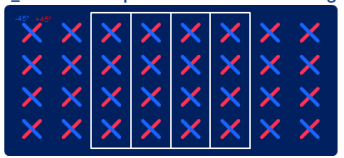
Simulation Scenarios	Inter Site Distance (ISD)	BS antenna height
Urban Macro (UMA)	500m	25m AGL
Urban Macro with Buildings (UMA-B)	500m	25m AGL
Urban Micro (UMI)	200m	10m AGL
Urban Micro with Buildings (UMI-B)	200m	10m AGL

Simulated panel split configurations

2_2_2 Cross Split



4_1 Horizontal Split with Extended Precoding



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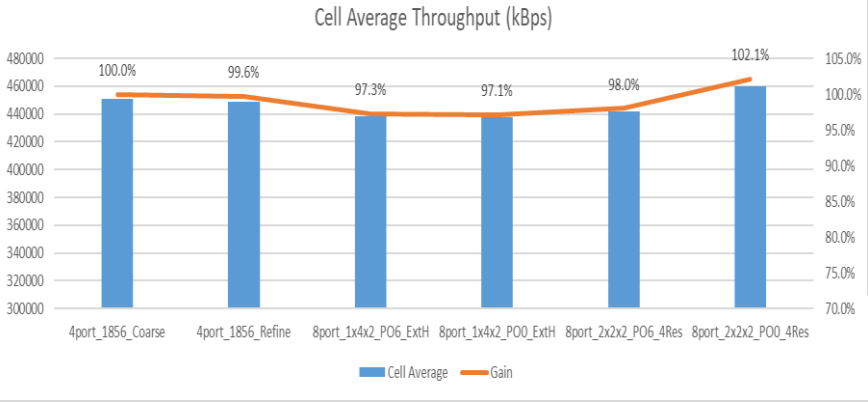
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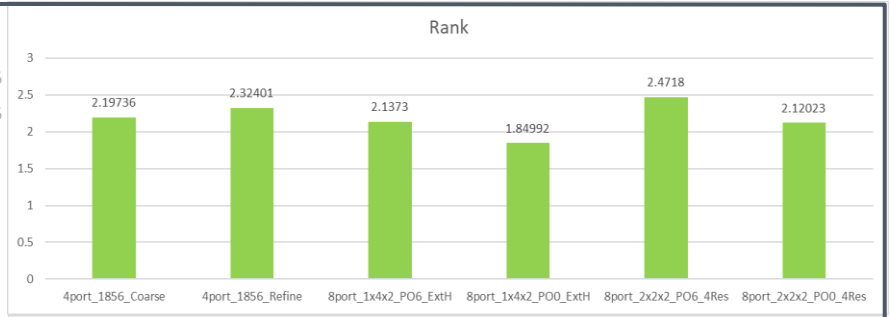
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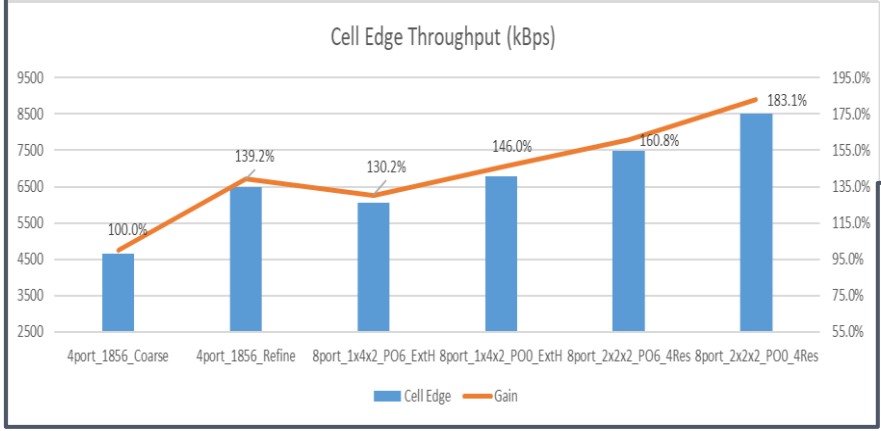
Simulation Scenario UMA

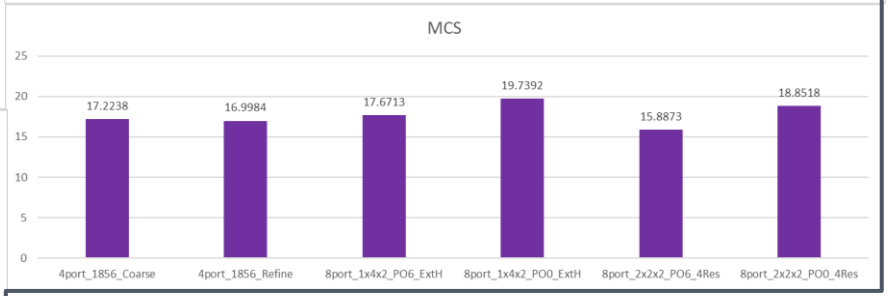
Radio Unit **AEQB** (64TRX)

- The results of simulations shown in this presentation are examples only.
- They demonstrate trends (not absolute values) expected after feature activation.
- The presented simulations should be analyzed with respect to the assumptions taken.
- They may differ from results achievable in real networks.









- No much difference between 1856 and 2200 in cell average TP
- For cell edge, 2x2x2_4Res has significant gain, compared with even 1856_refine
- PowerControlOffset 0dB is better than 6dB

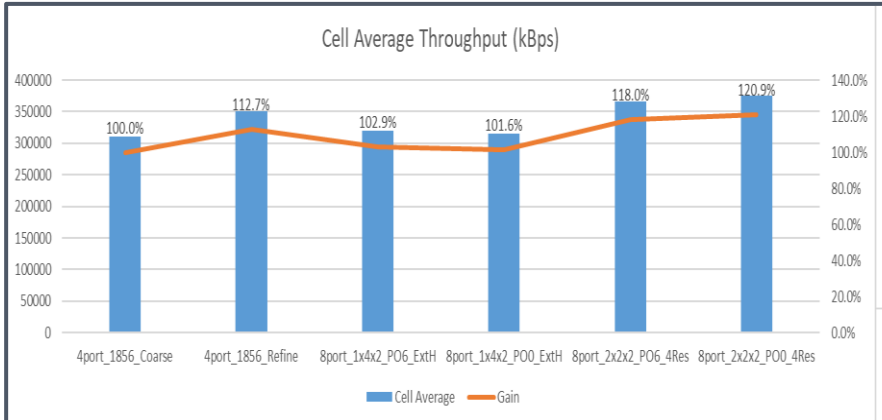
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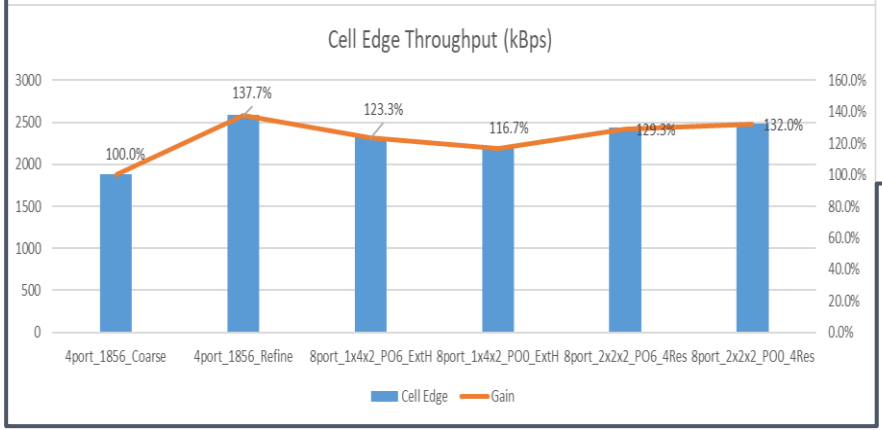
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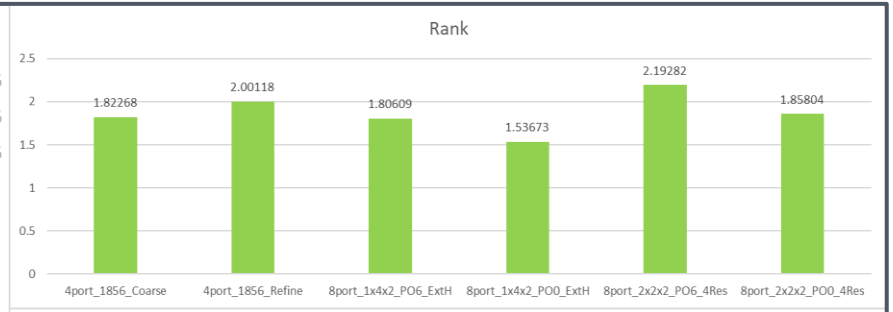
Simulation Scenario UMA-B

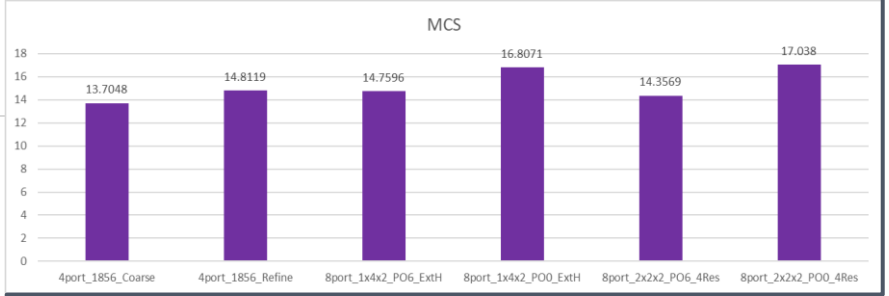
Radio Unit **AEQB** (64TRX)

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- 2x2x2_4Res has ~20% gain compared with 1856_coarse in cell average TP
- For cell edge, 2x2x2_4Res still has significant gain compared with 1856_coarse, but slightly worse than 1856_refine

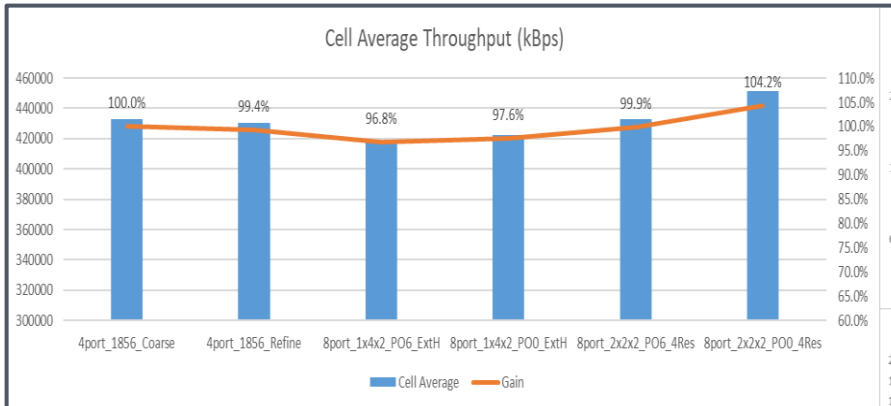
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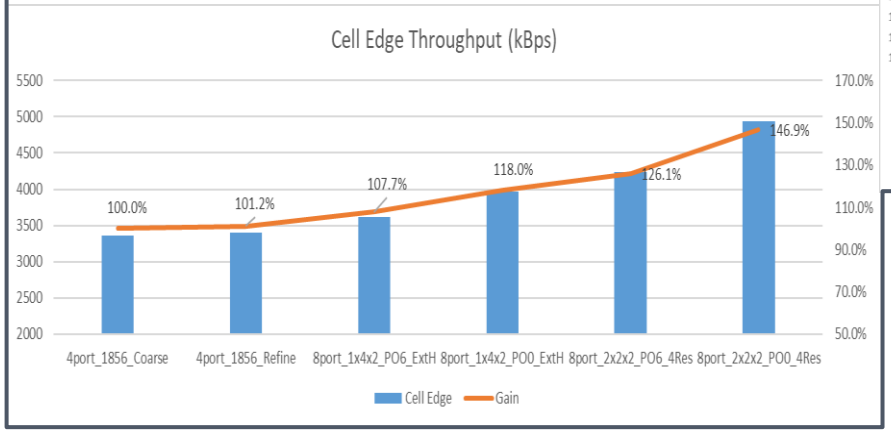
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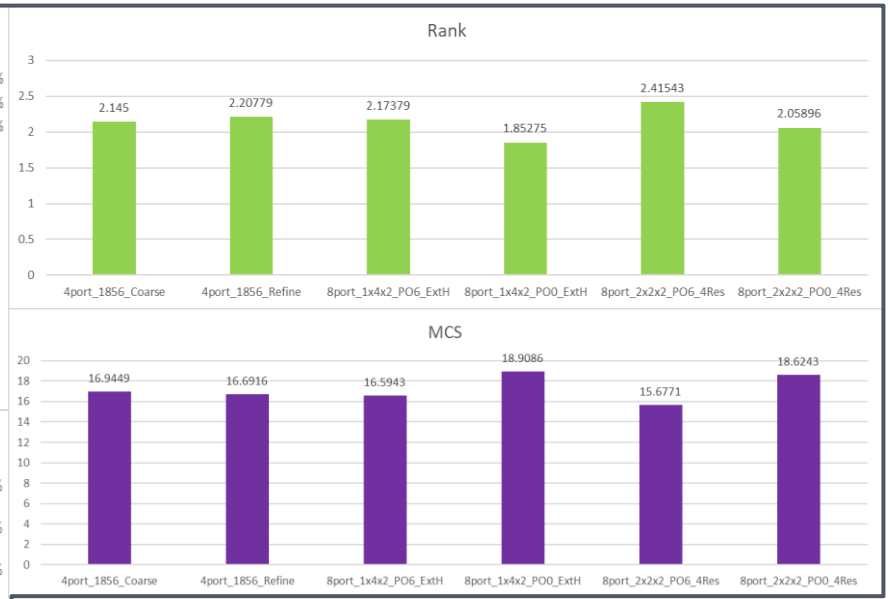
Simulation Scenario UMI

Radio Unit **AEQB** (64TRX)

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- They demonstrate trends (not absolute values) expected after feature activation.
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- They may differ from results achievable in real networks.







- 2x2x2_PO0_4Res has ~4% gain compared with 1856_coarse in cell average TP, while no much difference from other cases
- For cell edge, 2x2x2_PO0_4Res has significant gain, compared with even 1856_refine

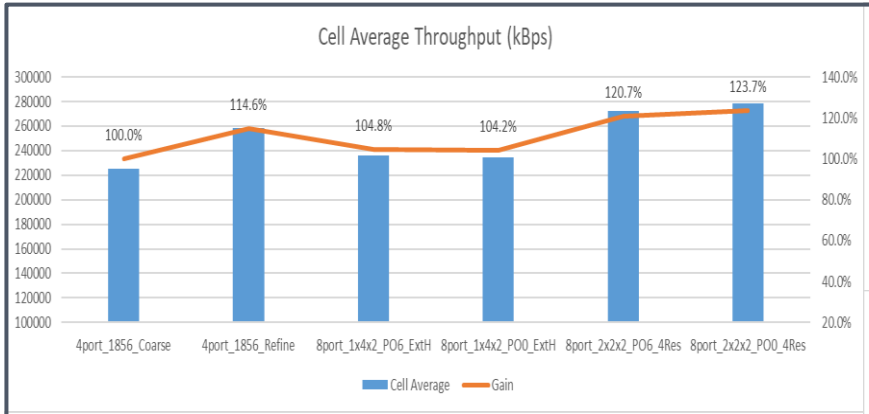
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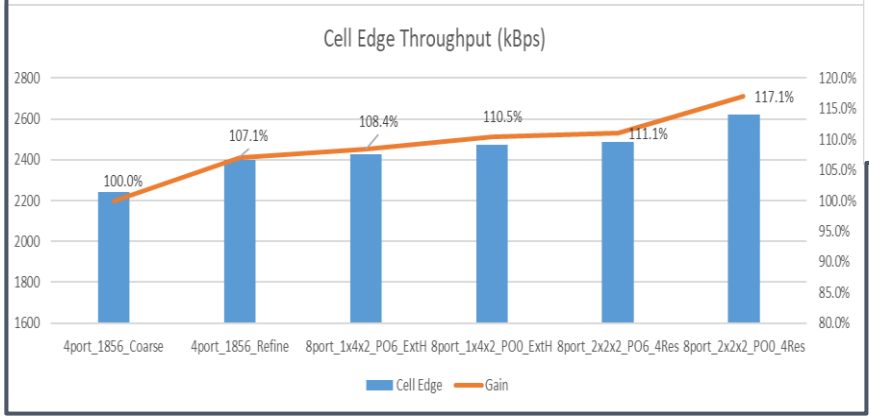
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Simulation Scenario UMI-B

Radio Unit **AEQB** (64TRX)

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- They demonstrate trends (not absolute values) expected after feature activation.
- The presented simulations should be analyzed with respect to the assumptions taken.
- They may differ from results achievable in real networks.







- 2x2x2_4Res has ~23% gain compared with 1856_coarse in cell average TP
- For cell edge, 2x2x2_4Res has 17% gain compared with 1856_coarse, as well as 10% compared with 1856_refine

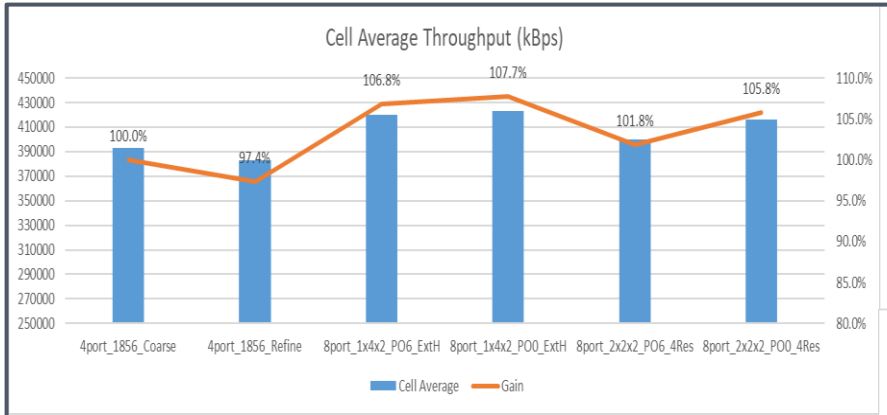
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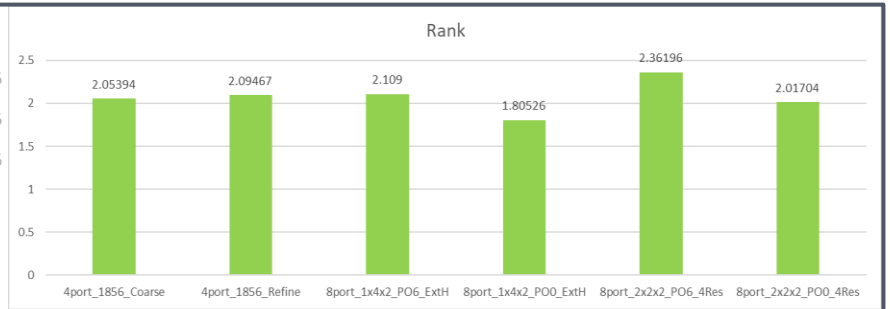
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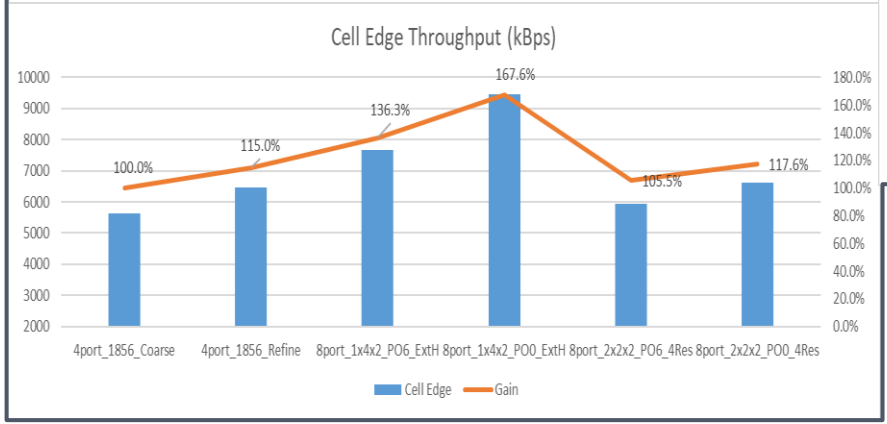
Simulation Scenario UMA

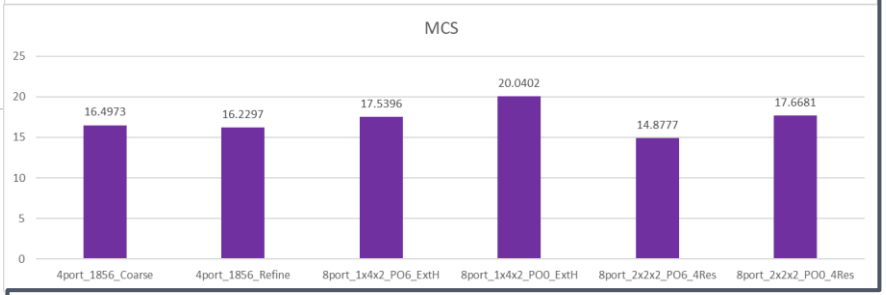
Radio Unit **AEQN** (32TRX)

- The results of simulations shown in this presentation are examples only.
- They demonstrate trends (not absolute values) expected after feature activation.
- The presented simulations should be analyzed with respect to the assumptions taken.
- They may differ from results achievable in real networks.









- 1x4x2_ExtH has 7.7% gain compared with 1856_coarse in cell average TP
- For cell edge, 1x4x2_ExtH has significant gain, compared with both 1856 cases
- PowerControlOffset 0dB is better than 6dB

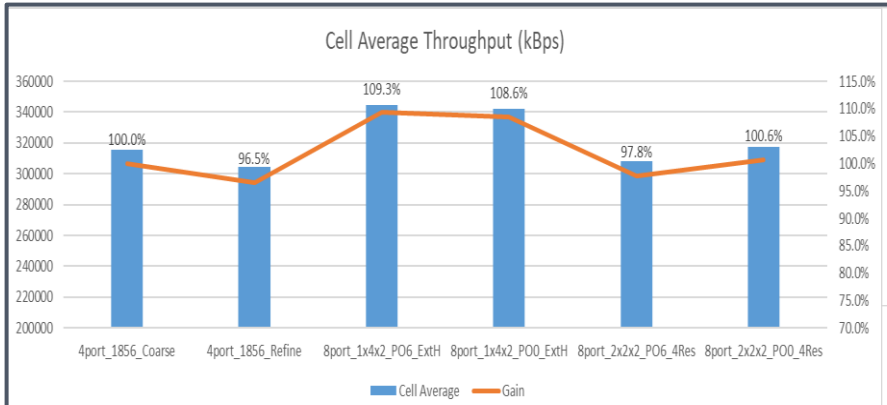
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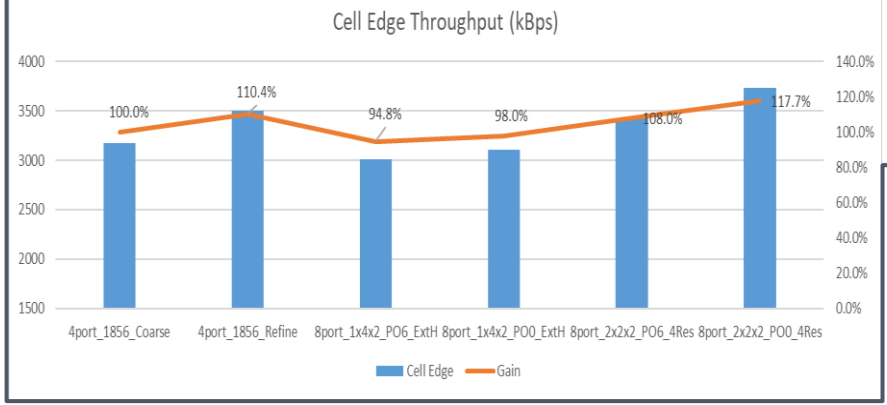
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Simulation Scenario UMA-B

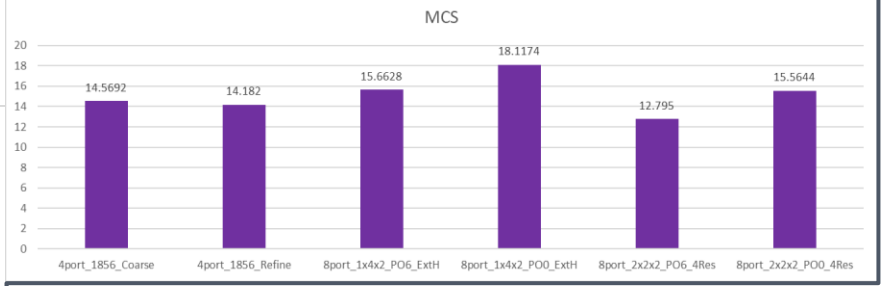
Radio Unit **AEQN** (32TRX)

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- 1x4x2_ExtH has ~9% gain compared with 1856_coarse in cell average TP
- For cell edge, 2x2x2_4Res has 7.7% gain compared with 1856_coarse
- 1x4x2_ExtH better than 2x2x2_4Res for cell average but worse in cell edge

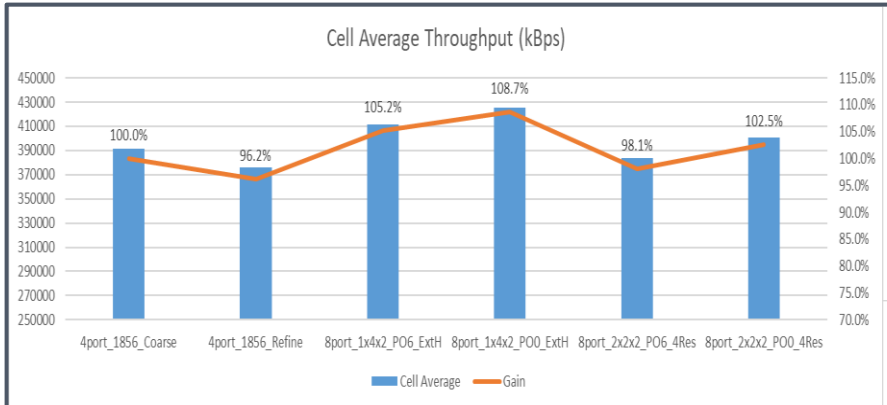
© Nokia 2021

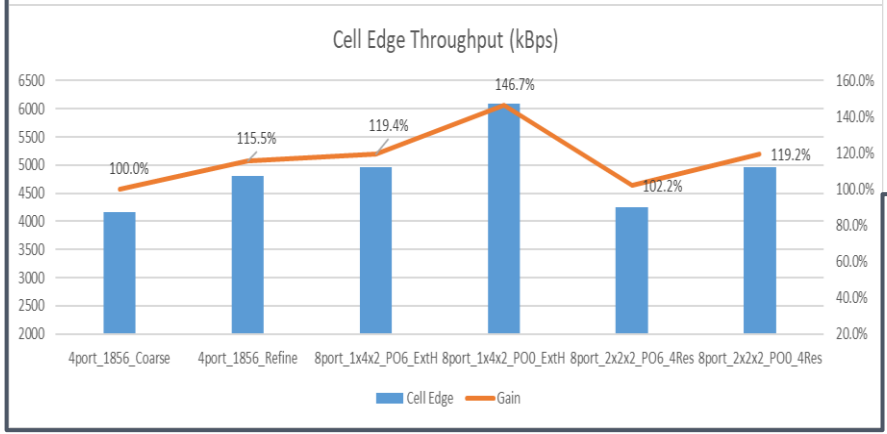
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Simulation Scenario UMI

Radio Unit **AEQN** (32TRX)

- The results of simulations shown in this presentation are examples only.
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- They may differ from results achievable in real networks.







- 1x4x2_PO0_ExtH has ~8.7% gain compared with 1856_coarse in cell average TP
- For cell edge, 1x4x2_PO0_ExtH has significant gain, compared with both 1856_coarse and 1856_refine

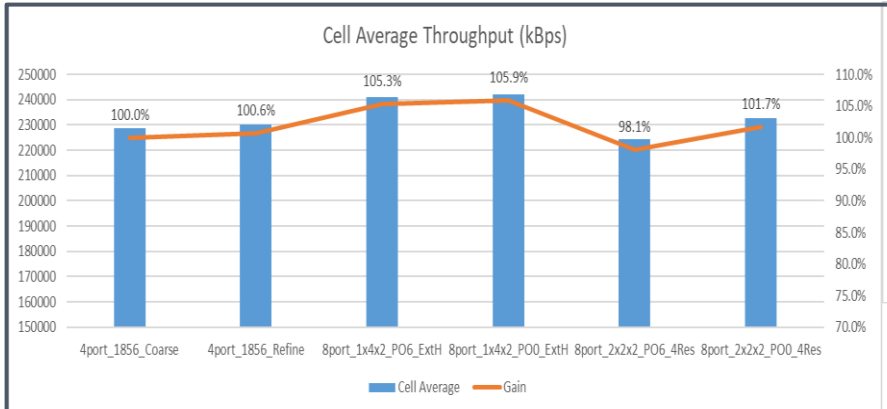
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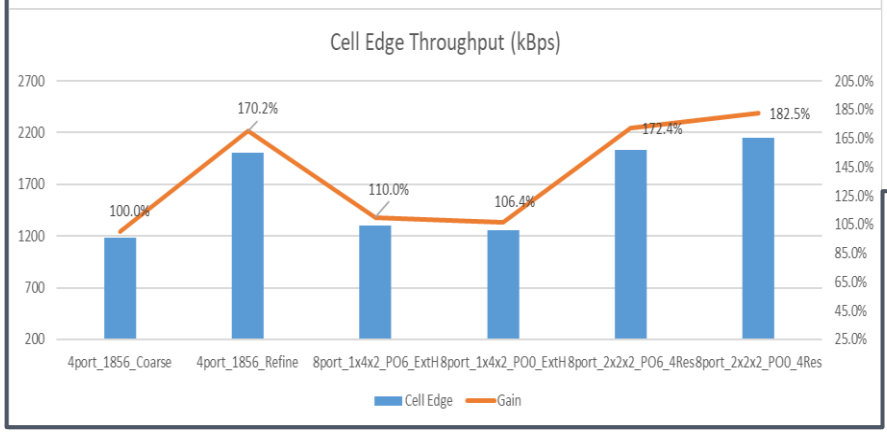
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Simulation Scenario UMI-B

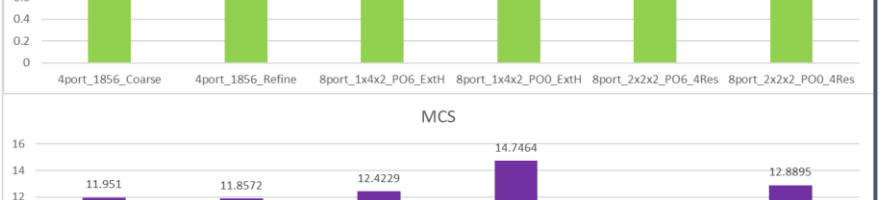
Radio Unit **AEQN** (32TRX)

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- 1x4x2_PO0_ExtH has 5.9% gain compared with 1856_coarse in cell average TP
- For cell edge, 2x2x2_4Res has significant gain compared with 1856_coarse
- 1x4x2_ExtH better than 2x2x2_4Res for cell average but worse in cell edge

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Simulation Results

Summary of general trends

- The results of simulations shown in this presentation are examples only.
- They demonstrate trends (not absolute values) expected after feature activation.
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- They may differ from results achievable in real networks.

AEQB Radio Unit:

- 2200 2x2 4Res performs best in all simulation scenarios, for both cell average and cell edge TP
  - Compared with 1856 coarse for cell average, benefit ranged from 2%~4% gain in the case w/o building, to >10% in building
  - For cell edge, more significant gains over coarse beam than cell average (>15%) in most cases. Most simulations show cell edge improvement over refined beams
- 2200 2x2 4Res always better than 4x1 ExtH, as it has better beam coverage in horizontal

AEQN Radio Unit:

- 2200 4x1 ExtH and 2x2 4 resources has gains for cell average and cell edge vs. coarse beams
  - Compared with 1856 coarse for cell average, 2.5%~8% gain
  - For cell edge, even significant gain in the case w/o building, ~18% for cell edge
- 2200 4x1 ExtH performs better than 2x2 4Res in most of the scenarios, which is different with single user simulation result. The reason is in vertical the subpanel beam of 2x2 covers large angle so that suffers from inter-cell interference

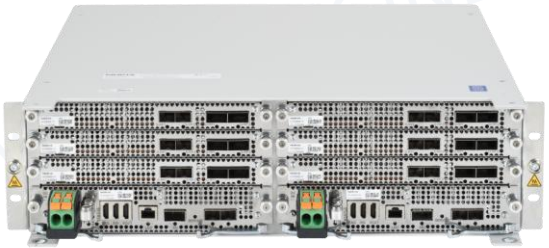
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Expected Benefits & Performance Impact

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Base Band



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RAP capacity

Features enhancing ‘RRC Connected UE’ cell capacity for FR1-TDD cells

5GC002200 Support 8 port CSI-RS codebook for SU-MIMO

- This feature primarily improves Spectral Efficiency for DL MIMO feature, but on the other side it removes bottleneck in cell UL RRC Connected UE capacity (removes the ‘number of UL CSI-RS report’ limit)
- It supports cell specific 8 ports CSI-RS configuration, the CSI-RS resource number for each UE will be 1/2/4 depend on UE capability, RU type and operator configuration
- Feature only supports Type I Single-Panel Codebook, with codebookMode = ‘1’. This feature only supports close loop SU MIMO, and report quantity cri-RI-LI-PMI-CQI is added

5GC2200 preconditions:

- 4x4 DL SU MIMO (5GC0605) feature activation is a precondition, however RRC Connected UE cell capacity gain is visible also for 2x2 DL MIMO configured UEs, as 2, 4 or 8 CSI-RS are configured in network depending on DL MIMO rank
  - nrCellType ‘4DL2UL’ or ‘4DL4UL’ or ‘8DL4UL’
  - BB Blocks with one or two C4 subcells per cell in DL are mandatory, i.e. BB Blocks with ‘4DL’ or ‘8DL’ cells
  - actBeamforming must be set to value ‘TRUE’
  - dlMimoMode ‘8x4’ or ‘ 4x4’,
    where NxM MIMO notation:
    N = number of CSI-RS antenna ports used for PMI per UE
    M = max number of PDSCH layers (max Rank) per UE

Remarks:

- 5GC2200 is not used with Subsectorization Mode (as of 5G20B and 5G21A at least)

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TENTATIVE CAPACITY FIGURES
Capacity figures are subject to Product Management approval.
Figures are pending verification and are subject to update.

RAP capacity

Max 5G RRC Connected UE capacity per cell / TDD-FR1 / 2-UL / 4-UL / ABIL

- Max number of RRC Connected UEs NSA [7] per one cell with 100MHz cell BW [6]

Max RRC Connected UEs per FR1 TDD cell A. 5GC1116_0ms B. 5GC1116_3ms C. 5GC1208 D. 5GC1070	baseline capacity	5GC1348	5GC2545	5GC2545 + 5GC2200
2UL actBeamforming 'disabled'	A. 80 B. 88 C. 120 D. 152	A. 96 B. 96 C. 128 D. 160	A. ~~320~~352 [1] B. 352 [1] C. 326 [1] [4] D. 294 [1] [2] [4]	N/A 5GC2200 requires actBeamforming = enabled
2UL actBeamforming 'enabled'	A. 40 B. 44 C. 60 D. 76	A. 48 B. 48 C. 64 D. 80	A. 160 B. 176 C. 240 D. 304 [3]	A. 160 B. 176 C. 240 D. 304
4UL [5] 2RX receiver actBeamforming 'enabled'	A. 80 B. 88 C. 120 D. 152	A. 96 B. 96 C. 128 D. 160	A. 320 B. ~~352~~360 C. 348 [4] D. 316 [4]	A. 320 B. ~~352~~360 C. 480 D. 608

[6] Cell BW<100MHz is limiting the 'Max RRC Connected UE capacity per cell' for some of the feature combinations

Remarks:
- RRC Connected UE capacity for 'SA' cell and 'both SA+NSA' cell is lower than for 'NSA only' cell for some of the feature combinations. Details are under investigation
- Optimal parameter settings required to reach Max number of RRC Connected UE - details

Reference: CFAM: *5GC1937 Performance and Capacity feature. - Classic gNB requirement version 8.0 dated 12.08.2020*
With following updates:
- 5GC2545 + 5GC1348 removed as features do not work at the same time [PR556794]

[1] 5GC2545 is supported with actBeamforming 'disabled' after fault PR543641 correction (5G20A_9.1800.1100 onwards)

[2] after fault PR547721 correction (5G20A_9.b1.xxxx 5G20B_10.b1.xxxx)

[3] Increased from 190 (220) after fault PR543641 correction

[4] max number of CSI-RS reports limit is reached. Limit is removed with 5GC2200 (beamforming is required)

[5] BB Blocks with two C2 subcells per cell in UL (2RX receiver): 5GC2222, 5GC2221 and 5GC2228
Note:
4RX feature for FR1-TDD comes in 5G21A with single C4 subcell per cell in UL with FD Beamforming. With single UL subcell per cell we loose concept of extra RRC capacity for baseline capacity, 5GC1348 and 5GC2545 cases

***Update 24.02.2021: 5GC1208 values were double checked with** Classic gNB requirement version 9,112*

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Expected Benefits & Performance Impact

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Coverage Aspects



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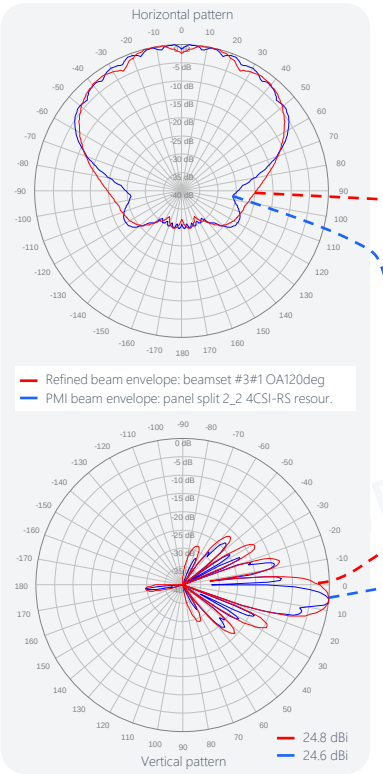
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illustrative picture

Coverage Aspects of PMI Beams and Refined Beams (2_2 #4 CSI-RS Resources)

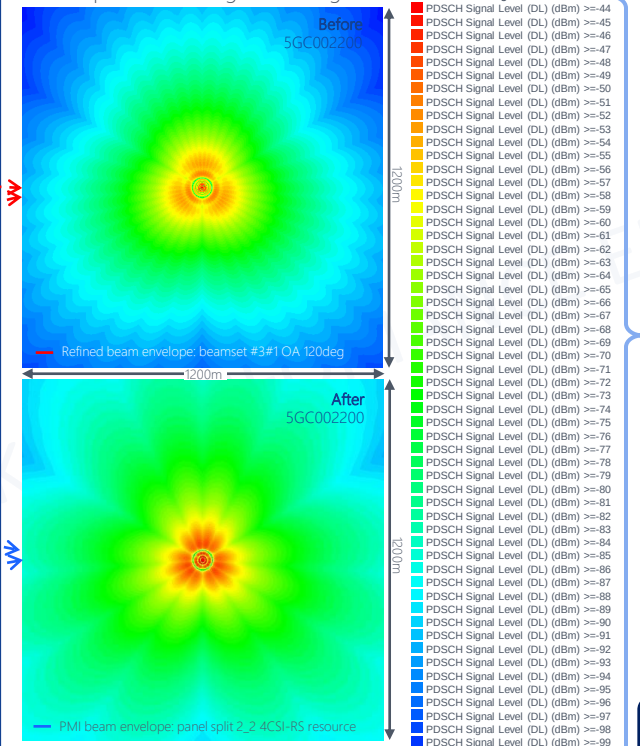
Example PDSCH composite coverage footprint relative comparison

PMI and Refined beam envelope patterns

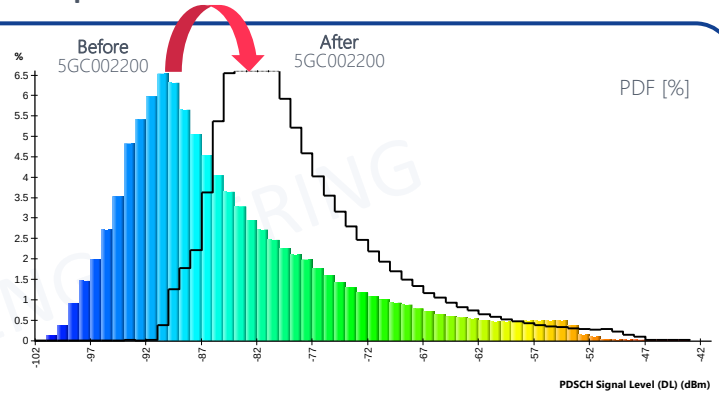


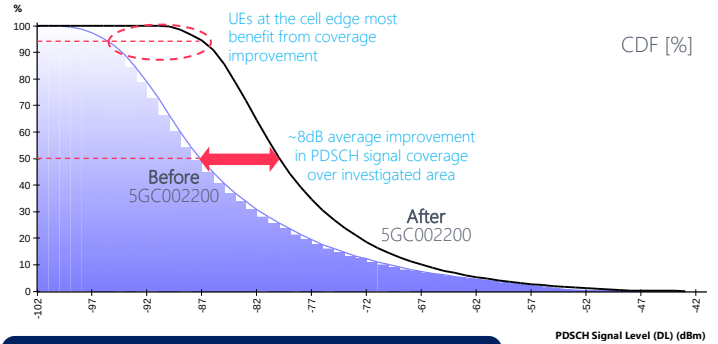
Radio Unit 64TRX, 192AE, Max power: 53dBm

Composit PDSCH signal coverage



3-sector site, azimuths: 0°,120°,240°, antenna height: 25m AGL, propagation model: SPM DU tuned at 3.5GHz, mapping data 5m: flat DTM 0m, homogenous land usage





Introduction of 5GC002200 may help to improve PDSCH coverage due to much more dense design of PMI beams, conveying data streams.

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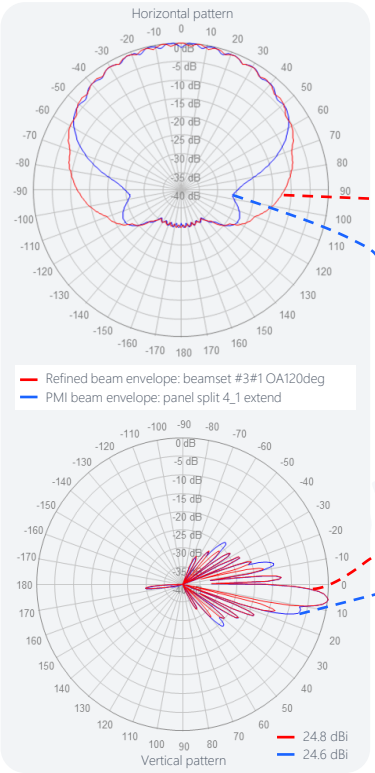
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illustrative picture

Coverage Aspects of PMI Beams and Refined Beams (4_1 #1 CSI-RS Resource)

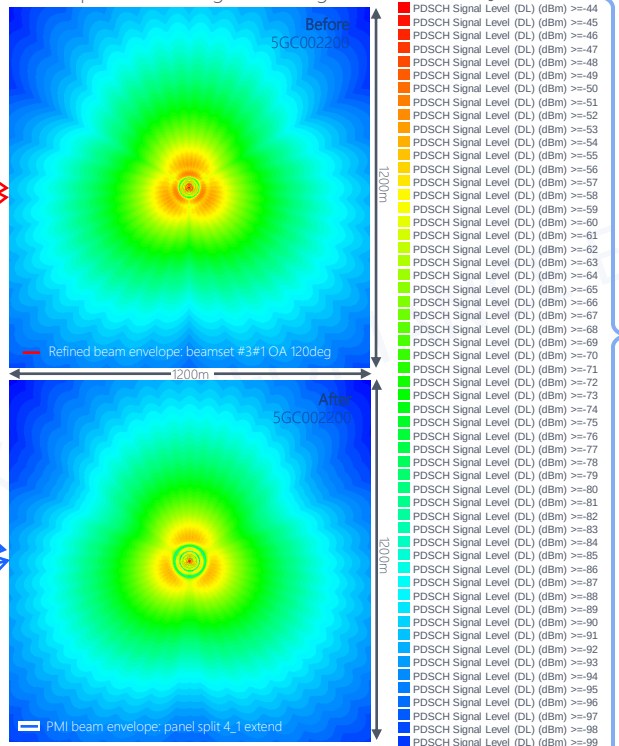
Example PDSCH composite coverage footprint relative comparison

PMI and Refined beam envelope patterns

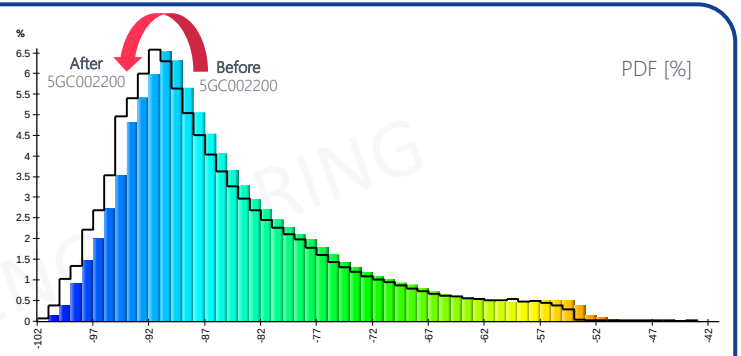


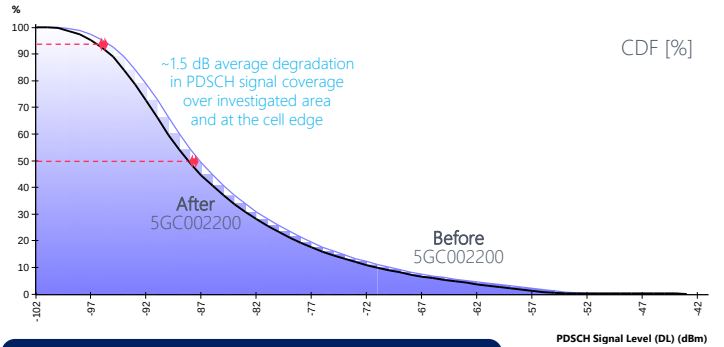
Radio Unit 64TRX, 192AE, Max power: 53dBm

Composit PDSCH signal coverage



3-sector site, azimuths: 0°,120°,240°, antenna height: 25m AGL, propagation model: SPM DU tuned at 3.5GHz, mapping data 5m: flat DTM 0m, homogenous land usage





In case of 4_1 extend config. introduces slight degradation PDSCH coverage due to less dense design of PMI beams comparing to 2_2 4CSI-RS.

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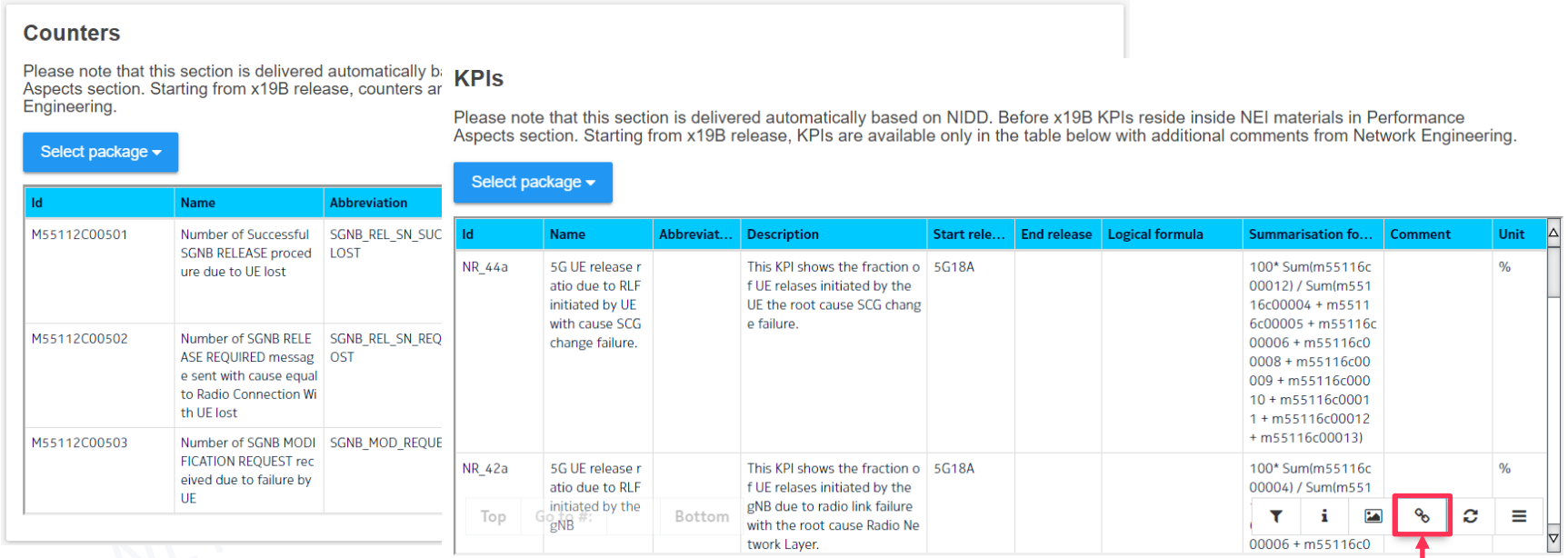
KPIs & Counters



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How to find feature related counters & KPIs in WebNEI?

- Under feature materials in WebNEI platform there are dedicated sections called Counters and KPIs, which lists all those related to given functionality



Counters

Please note that this section is delivered automatically b
Aspects section. Starting from x19B release, counters ar
Engineering.

Select package

Id	Name	Abbreviation
M55112C00501	Number of Successful SGNB RELEASE proced ure due to UE lost	SGNB_REL_SN_SUC LOST
M55112C00502	Number of SGNB RELE ASE REQUIRED messag e sent with cause equal to Radio Connection Wi th UE lost	SGNB_REL_SN_REQ OST
M55112C00503	Number of SGNB MODI FICATION REQUEST rec eived due to failure by UE	SGNB_MOD_REQUE

KPIs

Please note that this section is delivered automatically based on NIDD. Before x19B KPIs reside inside NEI materials in Performance Aspects section. Starting from x19B release, KPIs are available only in the table below with additional comments from Network Engineering.

Select package

Id	Name	Abbreviat...	Description	Start rele...	End release	Logical formula	Summarisation fo...	Comment	Unit
NR_44a	5G UE release r atio due to RLF initiated by UE with cause SCG change failure.		This KPI shows the fraction o f UE relases initiated by the UE the root cause SCG chang e failure.	5G18A			100* Sum(m55116c 00012) / Sum(m551 16c00004 + m5511 6c00005 + m55116c 00006 + m55116c0 0008 + m55116c00 009 + m55116c000 10 + m55116c0001 1 + m55116c00012 + m55116c00013)		%
NR_42a	5G UE release r atio due to RLF initiated by the gNB		This KPI shows the fraction o f UE relases initiated by the gNB due to radio link failure with the root cause Radio Ne twork Layer.	5G18A			100* Sum(m55116c 00004) / Sum(m551 00006 + m55116c0		%

Top Bottom

- If nesesery you can always download a copy of the KPIs/counters by pushing a button in right corner

PLEASE NOTE THE COUNTER SECTION CONTAINS ALSO COUNTERS THAT MAY PRODUCE UNPREDICTABLE RESULTS AFTER 5GC002200 ACTIVATION. PLEASE ALWAYS CHECK NetEng Comment TO VALIDATE POSSIBLE COUNTER IMPACT.

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Impact of Counters

5GC002200: Support 4x8 port CSI-RS codebook for SU-MIMO

- When the 5G20B feature *5GC002200: Support 4x8 port CSI-RS codebook for SU-MIMO* is taken into use, the PDSCH allocation will no longer use the Grid-of-Beams.
- A spillover feature CB007157 was supposed to redefine how the PMI beams are mapped to the existing beam related PM counter ecosystem.
- The feature was originally planned for 5G21A but recently was stopped (status on 01.04.2021)
- This means that the legacy counters are not updated for the PMI beams and may provide unreliable results.
- List of the affected counters:

PI ID	Measurement	Network element name
M55300C08001...2	NR DL HARQ	Successful PDSCH initial transmissions using 2TX and rank xx
M55300C09001...2	NR DL HARQ	Successful PDSCH transmissions by any HARQ transmissions using 2TX rank xx
M55300C10001...4	NR DL HARQ	Successful PDSCH transmissions by initial transmissions using 4TX rank xx
M55300C11001...4	NR DL HARQ	Successful PDSCH transmissions by any HARQ transmissions using 4TX and rank xx
M55301C04000...3	NR DL Signal Quality Level	PDSCH transmissions in Rank1 using PMI 0...3
M55301C05000...2	NR DL Signal Quality Level	PDSCH transmissions in Rank2 using PMI 0...2
M55301C09001...2	NR DL Signal Quality Level	PDSCH transmissions using 2TX transmission with rank xx
M55301C10001...4	NR DL Signal Quality Level	PDSCH transmissions using 4TX transmission with rank xx
M55301C11000...31	NR DL Signal Quality Level	PDSCH transmissions using 4TX in Rank1 using PMI 0...31
M55301C12000...31	NR DL Signal Quality Level	PDSCH transmissions using 4TX in Rank2 using PMI 0...31
M55301C13000...15	NR DL Signal Quality Level	PDSCH transmissions using 4TX in Rank3 using PMI 0...15.
M55301C14000...15	NR DL Signal Quality Level	PDSCH transmissions using 4TX in Rank4 using PMI 0...15.
M55305C00001...64	NR Beamforming	DL Serving Beam ID Histogram
M55305C02001...64	NR Beamforming	Serving Beam ID before HO Histogram
M55305C03001...64	NR Beamforming	Beam Change Histogram
M55305C04001...64	NR Beamforming	Failed Beam Change Histogram
M55305C07001...64	NR Beamforming	Best and Second Best Beam Histogram
M55305C08001...16	NR Beamforming	UE reported differential L1-RSRP of second best beam
M55305C09001...64	NR Beamforming	Beam Recoveries - Beam ID xx
M55305C10001...64	NR Beamforming	Succesful Beam Recoveries - Beam ID xx
M55305C11001...13	NR Beamforming	Time of "UE on same beam" histogram
M55305C12001...64	NR Beamforming	Number of UEs per beam
M55305C13001	NR Beamforming	Number of beam toggles
M55308C21001	NCELA	Accumulated number of utilized beams on PDSCH per slot
M55308C21002	NCELA	Maximum number of utilized beams on PDSCH per slot

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PM measurements, counters & KPIs

Top/Scorecard KPIs used by QoE for NPIs

- Following are selected KPIs called „scorecard" which are often used to describe overall performance within e.g. NPI project *(some KPIs are common between SA and NSA)*
- It can give good indication on main KPIs to be used per main areas

Standalone (SA) KPIs

KPI ID	KPI name
NR_5150a	5G Cell availability ratio
NR_5167a	5G SA RRC connection establishment success ratio
NR_5168a	5G SA Initial context setup success ratio
NR_5151a	5G SA Accessibility success ratio
NR_5088a	5G MAC PDU Cell throughput on active PDSCH data slots on initial HARQ Tx
NR_5089a	5G MAC PDU Cell throughput on active PUSCH slots using 64QAM on initial HARQ
NR_5177b	5G SA intra-gNB intra-DU handover total success ratio
NR_5183b	5G SA intra-frequency inter-gNB handover execution success ratio
NR_5160b	5G SA Total number of RRC connection establishment attempts
NR_5230a	5G SA Number of initial context setup requests received at gNB-CU
NR_5174b	5G SA intra-gNB intra-DU handover preparation attempts
NR_5182b	5G SA intra-frequency inter-gNB handover execution attempts
NR_5127a	5G SA Average number of SA non-GBR users
NR_5092a	5G SA PDCP SDU data volume transmited in downlink [MB]
NR_5093a	5G SA PDCP SDU data volume received in uplink [MB]

Non-Standalone (NSA) KPIs

KPI ID	KPI name
NR_5150a	5G Cell availability ratio
NR_5020d	5G NSA call accessibility, 5G side
NR_5005a	5G SgNB reconfiguration success ratio
NR_5032b	5G NSA SgNB triggered abnormal release ratio
NR_5088a	5G MAC PDU Cell throughput on active PDSCH slots on initial HARQ Tx
NR_5089a	5G MAC PDU Cell throughput on active PUSCH slots using 64QAM on initial HARQ Tx
NR_5049b	5G NSA Intra frequency intra DU PSCell change total success ratio
NR_5042b	5G NSA Intra gNB intra frequency PSCell change total success ratio
NR_5034a	5G NSA Inter gNB handover success ratio for NSA
NR_581a	5G Average MCS used in downlink for PDSCH with 64QAM table
NR_582a	5G Average MCS used in downlink for PDSCH with 256QAM table
NR_5003b	5G NSA Number of SgNB addition requests
NR_5046b	5G NSA Number of intra freq. intra DU PSCell change preparation attempts for NSA 3x
NR_5040b	5G Intra gNB intra frequency PSCell change preparation attempts
NR_5037a	5G NSA Inter gNB handover attempts for NSA
NR_5124a	5G NSA Average number of NSA users
NR_5080a	5G NSA PDCP SDU data volume transmitted without repetitions in DL
NR_5081a	5G NSA PDCP SDU data volume received without repetitions in UL

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Feature Configuration & Performance Evaluation



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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Feature Activation and Supported nrCellType Settings



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Performance Aspects <5GC002200>

Feature impact

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.dlMimoMode	Downlink MIMO Mode This parameter selects the DL MIMO mode for the PDSCH. NxM MIMO Notation: N = number of CSI-RS antenna ports used for PMI per UE; M = max number of PDSCH layers (max Rank) per UE. Configuring this parameter during cell addition might provoke gNB reset - gNB takes decision of reset necessity. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	Value	Name	[-]	30
		30	2x2 Closed Loop Spatial Multiplexing		
		40	4x4 or 4x2 Closed Loop Spatial Multiplexing		
		50	2x2 Open Loop Spatial Multiplexing		
		60	4x4 or 4x2 Open Loop Spatial Multiplexing		
		70	8x4 Closed Loop Spatial Multiplexing		

New entry introduced by 5GC002200 to enable the feature

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Performance Aspects <5GC002200>

Feature impact

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.nrCellType	NR Cell type This parameter allows the operator to specify the number of fronthaul streams of a New Radio Cell. This parameter is also used for identification of Supplemental downlink cell (FR2). Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	Value	Name	[-]	-
		20	2DL0UL		
		21	2DL1UL		
		22	2DL2UL		
		24	2DL4UL		
		42	4DL2UL		
		44	4DL4UL		
		82	8DL2UL		
		84	8DL4UL		
		88	8DL8UL		
		168	16DL8UL		
		1616	16DL16UL		

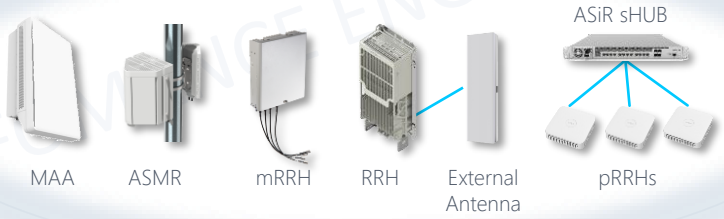
If NRCELL.dlMimoMode is equal to 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), then nrCellType=4DL2UL or nrCellType=4DL4UL or nrCellType=8DL4UL is allowed.

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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

What Radio Unit types are supported by 5GC002200?



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Feature Configuration & Performance Evaluation

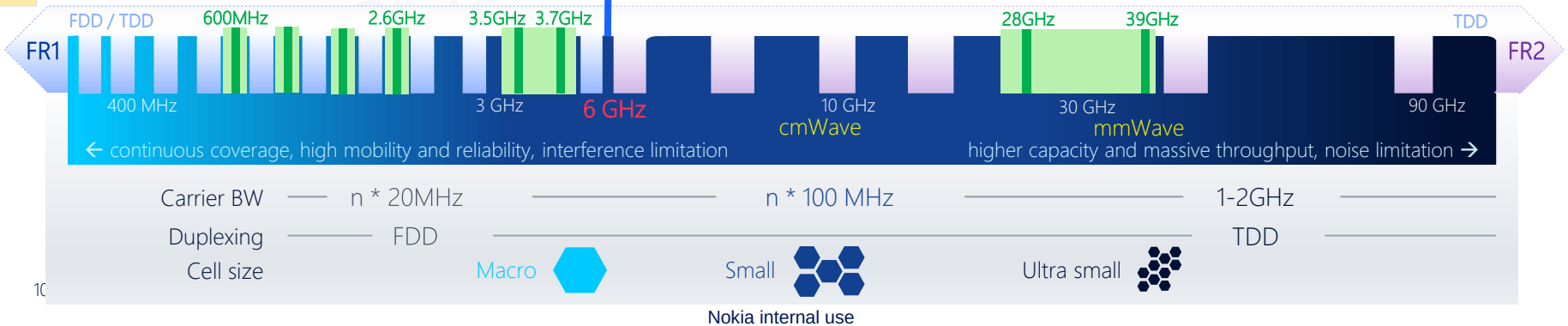
5G Radio Units families supported by 5GC002200

NO Beamforming
Analog Beamforming
Digital Beamforming

MAA - mMIMO Adaptive Antenna
ASMR - AirScale mmWave Radio
RRH - Remote Radio Head
mRRH - micro Remote Radio Head
pRRH - pico Remote Radio Head
ASiR - AirScale Indoor Radio

	SupportedTechnology	Radio Access	RU Type	No of TRXs	Antenna Type	Supported Technology	Radio Access	RU Type	No of TRXs	Antenna Type
Outdoor	5G	TDD	MAA	64	Internal Phased Array (192 AE)	5G	TDD	MAA	2	Internal 2xRFIC (2x256 AE)
	5G/LTE	TDD	MAA	32	Internal Phased Array (96 AE)	5G	TDD	MAA	2 or 8	Internal RFIC (2x256 AE)
	5G	TDD	MAA	8	External	5G	TDD	MAA	2*4	Internal RFIC (4x128 AE: main unit+extension)
	5G	FDD	RRH	2 or 4	External	5G	TDD	ASMR	2*3	Internal RFIC (3*192 AE: main unit+2*extensions)
	5G/LTE	FDD	RRH	4	External					
	5G	TDD	mRRH	2	Internal or External					
Indoor ASiR	5G/LTE	TDD	pRRH	4	Internal RFIC or External	5G	TDD	pRRH	product under planning	

RU families supported by 5GC002200



Nokia internal use

Feature Configuration & Performance Evaluation

5GC002200 vs 5GC001866 - interworking aspects

5GC002200 interwoking

- 5GC001866 supports beam weight download to RU as basic frame work, which is used by 5GC002200.
- 5G001866 is a prerequiste for 5GC002200 however **5GC001866 activation flag ActCustomBsr has not to be enabled as mandatory prerequisite** in order to make 5GC002200 up and running.
- If actCustomBsr=FALSE or actCustomBsr=TRUE but CustomBeamSetName is not configured then 5GC002200 **will work with SSB transmitted via beams defined by Nokia**.
- If actCustomBsr=TRUE and CustomBeamSetName is configured correctly then 5GC002200 **will work with SSB beams that customer inputs**.
- 5GC002200 cooperates with **Radio Units supported by 5GC001866**.

Nokia internal use

Feature Configuration & Performance Evaluation

Radio Units supported by 5GC001866/5GC002200 feature in 5G20B

Variant	Type	cmWave / mmWave	Protocol	Sales Item Name	RF HW family group	1866 Sub-feature mapping
AEQK	64T64R	cmWave	CPRI	AEQK AirScale MAA 64T64R 192AE 200W (CPRI)	cMAAr6.2_64T64R	1866-A
AEQV	32T32R	cmWave	CPRI	AEQV AirScale MAA 32T32R 192AE n77 100W (CPRI)	cMAAr6.1_32T32R	1866-A
AEQB	64T64R	cmWave	CPRI	AEQB AirScale MAA 64T64R 192AE n78 200W (CPRI)	cMAAr6.0_64T64R	1866-A
AEQC	32T32R	cmWave	CPRI	AEQC AirScale MAA 32T32 192AE n78 200W (CPRI)	cMAAr6.1_32T32R	1866-A
AEQE	64T64R	cmWave	CPRI	AEQE AirScale MAA 64T64R 192AE n78 200W (CPRI)	cMAAr6.0_64T64R	1866-A
AEQI	64T64R	cmWave	CPRI	AEQI AirScale MAA 64T64R 192AE n78 200W (CPRI)	cMAAr6.1_64T64R	1866-A
AEQP	64T64R	cmWave	CPRI	AEQP AirScale MAA 64T64R 192AE n78 200W (CPRI)	cMAAr6.1_64T64R	1866-A
AEQQ	32T32R	cmWave	CPRI	AEQQ AirScale MAA 32T32 192AE n78 200W (CPRI)	cMAAr6.1_32T32R	1866-A
AEQY	32T32R	cmWave	CPRI	AEQY AirScale MAA 32T32R 128AE n78 128W (CPRI)	cMAAr6.1_32T32R	1866-A
AEQZ	32T32R	cmWave	CPRI	AirScale MAA 32T32R 192AE n78 192W AEQZ (CPRI)	cMAAr6.1_32T32R	1866-A
AEHB	64T64R	cmWave	eCPRI	AEHB AirScale MAA 64T64R 192AE n41 320W	cMAArChile_64T64R	1866-B, 1866-D
AEHC	64T64R	cmWave	eCPRI	AEHC AirScale MAA 64T64R 192AE n41 320W	cMAArChile_64T64R	1866-B, 1866-D
AEQN	32T32R	cmWave	eCPRI	AEQN AirScale MAA 32T32R 96AE n78 80W (eCPRI)	cMAA_Gen4.0_32T32R	1866-B
AEQJ	64T64R	cmWave	CPRI	AEQJ AirScale MAA 64T64R 192AE n78 200W (CPRI)	cMAAr6.1_64T64R	1866-C
AAHF	64T64R	cmWave	CPRI	AAHF AirScale MAA 64T64R 128AE n41 120W	cMAA_Gen3.14_64T64R	1866-D
AAHJ	64T64R	cmWave	CPRI	AAHJ AirScale MAA 64T64R 128AE n41 120W	cMAA_Gen3.14_64T64R	1866-D
AAHF	64T64R	cmWave	CPRI	AAHF AirScale MAA 64T64R 128AE n41 120W	cMAA_Gen3.14_64T64R	1866-E
AAHJ	64T64R	cmWave	CPRI	AAHJ AirScale MAA 64T64R 128AE n41 120W	cMAA_Gen3.14_64T64R	1866-E
AZQO	8T8R	cmWave	CPRI	AZQO AirScale RRH 8T8R n77 160W (CPRI)	cMAAr6.1_8T8R	1866-J
AZQI	8T8R	cmWave	CPRI	AZQI AirScale RRH 8T8R n78 320W (CPRI)	cMAAr6.1_8T8R	1866-J
AZQJ	8T8R	cmWave	CPRI	AZQJ AirScale RRH 8T8R n78 320W (CPRI)	cMAAr6.1_8T8R	1866-J
AZHL	8T8R	cmWave	CPRI	AirScale RRH 8T8R n41 320W (CPRI)	cMAAr6.1_8T8R	1866-J

If the considered Radio Unit is not on the list and has not been moved to 5G21X release by FGCR-7659 please contact authors for further clarification.

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Feature Configuration & Performance Evaluation

FGCR-7659 – 5G20B Operability descopes (move content from 5G20B to 5G21A/5G21B)

5GC001866 support for listed below Radio Units is descoped from 5G20B to 5G21A/5G21B

5GC001866 can't be supported for all requested RU types in 5G20B. With this CR support of following RU is postponed to 5G21A:

Varinat	Chip	Program	Type	Ext antennas	cmWave / mmWave	Protocol	Sales Item Name	RF HW family group	1866 Sub-feature mapping
AEQN	Beamer FPGA	AEQN	32T32R	NA	cmWave	CPRI	AEQN AirScale MAA 32T32R 96AE n78 80W (CPRI)	cMAA_Gen4.0_32T32R	1866-A
AEHA	**	5G Gen3 Sub6 GHz	64T64R	NA	cmWave	CPRI	AEHA AirScale MAA 64T64R 128AE n41 200W	5Ggen3_LowF_64T64R	1866-E
AEQD*	**	5G Gen3 Sub6 GHz	64T64R	NA	cmWave	CPRI	AEQD AirScale MAA 64T64R 128AE n78 200W	5Ggen3_LowF_64T64R	1866-E
AZQG	**	Pulsar	8T8R	8	cmWave	CPRI	AZQG AirScale RRH 8T8R n78 320W (CPRI)	cMAA_Gen4.0_8T8R	1866-J
AZQH	**	Pulsar	8T8R	8	cmWave	CPRI	AZQH AirScale RRH 8T8R n78 320W (CPRI)	cMAA_Gen4.0_8T8R	1866-J
AZQL	**	Pulsar	8T8R	8	cmWave	CPRI	AZQL AirScale RRH 8T8R n78 320W (CPRI)	cMAA_Gen4.0_8T8R	1866-J

note that for some units, only CPRI mode is postponed while eCPRI is kept.

Link to the CR in JIRA: https://jira3.ext.net.nokia.com/browse/FGCR-7659

* - see FGCR-11190 on next slide

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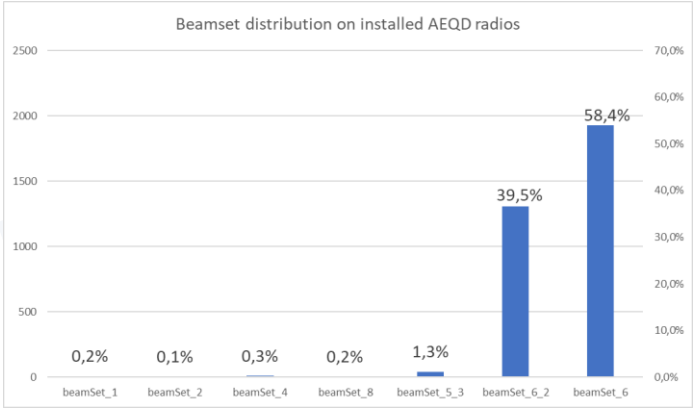
Feature Configuration & Performance Evaluation

FGCR-11190 enabling 5GC002200 4x8 CSI-RS Codebook based beamforming on AEQD

5GC002200 Support 4 x 8 port CSI-RS codebook for SU-MIMO requires that RUs support 5GC001866 framework.
In 5G20B timeframe, AEQD is not supporting 5GC001866.
To enable 5GC002200 on AEQD in 5G20B a workaround is implemented in CR

Implementation of this workaround solution would **overwrite any beamset** definitions **above beamset #6#2**. According to the field data, there are no beamsets above #6#2 deployed with AEQD radios, so there sould be no impact on the existing configurations.

The solution is targeted only for **specific customer configuration** and has **many limitations**. It works only with **few fixed beamsets** and **tiltOffset value 0** which meets the target customer requirement.



With 5GC002200 active following fixed requirements apply:

- basicBeamSet= beamSet_6_2
- leftEdgeAngle=60
- rightEdgeAngle=60
- tiltOffset= 0 or not configured
- Multiple cells must not be configured on AEQD

With FGCR-11190 implemented, following beamsets are supported on AEQD:

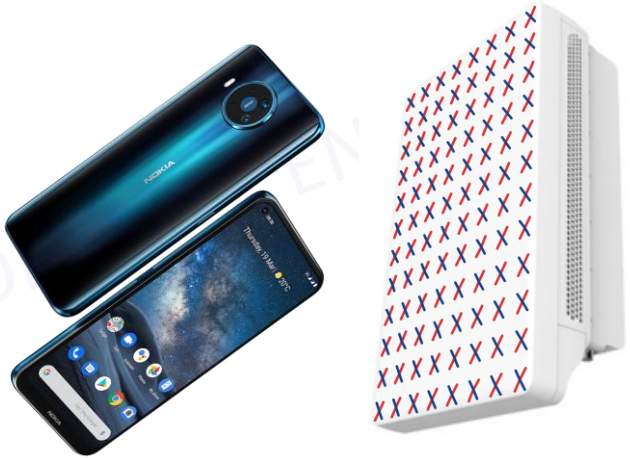
- #1
- #2
- #4
- #6
- #8
- #4#4
- #5#3
- #6#2

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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Radio Unit and UE capability aspects



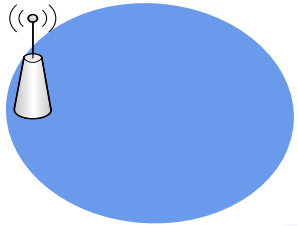
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Feature Configuration & Performance Evaluation

UE capabilities aspects

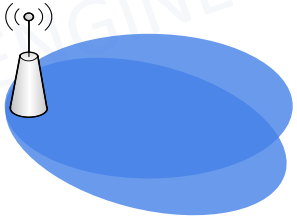
- Combined reporting of sub-panel beam (CRI) with legacy CSI (CQI, RI and PMI) is **subject of UE capability**.
- UE supports either **1, 2 or 4 CSI-RS resources** and respective CRI reporting.
- Different resources are defined for the different sets of UEs:
  - **Single cell wide sub-panel beam** for UEs supporting only a single resource.
  - 2 resources, **each covering half of the cell** for UEs supporting 2 resources.
  - 4 resources **with quarter-sector beams** for UEs supporting 4 resources.
- If all variants are supported in a cell, **7 CSI-RS resources** are provided by the cell.

CSI-RS resource set 1



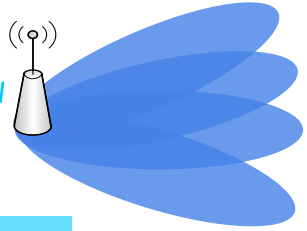
Resource set 1:
- 1CSI-RS resource
- Panel split 4,1
- 16 PMI beams

CSI-RS resource set 2



Resource set 2:
- 2 CSI-RS resources
- Panel split 2,2
- 128 PMI beams

CSI-RS resource set 3



Resource set 3:
- 4 CSI-RS resources
- Panel split 2,2
- 256 PMI beams

Example UE capability (Samsung S20)

```
codebookParameters
    type1
        singlePanel
            supportedCSI-RS-ResourceList: 3 items
                Item 0
                    SupportedCSI-RS-Resource
                        maxNumberTxPortsPerResource: p32 (6)
                        maxNumberResourcesPerBand: 2
                        totalNumberTxPortsPerBand: 64
                Item 1
                    SupportedCSI-RS-Resource
                        maxNumberTxPortsPerResource: p8 (2)
                        maxNumberResourcesPerBand: 8
                        totalNumberTxPortsPerBand: 64
                Item 2
                    SupportedCSI-RS-Resource
                        maxNumberTxPortsPerResource: p16 (4)
                        maxNumberResourcesPerBand: 2
                        totalNumberTxPortsPerBand: 32
            modes: mode1 (0)
            maxNumberCSI-RS-PerResourceSet: 4
```

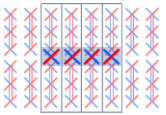
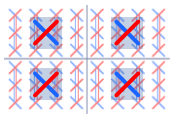
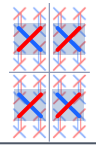
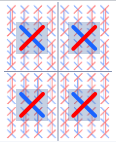
Defines max number of supported CSI-RS resources

UE interworking issues

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Feature Configuration & Performance Evaluation

RU type specific panel split configurations supported by 5GC002200

N1, N2 configurations used per RU type and configured numer of resources								
RU	Rows	Colums	rdTypeSinglePanelN1N2	NumRes can be controlled via NRCELL.periodicCsiAcquisition.**maxCsirsNumber**			Radio Unit Panel Split & CSI-RS port mapping	
				1	2	4	(4, 1)	(2, 2)
8TRX	1	4	0	4, 1	-	-		
			1	4, 1	-	-		
			2	4, 1	-	-		
32TRX	2	8	0	4, 1	2, 2	2, 2		
			1	2, 2	2, 2	2, 2		
			2	4, 1	4, 1	4, 1		
	4	4	0	2, 2	2, 2	-		
			1	2, 2	2, 2	-		
			2	4, 1	4, 1	-		
64TRX	4	8	0	4, 1	2, 2	2, 2		
			1	2, 2	2, 2	2, 2		
			2	4, 1	4, 1	4, 1		

Recommended panel split and corresponding *maxCsirsNumber* settings

Combinations available in 5G20B

CSI-RS ports

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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

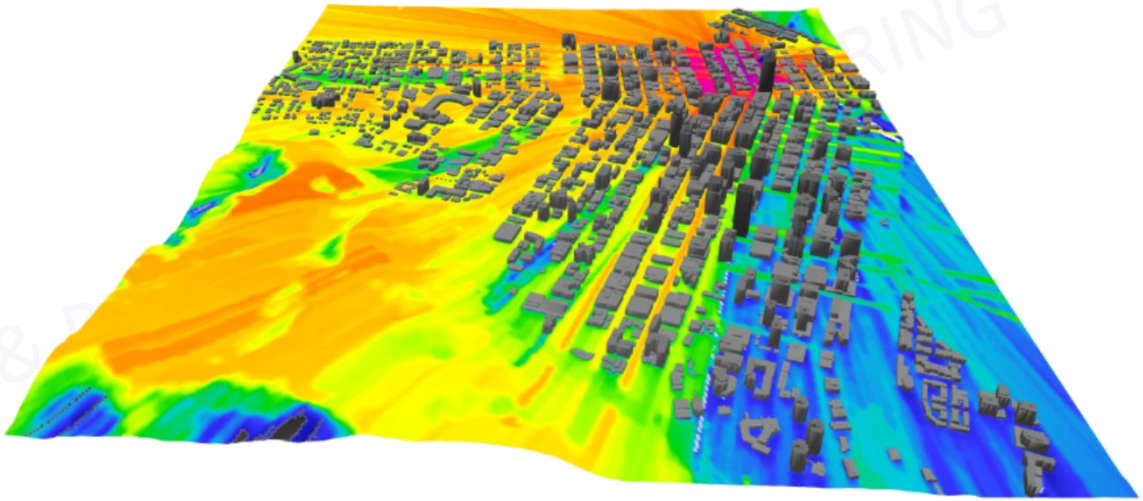
MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.periodicCsiAcquisition.maxCsirsNumber	Maximum CSI-RS number This parameter defines the maximum number of cell-specific CSI-RS for channel acquisition per UE. Note that, for each UE, the number of CSI-RS actually configured is also limited by the capability of the UE. Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	1...4 [decimal]	[-]	-

If NRCELL.dlMimoMode is 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be configured and shall be equal to 1,2 or 4 according to RU/UE capability.

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DL Power Aspects



To analyse impact of those parameters on the downlink power it is recommended to use Excel based tool **5G DL Power Calculator**, available under following LINK



Nokia internal use

Feature Configuration & Performance Evaluation

NRCELL.csirsPower Value (5GC002200 is enabled)

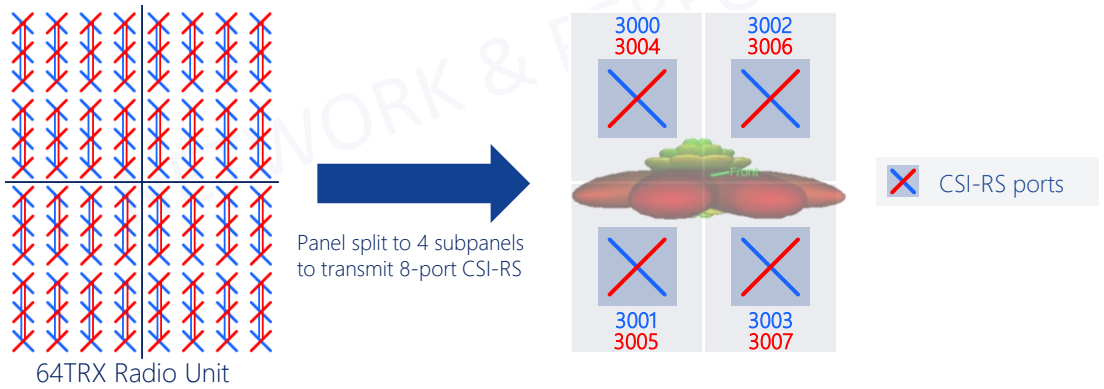
5G DL power calculator Tool output: NRCELL.csirsPower configuration value w.r.t powerControlOffsetSS and NRCELL.ssPbchBlockPower configuration

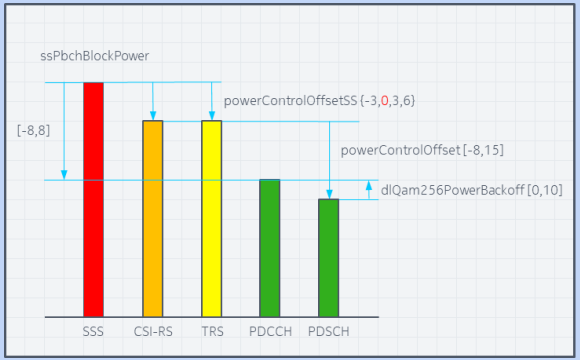
Once **NRCELL.dlMimoMode=70** (8x4 Closed Loop Spatial Multiplexing) the 5GC002200 Support 4 x 8 port CSI-RS codebook for SU-MIMO is enabled:

- CSI-RS for measurement EPRE can be configured via parameter **NRCELL.csirsPower** according to the following equation:

$$NRCELL.csirsPower = powerControlOffsetSS + (NRCELL.ssPbchBlockPower + 3dB)$$

  - **5GC002200** introduces parameter **NRCELL.csirsPower** configuration based on **powerControlOffsetSS** that can be selected from the allowed values **{-3,0,3,6} [dB]**
  - where: increasing powerControlOffsetSS values will lead to decrease Max_ssPbchBlockPower values





Nokia internal use

Feature Configuration & Performance Evaluation

NRCELL.csirsPower Value vs Max ssPbchBlockPower Value (5GC002200 is enabled)

NRCELL.csirsPower = powerControlOffsetSS + (NRCELL.ssPbchBlockPower + 3dB)

- 5GC002200 introduces parameter NRCELL.csirsPower configuration based on powerControlOffsetSS that can be selected from the allowed values {-3,0,3,6} [dB]

where: increasing powerControlOffsetSS values will lead to decrease Max_ssPbchBlockPower values

2x2 MIMO

Before 5GC002200

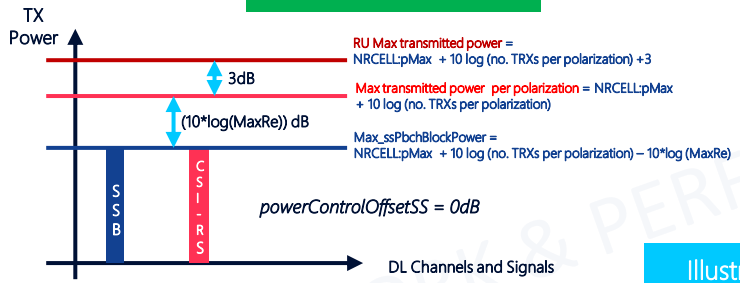


Illustration 1

After 5GC002200

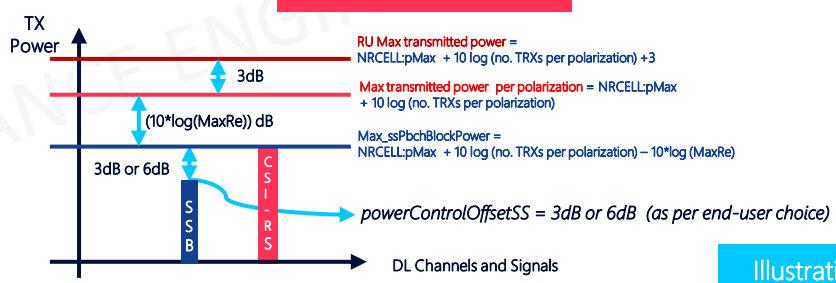


Illustration 3

4x4 MIMO

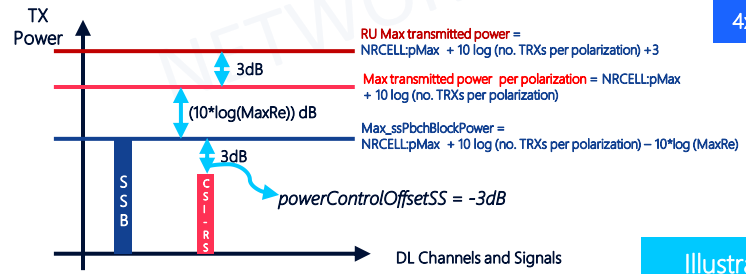


Illustration 2

Following 5GC002200 and after the introduction of parameter NRCELL.csirsPower, end-user has the choice to select powerControlOffsetSS value (communicated to UE) as per the following cases:

- powerControlOffsetSS = -3dB (as in illustration 2)
- powerControlOffsetSS = 0dB (as in illustration 1)
- powerControlOffsetSS = 3dB
- powerControlOffsetSS = 6dB

} (as in illustration 3)

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Feature Configuration & Performance Evaluation

NRCELL.csirsPower vs ssPbchBlockPower – Example UE logs

NRCELL.csirsPower = powerControlOffsetSS + (NRCELL.ssPbchBlockPower + 3dB)

- 5GC002200 introduces parameter NRCELL.csirsPower configuration based on powerControlOffsetSS that can be selected from the allowed values {-3,0,3,6} [dB]
- where: increasing powerControlOffsetSS values will lead to decrease Max_ssPbchBlockPower values

```
        dl-UL-TransmissionPeriodicity ms5,
        nrofDownlinkSlots 3,
        nrofDownlinkSymbols 6,
        nrofUplinkSlots 2,
        nrofUplinkSymbols 4,
        dl-UL-TransmissionPeriodicity-v1530 ms3
      ,
      pattern2
        dl-UL-TransmissionPeriodicity ms2,
        nrofDownlinkSlots 4,
        nrofDownlinkSymbols 0,
        nrofUplinkSlots 0,
        nrofUplinkSymbols 0
      ,
    ss-PBCH-BlockPower 5
  ,
  newUE-Identity 61,
  t304 ms2000
,
rlf-TimersAndConstants setup :
    t310 ms2000,
    n310 n10,
    n311 n1,
    t311 ms30000
,
spCellConfigDedicated
  downlinkBWP-ToAddModList
```

csirsPower=5dB
ssPbchBlockPower=2dB
Expected UE received ss-PBCH-BlockPower=5dB
powerControlOffsetSS=0dB
SSB power has not to be adjusted.

csirsPower=11dB
ssPbchBlockPower=2dB
Expected UE received ss-PBCH-BlockPower=5dB
powerControlOffsetSS=6dB
SSB power decreased by 4dB

```
        nrofUplinkSymbols 4,
        dl-UL-TransmissionPeriodicity-v1530 ms3
      ,
      pattern2
        dl-UL-TransmissionPeriodicity ms2,
        nrofDownlinkSlots 4,
        nrofDownlinkSymbols 0,
        nrofUplinkSlots 0,
        nrofUplinkSymbols 0
      ,
    ss-PBCH-BlockPower 1
  ,
  newUE-Identity 61,
  t304 ms2000
,
rlf-TimersAndConstants setup :
    t310 ms2000,
    n310 n10,
    n311 n1,
    t311 ms30000
,
spCellConfigDedicated
  downlinkBWP-ToAddModList
    bwp-Id 1,
    bwp-Common
```

```
      pattern2
        dl-UL-TransmissionPeriodicity ms2,
        nrofDownlinkSlots 4,
        nrofDownlinkSymbols 0,
        nrofUplinkSlots 0,
        nrofUplinkSymbols 0
      ,
    ss-PBCH-BlockPower 4
  ,
  newUE-Identity 61,
  t304 ms2000
,
rlf-TimersAndConstants setup :
    t310 ms2000,
    n310 n10,
    n311 n1,
    t311 ms30000
,
spCellConfigDedicated
  downlinkBWP-ToAddModList
    bwp-Id 1,
    bwp-Common
```

csirsPower=8dB
ssPbchBlockPower=2dB
Expected UE received ss-PBCH-BlockPower=5dB
powerControlOffsetSS=3dB
SSB power decreased by 1dB

- From A&S definition, with pMax 25.8 dBm, if (csirsPower-ssPbchBlockPower) > 3dB, total power cannot support SSB/CSI-RS as expected power, then **SSB power will be decreased to match pMax**.
- As noticed by A&S experience:
  - if (csirsPower-ssPbchBlockPower) = 3dB, SSB/CSI-RS align with definition
  - if (csirsPower-ssPbchBlockPower) = 6dB, SSB power will be 1dB lower than expected.

Nokia internal use

Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.periodicCsiAcquisition.csirsPower	CSI-RS power This parameter indicates the power of the cell-specific CS-RS for CSI acquisition in terms of EPRE. Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	-60...50 [decimal]	[dBm]	-

If NRCELL.dlMimoMode is equal to 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be equal to powerControlOffsetSS+(NRCELL.ssPbchBlockPower+3dB, with powerControlOffsetSS has values [-3,0,3,6]dB, else this value is invalid. powerControlOffsetSS=0 is recommended.

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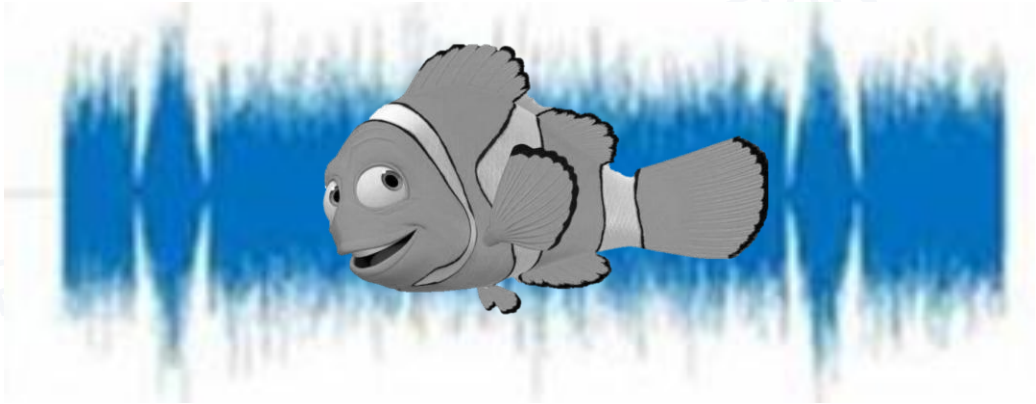
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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Power Back Off

Why PAPR matters and how to overcome fish effect?



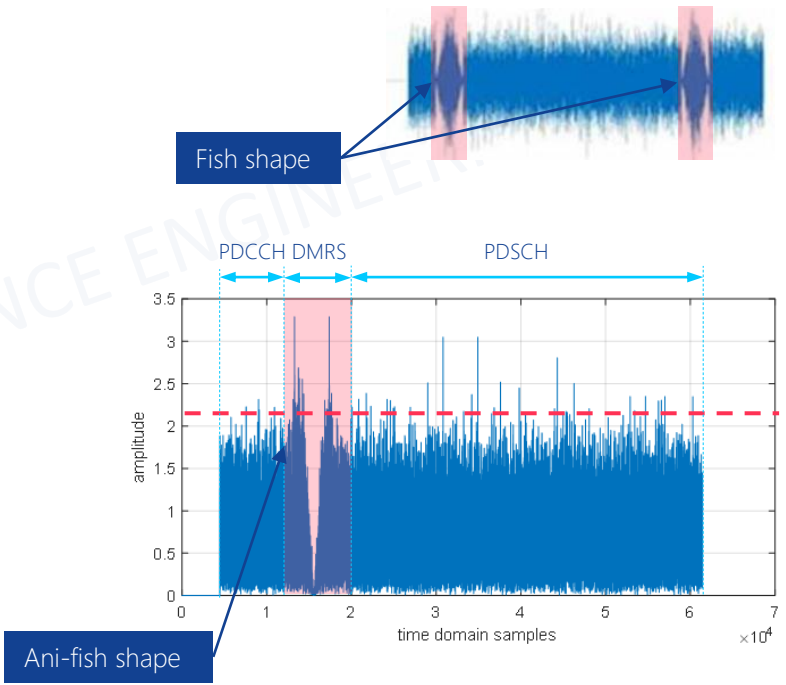
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DMRS PAPR Issue
Background information

- PA: Power Amplifier
- PAPR: Peak-to-Average Power Ratio
- EVM: Error Vector Magnitude

illustrative picture

- In 3GPP Rel 15, OFDM symbol(s) with DMRS for PDSCH experience time domain amplitude modulation **leading to a higher PAPR comparing to other symbols**. This can lead to '**Fish**' and '**Anti-Fish**' shapes in time domain, i.e., high PAPR signals.
- More Information:
  - Root cause of Fish/Anti-Fish effect **is transmission of DMRS sequence from different DMRS CDM groups in the same OFDM symbol, using same PA**
  - The DMRS PAPR issue **manifests with rank3 and rank4 MIMO transmission** (problematic DMRS ports for rank2 are avoided by scheduler algorithms).
  - Both Closed Loop (CL) and Open Loop (OL) MIMO schemes **experience DMRS PAPR issue**, except combinations of OL and beamforming.
  - **Any antenna configs that lead to rank3 or rank4 transmission** including CL 8x4 MIMO introduced by 5GC002200.
  - If no precoding is applied, then **DMRS PAPR issue doesn't appear**.



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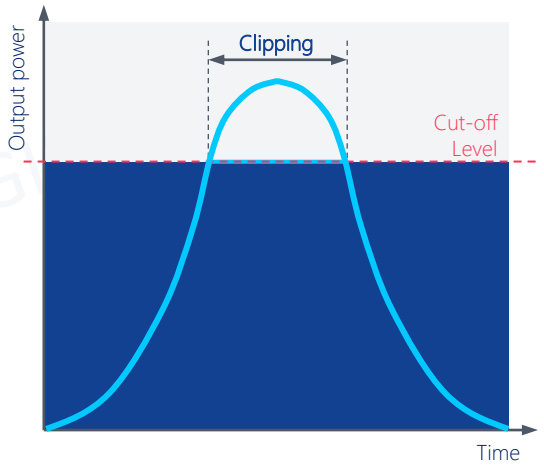
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illustrative picture

DMRS PAPR Issue

Clipping and Power Backoff

- In order to limit PA operation in non-linear region, **a clipper is implemented to clip signal amplitude**.
- When DMRS symbol with Fish/Anti-Fish shape passes through clipper, signal amplitude above clipper level is cut off, this **leads to distortion in DMRS received by UE**.
- The distortion in DMRS translates to demodulation issues, which can cumulatively be expressed using **E**rror **V**ector **M**agnitude (EVM) and clipping increase EVM, though it is still better than non-linear operation/PA saturation.
- One method of combating this DMRS PAPR issue & associated clipping is **to use more power back off in configurations** that are known to be prone to this issue, i.e., CL-MIMO including 5GC002200.
- Hence when 5GC002200 is turned ON, recommended that **dlQam256PowerBackoffSub6** be set to **3 dB**



- PA: Power Amplifier
- PAPR: Peak-to-Average Power Ratio
- EVM: Error Vector Magnitude

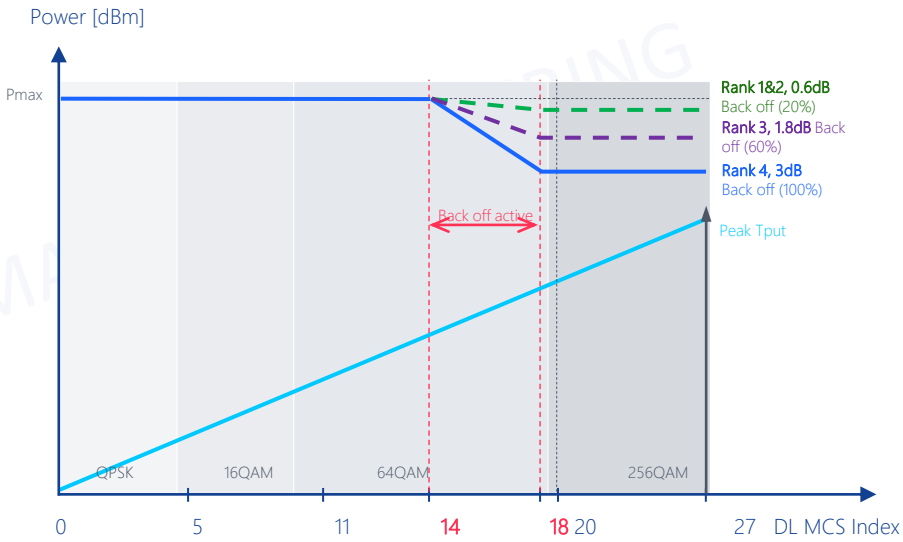
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illustrative picture

5GC002200 256QAM Power Back Off

Current algorithm overview

- To maximize DL Throughput performance, **an automatic power reduction is applied on cells**, where **256QAM modulation scheme is enabled**.
- The algorithm utilizes **a gradual power backoff** starting at MCS14 and completing at MCS18.
- At MCS 19 and above Maximum power back off is applied.
- Maximum power back off is determined by parameter **dlQam256PowerBackoffSub6**
- Algorithm was enhanced in 20B tailoring power back off per rank:
  - **Rank 4:** full back off per parameter dlQam256PowerBackoffSub6
  - **Rank 3:** 60% of full back off value
  - **Rank 1 and 2:** 20% of full back off value



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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.dlQam256PowerBackoffSub6	Power backoff in DL bands below 6GHz This parameter defines the power backoff applied for UEs supporting the 256QAM scheme in DL bands of below 6GHz	0...10 [decimal]	[-]	1.5

If NRCELL.dlMimoMode is 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be configured and shall be equal to recommended value 3.

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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Interaction with DRX



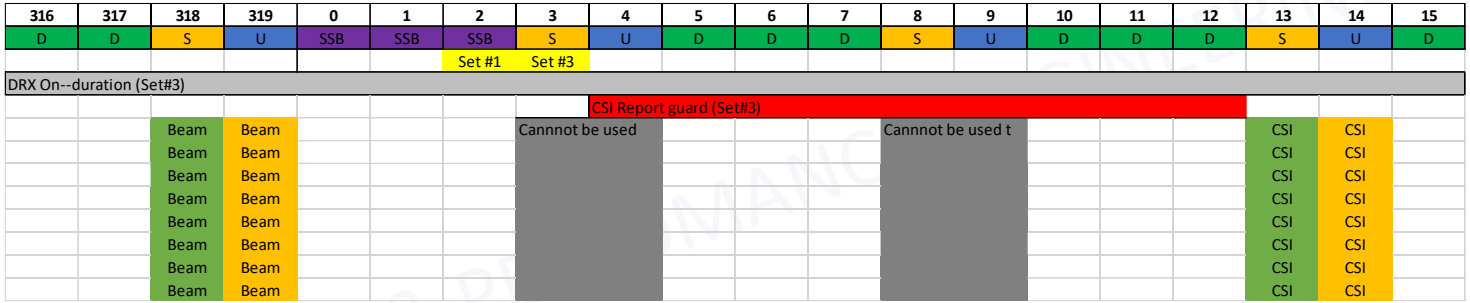
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Feature Configuration & Performance Evaluation

cDRX and Number of CSI-RS resources configured to the UE

- During the DRX onDuration of 10 ms the **following must be satisfied**:
  - The UL SSB beam report transmission occasion
  - The DL CSI-RS transmission occasion
  - The UL CSI report occasion (e.g. with PMI beams)
  - UE guard for CSI-RS processing



- Given the constraints of accommodating the beam reports and CSI reports among the available PUCCH resources within the DRX ON duration **limits the number of UEs that can be accommodated with 4 CSI-RS resources**.
- The number of UEs that can accommodate 2 and 4 CSI-RS resources **is dependent on**:
  - Frame format: tdLTE uses two consecutive UL subframes which is more limiting
  - 5GC002545 (multiplexing of CSI reports on PUCCH). If 2545 is enabled, more UEs can be supported.
    Note: Enabling 2545 for CPRI radios may impact UL coverage (due to PUCCH being on sector wide beam versus SSB)
  - 1 UL sub-cells vs. 2 sub-cells configuration
  - DRX ON duration

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Feature Configuration & Performance Evaluation

cDRX and Number of CSI-RS resources configured to the UE

- Based on a 10ms DRX ON duration, 160 ms CSI-RS transmission period and CSI/Beam report period of 160 ms, the number of UEs that can support 4, 2 and 1 CSI-RS resource sets is illustrated by the tables below.
- As an example, if using the 1208 frame structure with CPRI radio and small inter-site distances (e.g. not concerned about loss of PUCCH coverage) then
  - The first 16 RRC connected UEs can be configured with 4 CSI-RS resources
  - The next 56 RRC connected UEs can be configured with 2 CSI-RS resources
  - The next 224 RRC connected UEs can be configured with 1 CSI-RS resources

  Note if you exceed the limit of RRC connected UEs which can support 1 CS-RS resource than DRX is turned off for remaining RRC connected UEs.

1208 frame format (i.e., Korea)				
	Number of rrc connected UEs			
# of CSI_RS resources	2545 enabled w/ 2 subcells	2545 enabled w/ 1 subcell	2545 disabled w/ 2 subcell	2545 disabled, w/1 subcell
4	16	8	4	2
2	56	28	14	7
1	224	112	56	28

CPRI & small ISDs | eCPRI | CPRI AEQA, AEQD & AEHA

tdLTE frame format (i.e., Vodafone, Orange, TMO)				
	# of rrc connected UEs			
# of CSI_RS resources	2545 enabled w/ 2 subcells	2545 enabled w/ 1 subcell	2545 disabled w/2 subcells	2545 disabled, 1 subcell
4	8	4	2	1
2	36	18	9	4
1	192	96	48	24

eCPRI | CPRI AEQA, AEQD & AEHA

- 01797494* is open to improve this functionality increasing the above numbers by a factor of 4 by repeating CSI-RS transmissions. Note that this increases the CSI-RS overhead and can slightly decrease DL throughput.
- 2 sub cells for CPRI radios is supported for AEQA, AEQD and AEHA radios.

* - Interworking issues among 5GC002200 and CDRX

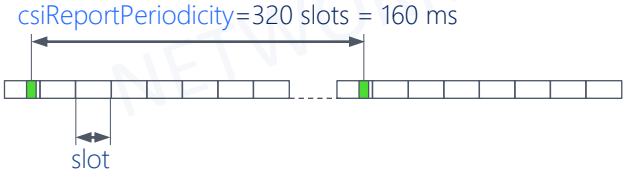
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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step		Unit	Default
NRCELL.periodicCsiAcquisition.csiReportPeriodicity	CSI reporting periodicity This parameter indicates the CSI reporting periodicity	Value	Name	[-]	320
		20	slots20		
		40	slots40		
		80	slots80		
		160	slots160		
		320	slots320		



For each NRCELLGRP and each NRCELL instances belonging to that NRCELLGRP instance (known via NRCELLGRP.nrCellList),
- All these NRCELL instances shall have the same value for NRCELL.periodicCsiAcquisition.csiReportPeriodicity
- If NRCELLGRP.scs = 30 (corresponding to 30 KHz) and NRCELL.prachConfigurationIndex = 90, 91, 94 or 98, this parameter shall be greater or equal to Max (NRCELLGRP.ssBurstSetPeriod * 2, 20)
- If NRCELLGRP.scs = 120 (corresponding to 120 KHz) and NRCELL.prachConfigurationIndex = 38, this parameter shall be greater or equal to Max (NRCELLGRP.ssBurstSetPeriod * 8, 160)

This parameter shall be a multiple of NRCELL.drxProfile1.drxLongCycle (NRCELL.periodicCsiAcquisition.csiReportPeriodicity(ms) = N* NRCELL.drxProfile1drxLongCycle with N=1, 2,...16) if NRCELL.actCDrx is equal to TRUE and
- If NRCELL.nrarfcn shall be<= 800000 (FR1) and NRCELL.drxProfile1.drxLongCycle is smaller or equal to 7 (corresponding to 80ms)
- If NRCELL.nrarfcn>= 2016667 (FR2) and NRCELL.drxProfile1.drxLongCycle is smaller or equal to 3 (corresponding to 40ms)

Note: (ms) above denote the corresponding parameter values converted from slot to ms as follows:
- If NRCELLGRP.scs is equal to 15 (corresponding to 15 KHz), 1slot is equal to 1ms
- If NRCELLGRP.scs is equal to 30 (corresponding to 30 KHz), 1slot is equal to 0.5ms;
- If NRCELLGRP.scs is equal to 120 (corresponding to 120 KHz), 1slot is equal to 0.125ms

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Configuration Management <5GC002200>

New parameters

Online Modifable | Requires Object Locking | Requires BTS Restart

MOC.{Structure}.Abbreviated name	Full name/Description	Range and Step	Unit	Default
NRCELL.periodicCsiAcquisition. csirsNumberWithinReportPeriod_Set1	CSI-RS number for CSI-RS resource set1 within one report period This parameter defines the number of CSI-RS for CSI-RS resource set1 (1 CSI-RS resource in this set that is configured to UE) within one report period. Configuring this parameter during cell addition will not provoke gNB reset. Updating this parameter will provoke gNB reset. Removal of this parameter during cell deletion will not provoke gNB reset.	1...5 [decimal]	[-]	-

If NRCELL.dlMimoMode is 70 (corresponding to 8x4 Closed Loop Spatial Multiplexing), this parameter shall be configured and shall be equal to recommended value 5.

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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Interworking with other features



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5GC002200 Interworking Matrix

Feature deployment with CPRI or eCPRI Radio Units

- 5GC002200 can be enabled
- 5GC002200 can be enabled with restrictions
- 5GC002200 cannot be enabled
- 5GC002200 enabled together delivers high gain in field

	Feature ID	533	534	772	2545	2534	1348	2015	679	2537
	Feature Name	Digital Beamforming for CPRI based RUs	Digital Beamforming for eCPRI based RUs	Common DRX	FDM of period. CSI reports on PUCCH for FR1 cells	PDSCH at SSB slots	Uplink channel multiplexing	PDSCH FDM on DMRS	UL/DL FDM scheduling	FDM allocation enhancem.
CPRI	Feature can be enabled?	Yes, w/ restrictions	No	Yes, w/ restrictions	Yes	Yes	Yes	Yes	Yes	No
CPRI	Comments	Beam refinement cannot be fully enabled w/ SDX55. Please refer to „Known Issues" subchapter.		Fall back to #2 CSI-RS resources observed . Please refer to „Interaction with cDRX" subchapter.	Trade-off: UL coverage degradation vs decrease fallback to 2CSI-RS res.* „Interaction with cDRX" subchapter.	Beam refinement cannot be activated together with 2534				Not supported.
eCPRI	Feature can be enabled?	No	Yes, w/ restrictions	Yes, w/ restrictions	Yes	No	Yes	Yes	Yes	Yes
eCPRI	Comments		Beam refinement cannot be fully enabled w/ SDX55. Please refer to „Known Issues" subchapter.	Fall back to #2 CSI-RS resources observed . Please refer to „Interaction with cDRX" subchapter.	Trade-off: UL coverage degradation vs decrease fallback to 2CSI-RS res.* „Interaction with cDRX" subchapter.	No supported in 5G20B				

*-under investigation

Please note the matrix is created based on live network observations (drive tests) and specific software version and may not reflect current status of the product (status: 09/04/2021)

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Feature Configuration & Performance Evaluation

Potential impact of FDM features on 5GC002200 performance

	679	2200	2200+679	679+2537	2200+679+2537
CPRI/eCPRI fronthaul	CPRI			eCPRI	
mutliplexing capability in the same slot?	yes	no	yes	yes	yes
mulitplexing capability in different beams?	no	no	no	yes	yes
max number of beams	32	256	256	32	256
FDM scheduling opportunities	medium	no	low	high	high

5GC000679 UL/DL FDM scheduling
5GC002537 FDM allocation enhancements

- Number of FDM scheduling oportunities decreases with higher number of beams in 5GC002200
- Not an issue with eCPRI 2537.

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20B TDD Key Features, Dependencies and Limitations

DL Feature	Gain	Limitations
CB006555 Adaptive PDCCH	**1.5 to 4%** for low number of users, dependent on frame configuration	For CPRI, >3 RRC connected Ues gain is 0 automatically enabled
5GC002015 PDSCH on DMRS	**0 to 4%** dependent on percent of rank 1 &2	none
5GC002534 PDSCH on SSB Slots	**4 to 8%** dependent on #configuration	CPRI only, beam refinement (5GC000533) must be off
5GC002200 8 port PMI BF	**4 to 10%** versus GoB. Less if compared to refined beams	Requires Exynos SW upgrade Requires SDX55 SW upgrade if used with refined beams.

UL Feature	Gain	Limitations
5GC001348 PUSCH on PRACH	**0 to 14%** dependent on radio and configuration	Can not get full gain with CPRI & MADE/PULSAR radios

Feature	Gain	Limitations
5GC002537 FD Enhancements	Improves multi-UE FDM by removing same beam limitation	eCPRI Only

Key UE compatibility Items with 2200

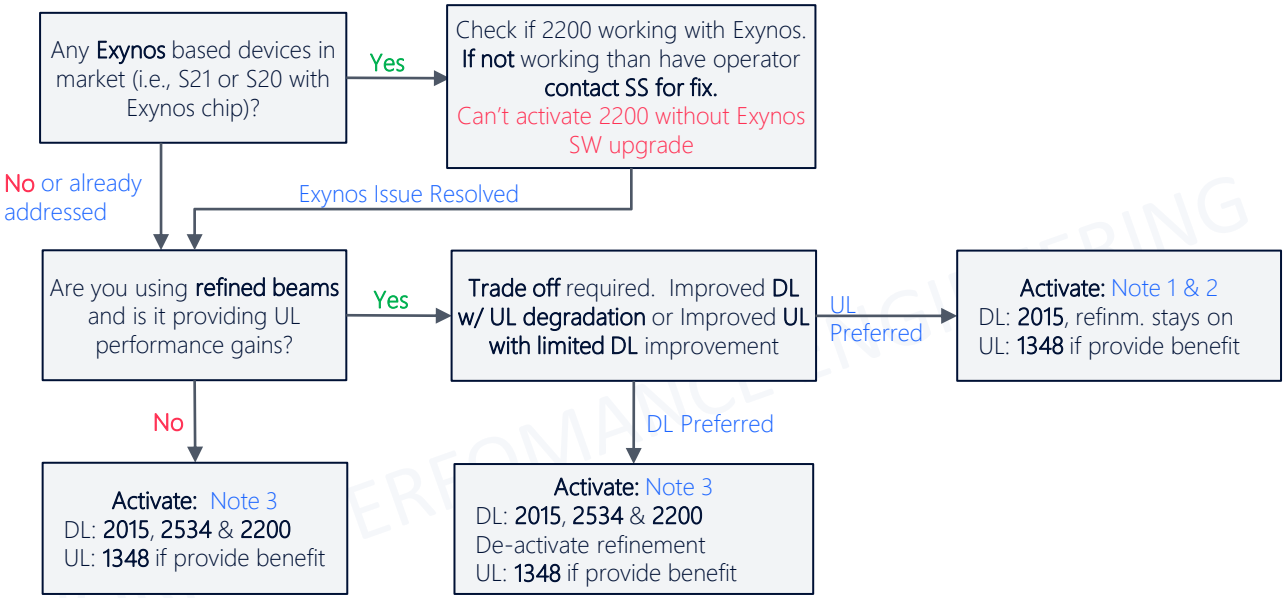
1. Samsung S20 & S21 based on Exynos devices does not work with 2200 with older Exynos SW
2. SDX55 based devices with older SW can't support 2200 w/ 4CSI_RS resources and refined beams

Change with 20B: In 20B we lose SRS based refinement and some UEs do not support CRI refinement. i.e., SS Exynos based devices (S20 and S21) only support up to 16 refined beams (4 SSBs)

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20B TDD Performance Decision Matrix for CPRI Radios



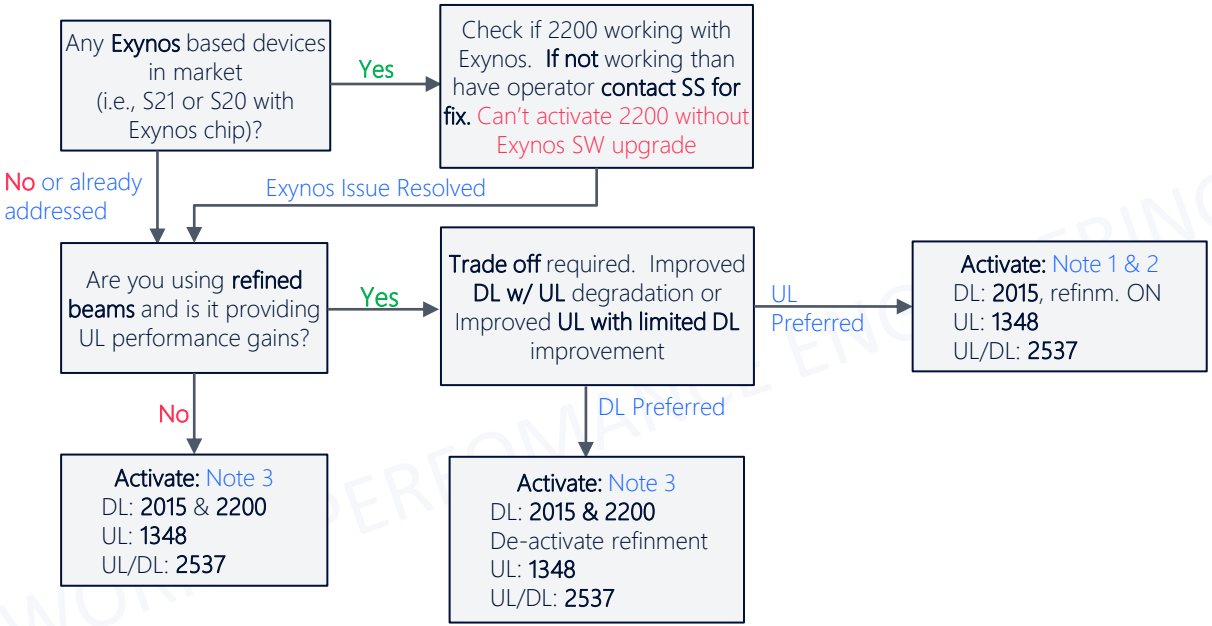
Note 1: Current assumption is 2200 can not be activated with refined beams as its likely SDX55 based devices without fix deployed.
Re-visit if/when we validate 2 resource case provides gains and/or if we think its realistic a market will have updated SW for all SDX55 devices

Note 2: Option for investigation: Test with 4 SSBs plus refinement and compare UL coverage and performance versus currently deployed SSB configuration. If 4 SSBs provides acceptable performance this configuration would allow CRI refinement for Exynos devices which can only support up to 16 refined beams

Note 3: If Exynos Issue not resolved you can move forward with other 20B features and disable 2200.

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20B TDD Performance Decision Matrix for eCPRI Radios



Note 1: Current assumption is **2200 can not be activated with refined beams** as its likely SDX55 based devices without SW fix deployed. Re-visit if/when we validate 2 resource case provides gains and/or if we think its realistic a market will have updated SW for all SDX55 devices

Note 2: Option for investigation: **Test with 4 SSBs plus refinement** and compare UL coverage and performance versus currently deployed SSB configuration. If 4 SSBs provides acceptable performance this configuration would allow CRI refinement for Exynos devices which can only support up to 16 refined beams

Note 3: If Exynos Issue not resolved you can **move forward with other 20B features and disable 2200**.

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Feature Configuration & Performance Evaluation

5GC002200 Support 4x8 port CSI-RS codebook for SU-MIMO

Known Issues



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Feature Configuration & Performance Evaluation

Known compatibility issues (and proposals of solution) reported by via SSD Central

- UE Samsung S20/S21 (Exynos chipset)
  - S20/S21 UE based on Exynos does not work well with the 5GC002200 feature enabled, always reporting rank 1. Samsung may have provided a SW fix for this. No immediate solution in short term.
    It is thus not recommended to activate 2200 in commercial networks with S20/S21 UE (exynos chipset) as the Tput degradation is too big (divided by 3).
- UE SDX55 does not support both 2200 and 533 (CSI-RS beam refinement)
  - Endless loops of Sgnb release/addition. PDL rule has been implemented in MP1 to avoid configuring both 2200 and 533. However it has been seen after that SDX55 UE would be able to work with 2 CSI-RS and beam refinement. Therefore, a new rule will be implemented in MP2 to allow 533 + 2200 with 2 CSI-RS only. This should allow to keep both features together and thus recover UL Tput with 533 CSI-RS beam refinement. QCT acknowledged they have an issue and they will fix it, but deployment in all UEs will take long time.

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Please note recommendations are based on live network observations and specific UE/product software version and may not reflect current status of the product (status: 15/04/2021)

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Observed Performance Results

- Status: 26.02.2021
- The measurement results are based on specific software version.
- The presented problems may not reflect a current status of the product.



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5GC002200 Testing Summary

Over-The-Air (OTA) testing

- RIT test area in Hangzou, China — Type: Drive Test — RU: AEQV 32TRX

Main results:

↪ Gains in rank1 and rank2 area around 20%

↪ Almost No gain in rank3 and rank4 regime

- FiVe Espoo — Type: Drive Test — RU: AEQE 64TRX

Main results:

↪ Same as RIT

- Customer Network — Type: Drive Test+Field KPI — RU: AAHF split mode

Main Results:

↪ With tilt 0^0 settings around 11% gain observed with average rank2

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5GC002200 OTA test description with AEQV

32 TRX radio, band n77, 100 MHz BW, beamset_6

OTA testing

- RIT test area in Hangzou, China
- Two drive test routes
  1. Inside adjacent parking lot; line-of-site
  2. Outside parking lot; mostly non-LOS

Testing parameters

- 5GC2200 disabled – for baseline; PDSCH uses SSB beams
- 5GC2200 enabled with 4 CSI-RS resources
  - Supported by UE's with SDX55, SDX60 and later, Media-TEK
- 5GC2200 enabled with 2 CSI-RS resources
  - Supported by SDX50



(1) Inside parking lot



(2) Outside parking lot

RF cell configuration	
Antenna height	23m
Azimuth	275^0
Mechanical down tilt	5°
Electrical downtilt	6°

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5GC002200 OTA test results with AEQV

32 TRX radio, band n77, 100 MHz BW, beamset_6

Line of Site Route

- Smaller gains for 5GC2200
- Slight improvement in MCS
- Similar rank

Test case	NR PHY Tput (Mbps)	NR PHY improvement	Rank	MCS
2200 disabled	727	baseline	3.5	17.4
2200 enabled 2 CSI-RS resources	732	1%	3.4	18.4
2200 enabled 4 CSI-RS resources	755	4%	3.5	18.4

Non-LOS Route

- Larger gains for 5GC2200
- Larger improvement in MCS
- Similar rank

Test case	NR PHY Tput (Mbps)	NR PHY improvement	Rank	MCS
2200 disabled	363	baseline	2.7	10
2200 enabled 2 CSI-RS resources	407	12%	2.45	13.8
2200 enabled 4 CSI-RS resources	456	26%	2.7	13.8

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References and acknowledgments

Reference	Version	Author(s)	Link	Date of access
5GC002200 CFAM	1.9	Xukai Gao, CFAM Team	Link	
5G counters and KPIs	5.0	Poul Larsen		
5GC002200 _MoRSE_simulation_report_renew	0.4	Jacob Tao		
LTE430416TRX FDD mMIMO: DL SU-MIMO 2 layer with Rel13 codebook PoC	0.2	Adam J. Pomianek	Link	
FSY#0682020 – Rel-15 DMRS optimization for BF/MIMO	1.0	Michal Maternia and co-authors		
Power Control	1.0	George Samuel		
TMO US 5GC002200 testing reports and raw data sharing	1.0	Arvind Krishnamoorthy MN RAN Spec Field Performance		
Feature Configuration & Performance Evaluation - additional inputs	1.0	MN RAN Spec Cap&Perf MN RAN Spec Field Performance		

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